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REV

DESCRIPTION

DATE

APPROVED

A

INITIAL RELEASE

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QUALCOMM

2025-04-21 10:44 PDT

tingweiz

HAMOA IPC BGA MODULE, P2

CONTRACT NO.

QUALCOMM TECHNOLOGIES, INC.

DRAWN BY
QSIP_HW

DATE
04/21/2025

ENGINEER

PROJECT ENGINEER

QUALITY ASSURANCE

CONFIGURATION

DESIGN ACTIVITY APPROVAL

DESIGN APPROVAL

TITLE
SCHEMATIC, CCA, HAMOA SIP-A MODULE

SIZE
D

DRAWING NO
LD20-90411-764

REV
A

SCALE
NONE

RELEASE DATE

SHEET 1 OF 62

TITLE

8

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Qualcomm
2026-04-01 00:53:44 PDT
tingweiz

BLOCK_DIAGRAM

SIZE		DRAWING NO	REV
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SCALE	NONE	SHEET 3	OF 62

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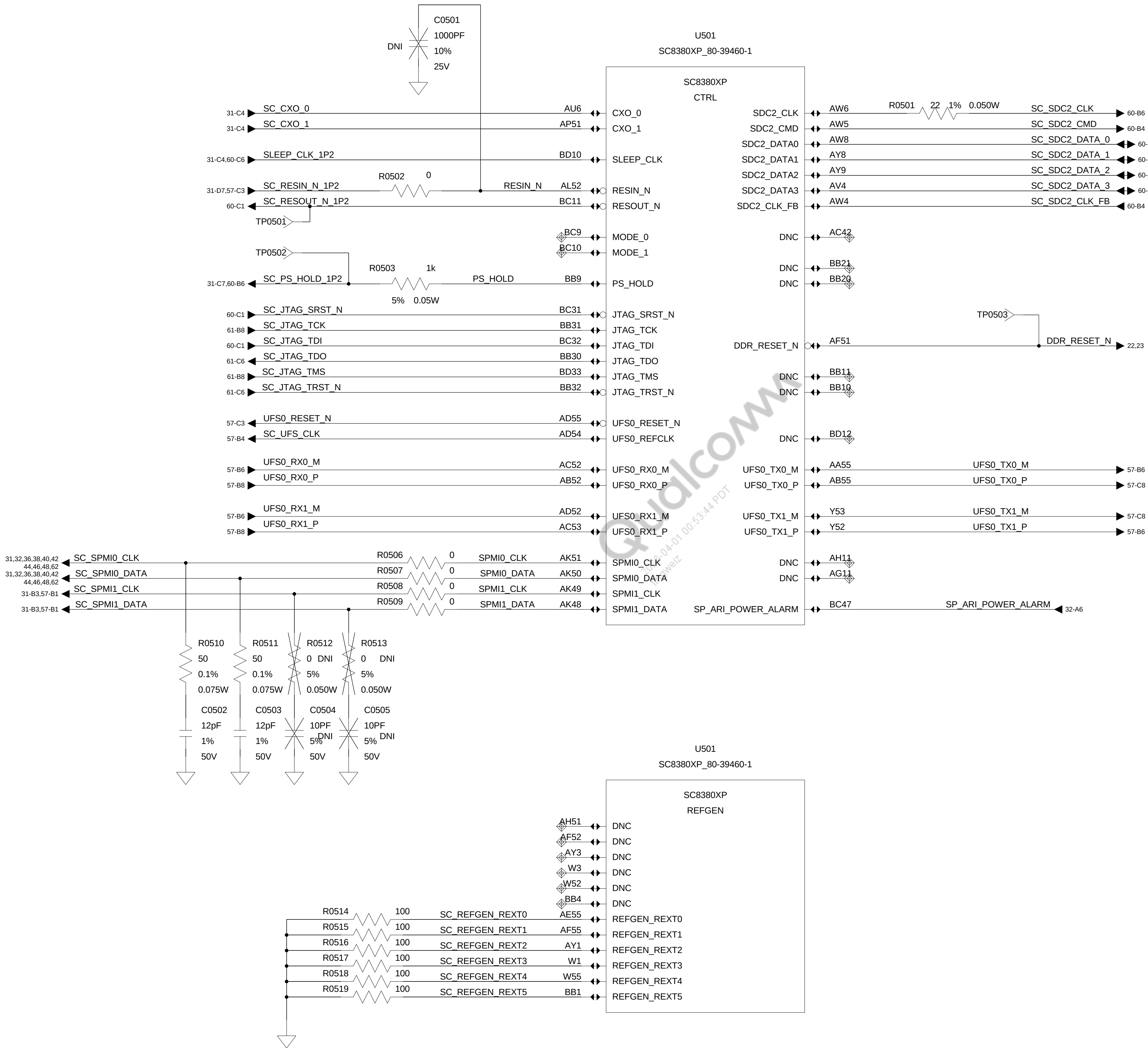
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RESERVED_01

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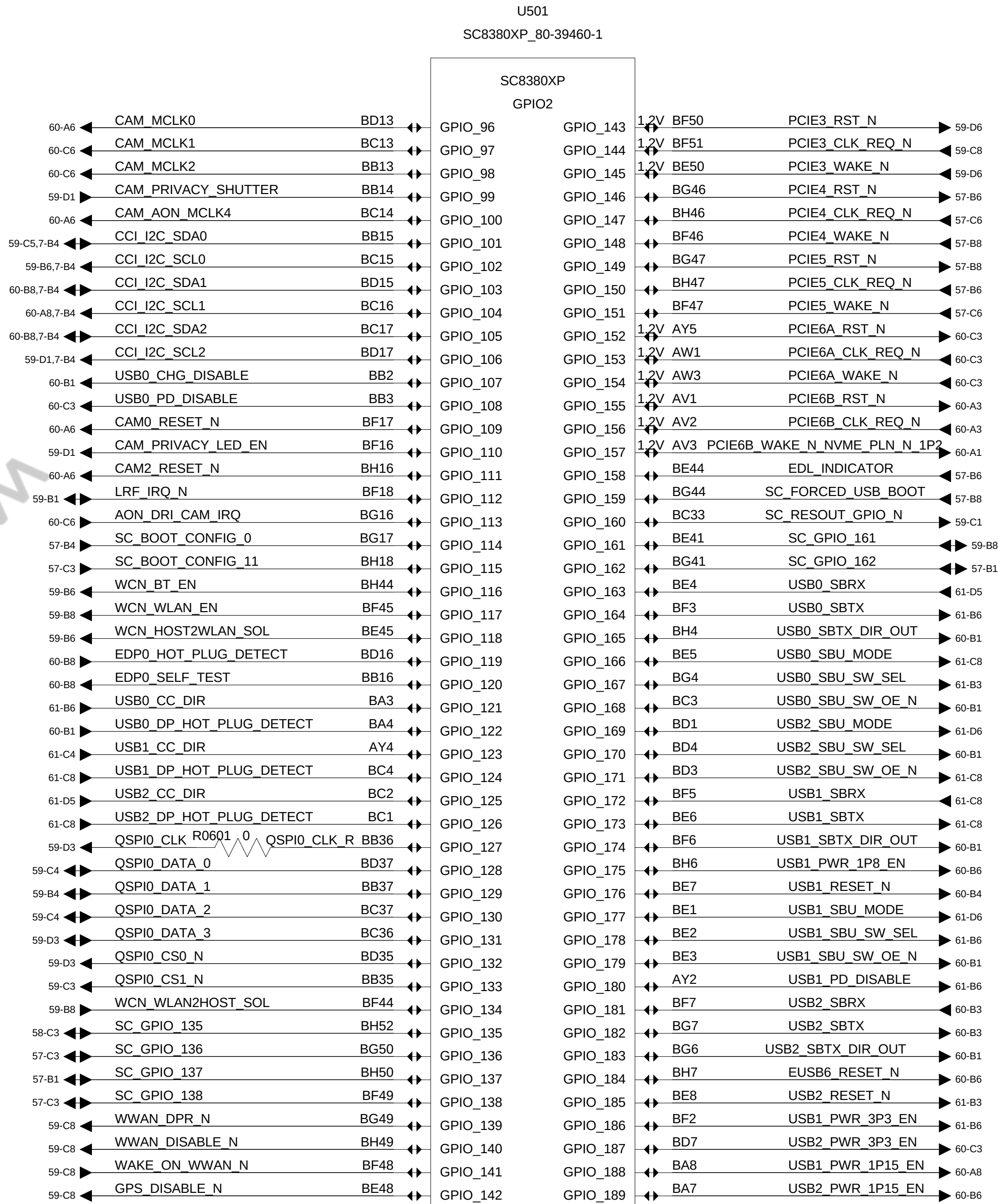
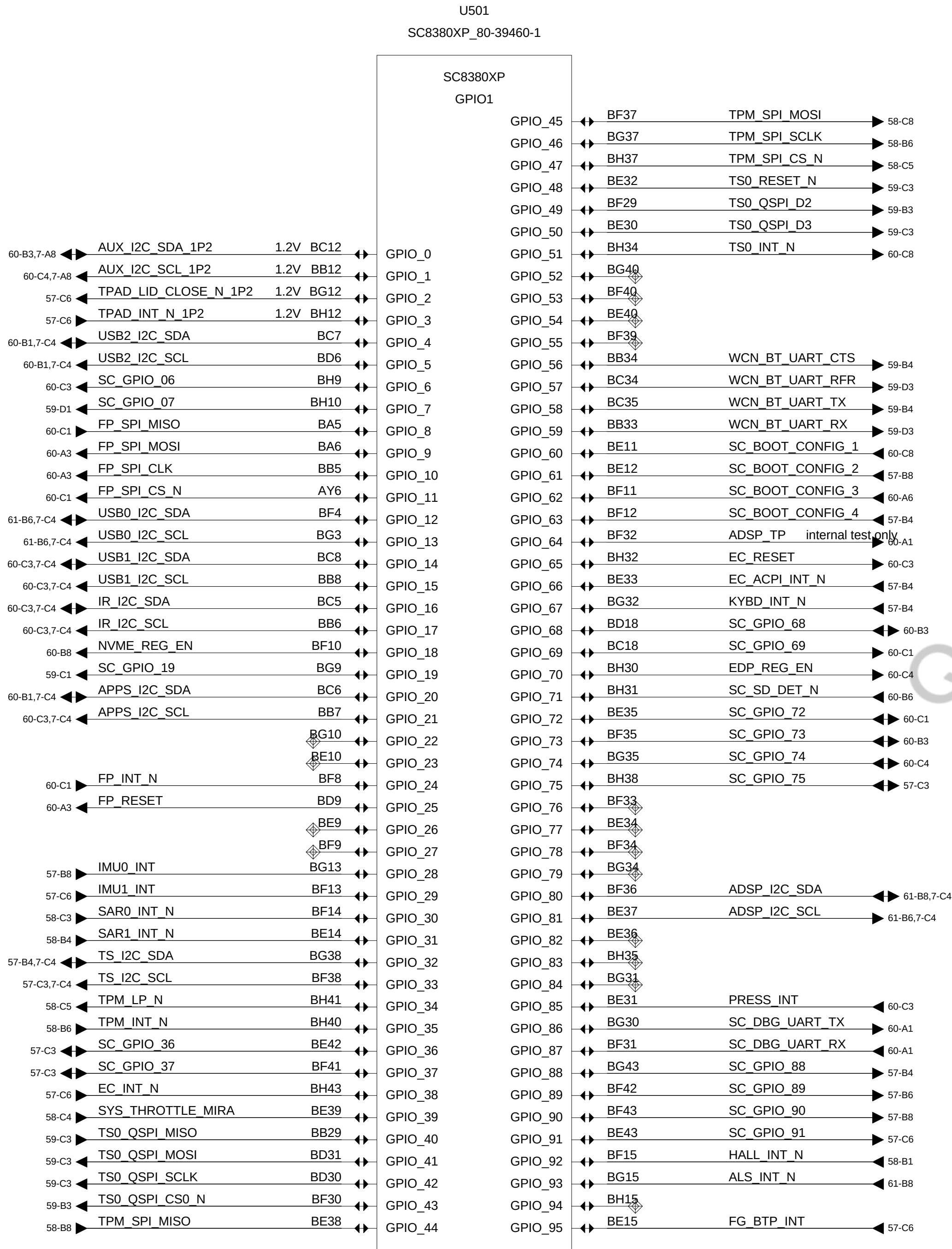
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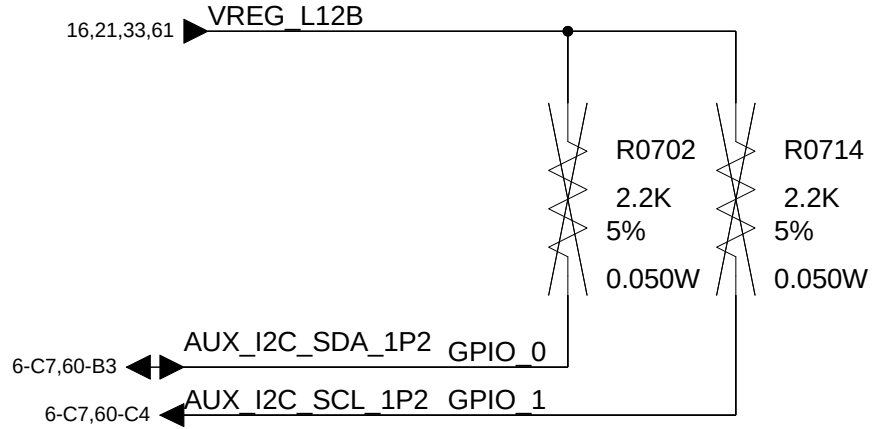
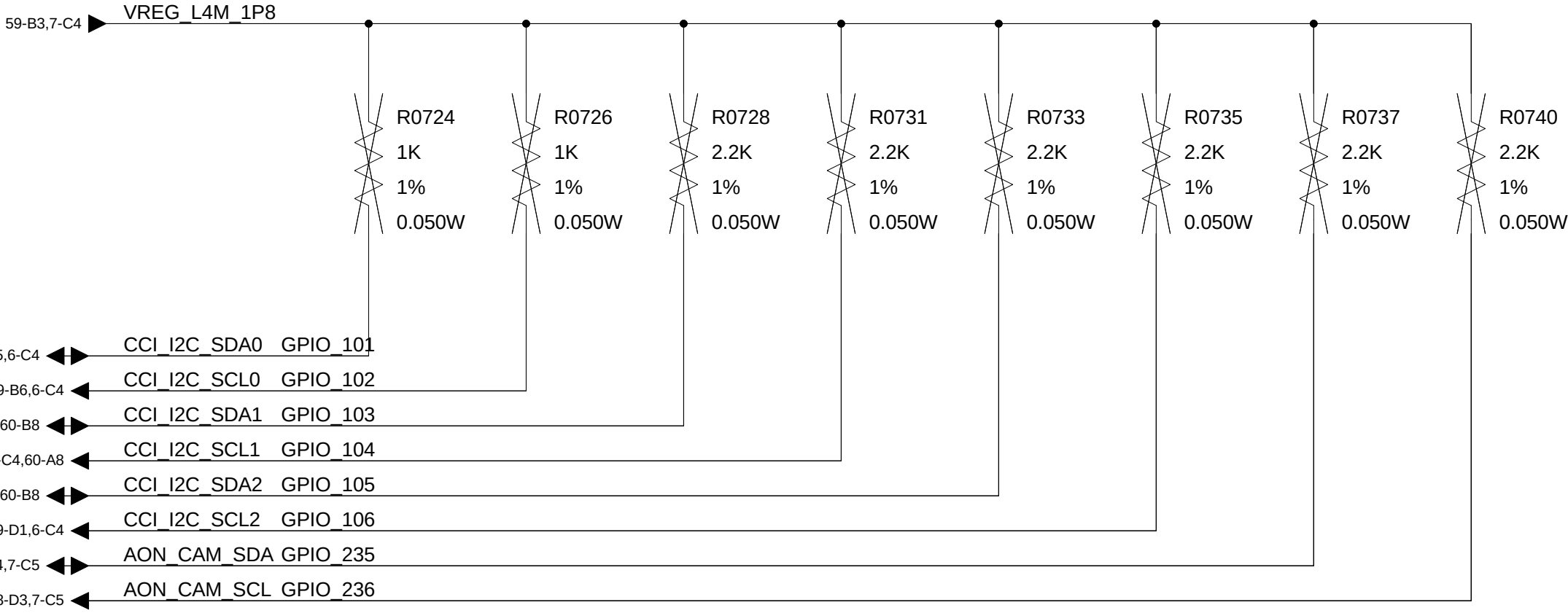
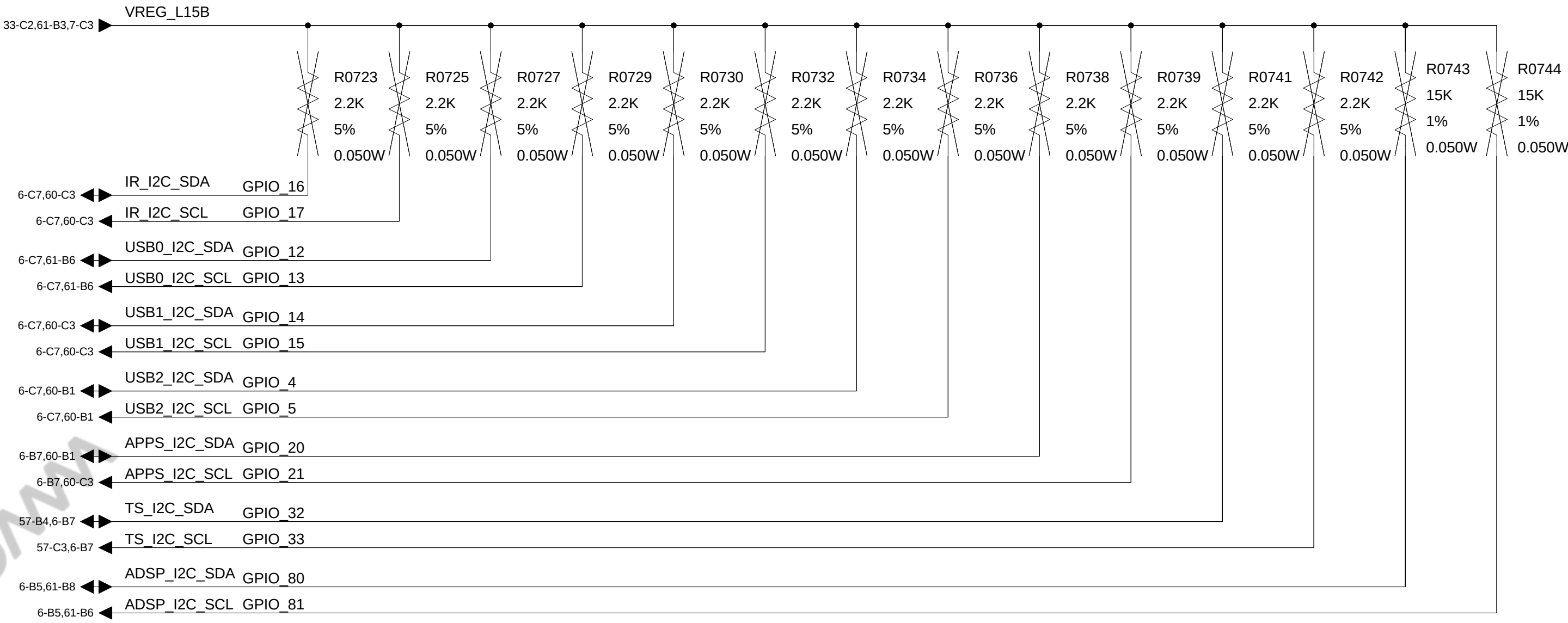
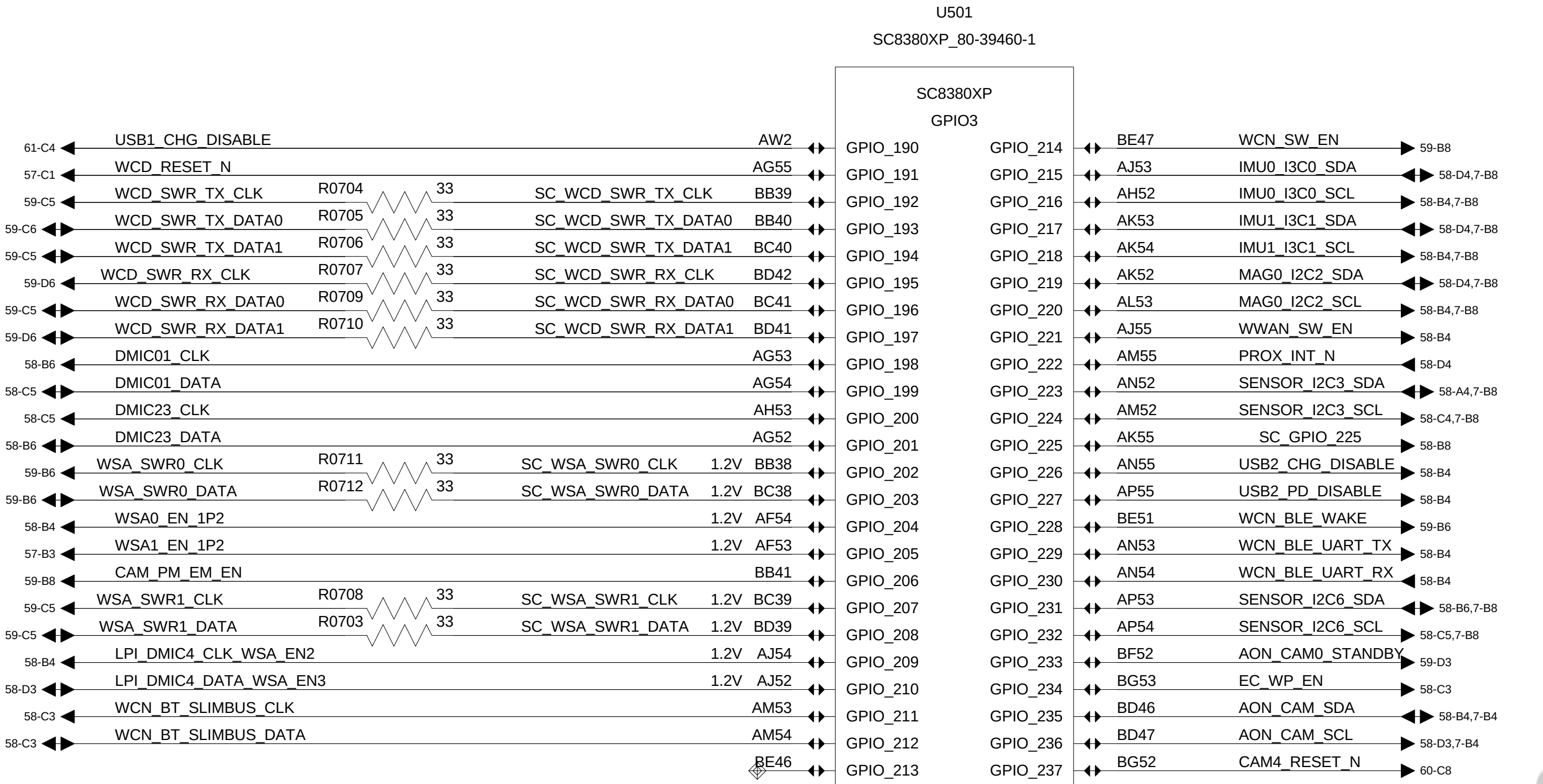
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REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		



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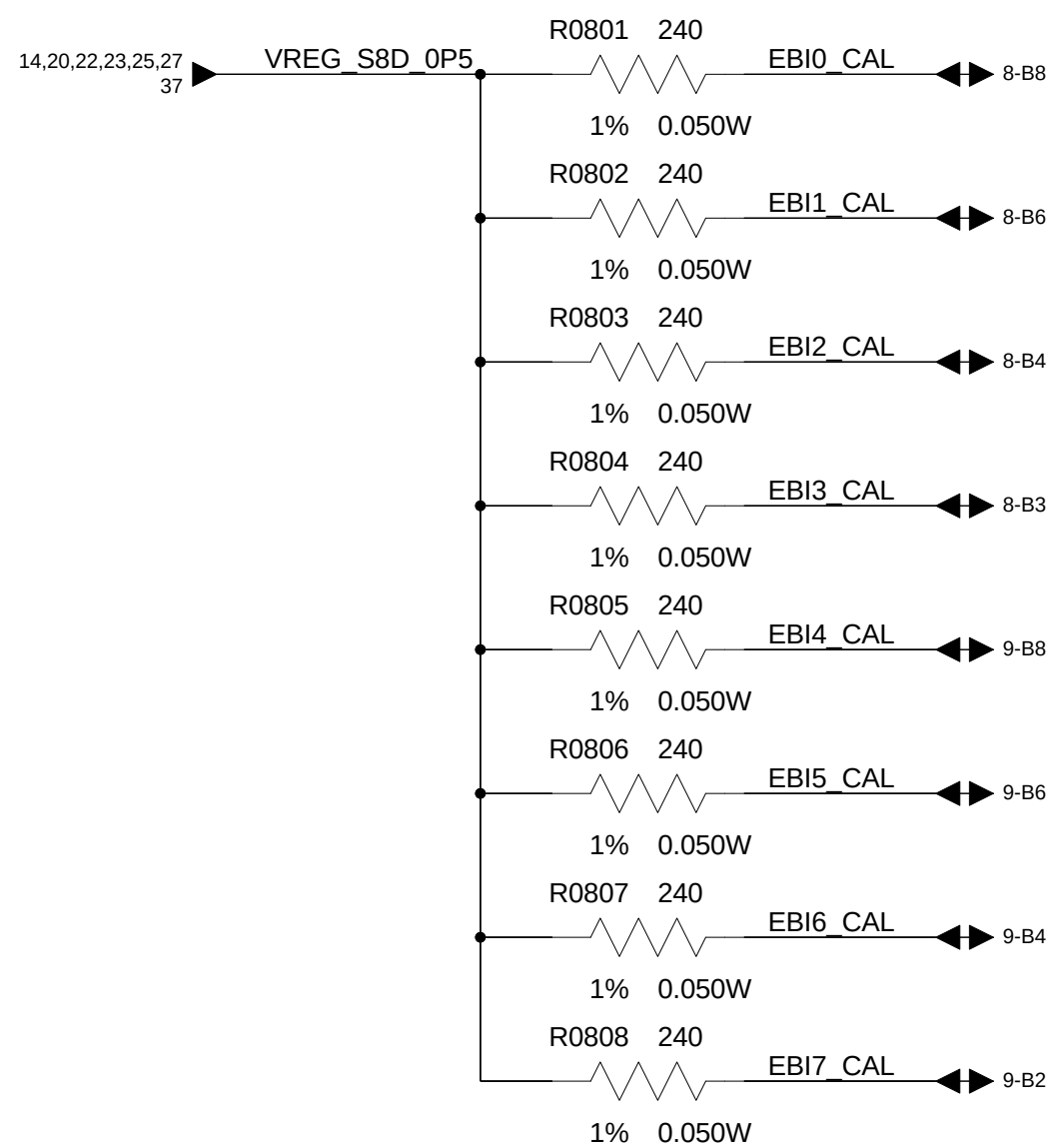
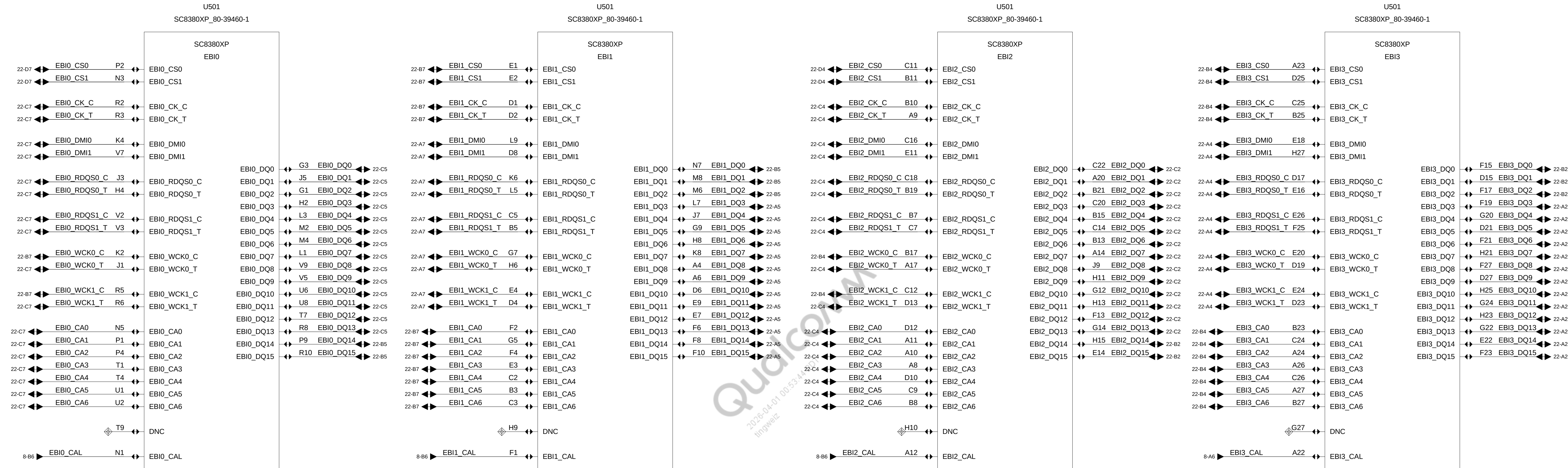
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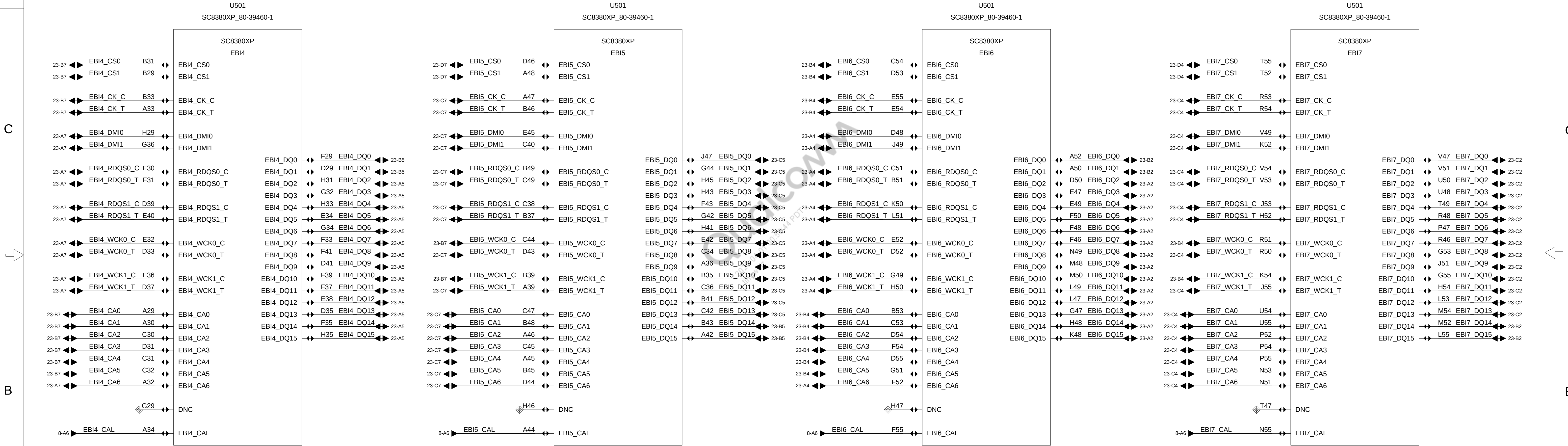
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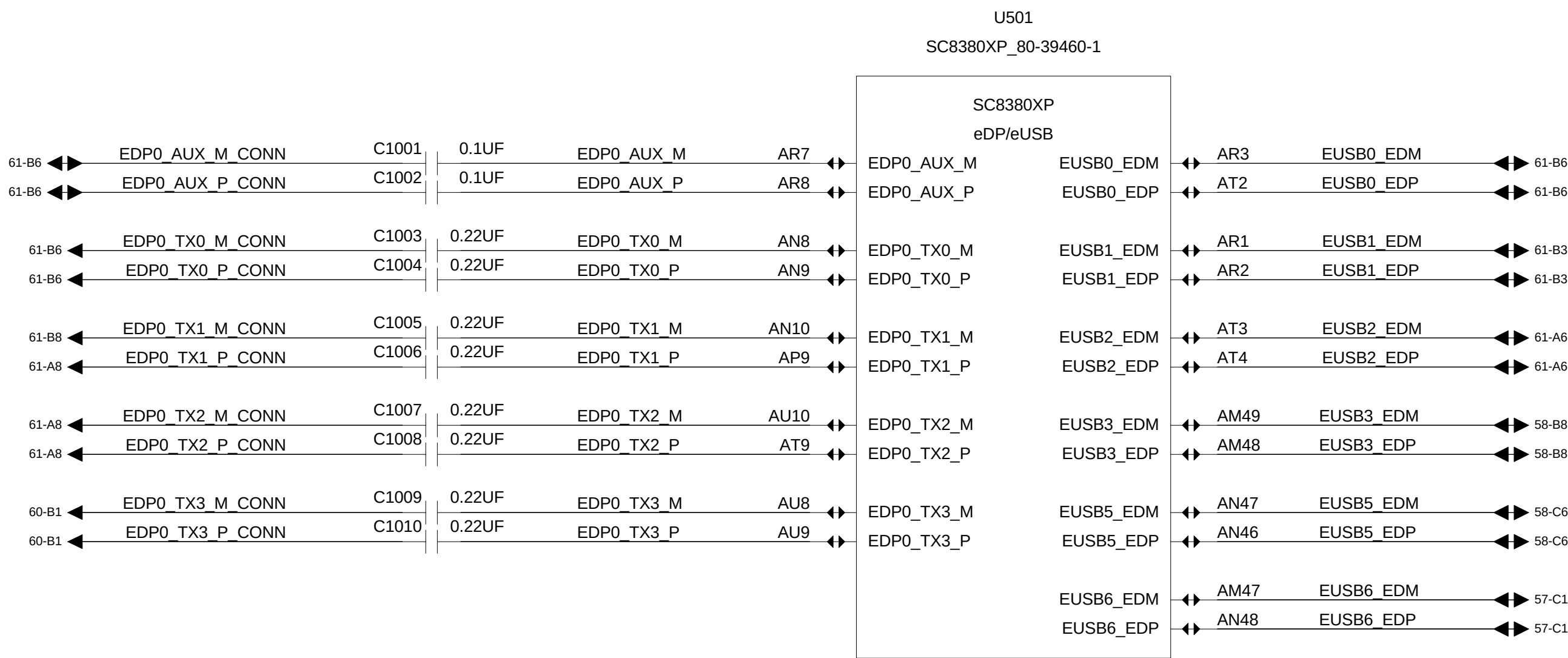
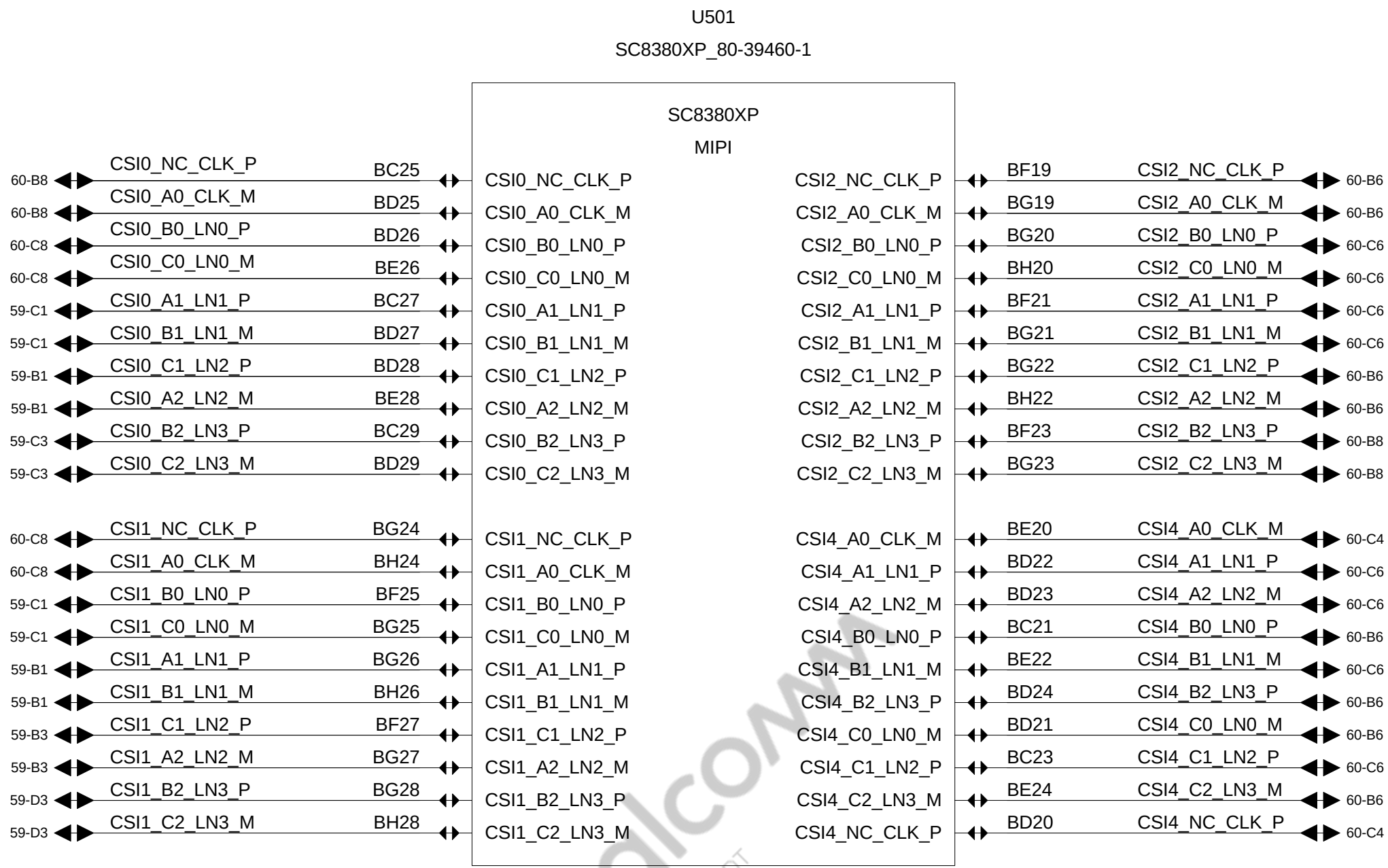
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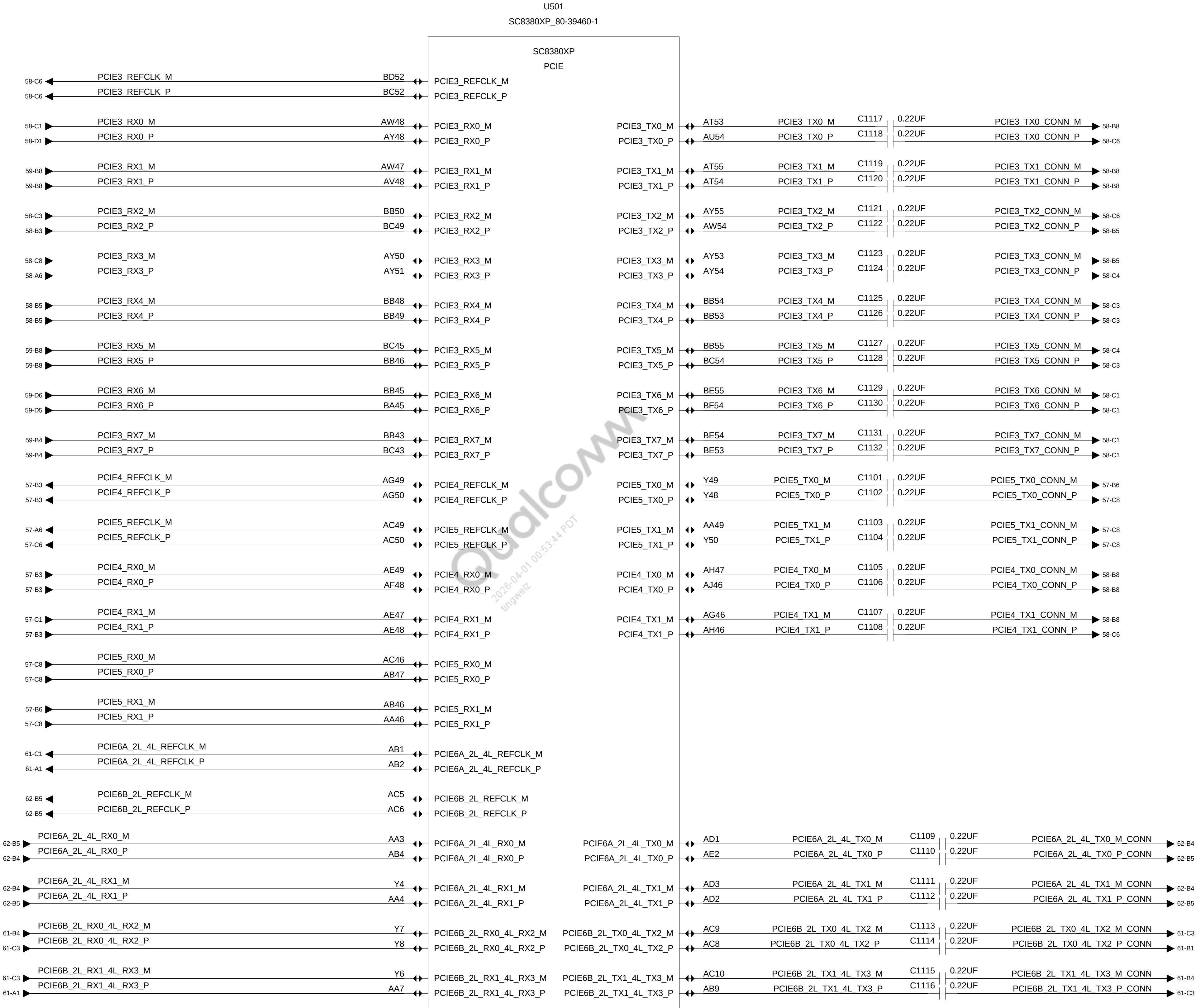
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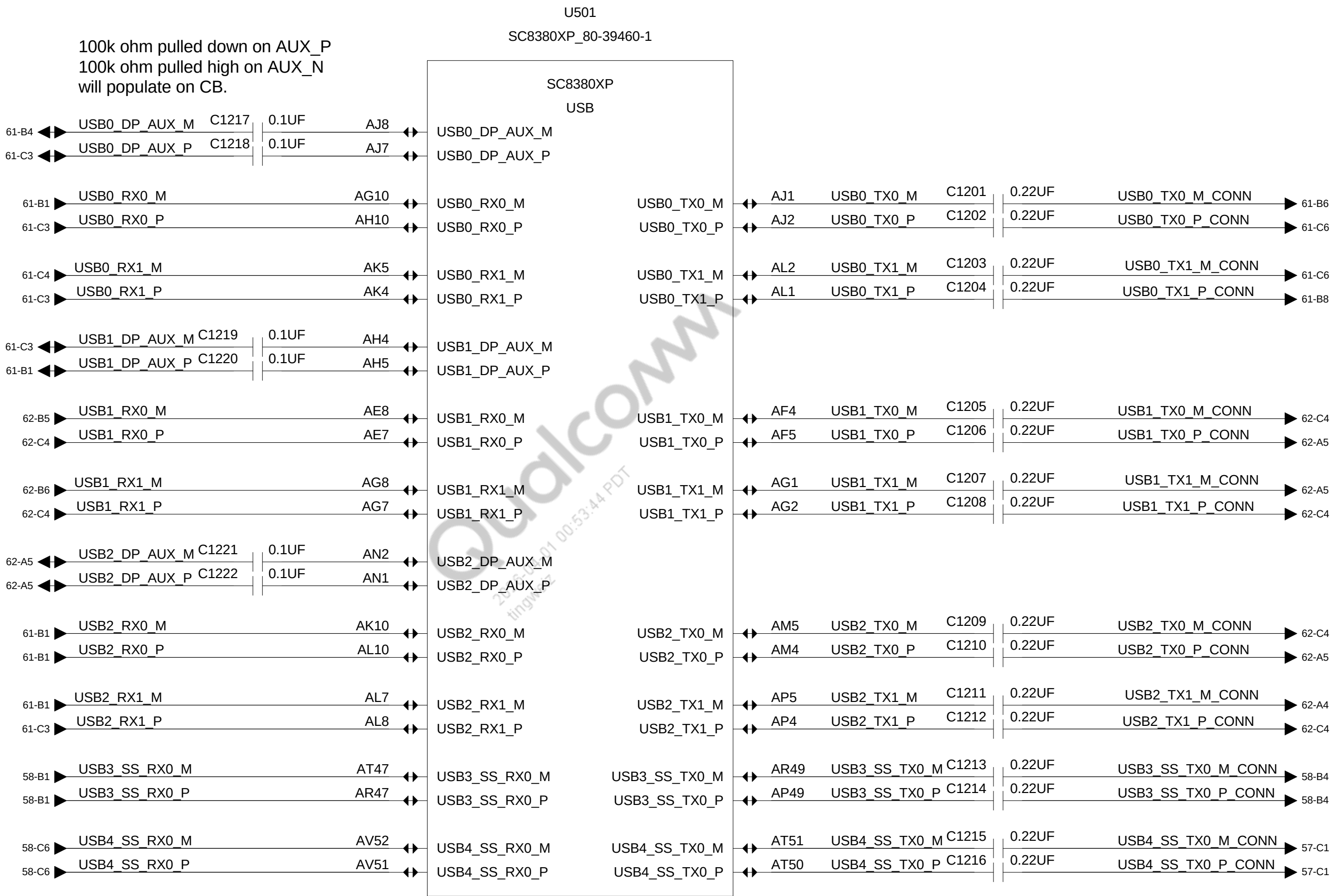


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LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		



REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

D

VDD_APC0

19-D8,28-D8,39-C2 VREG_S2345678G_0P7

VDD_GFX

VREG_S12345678E_S1G_0P7

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VDD_APC1

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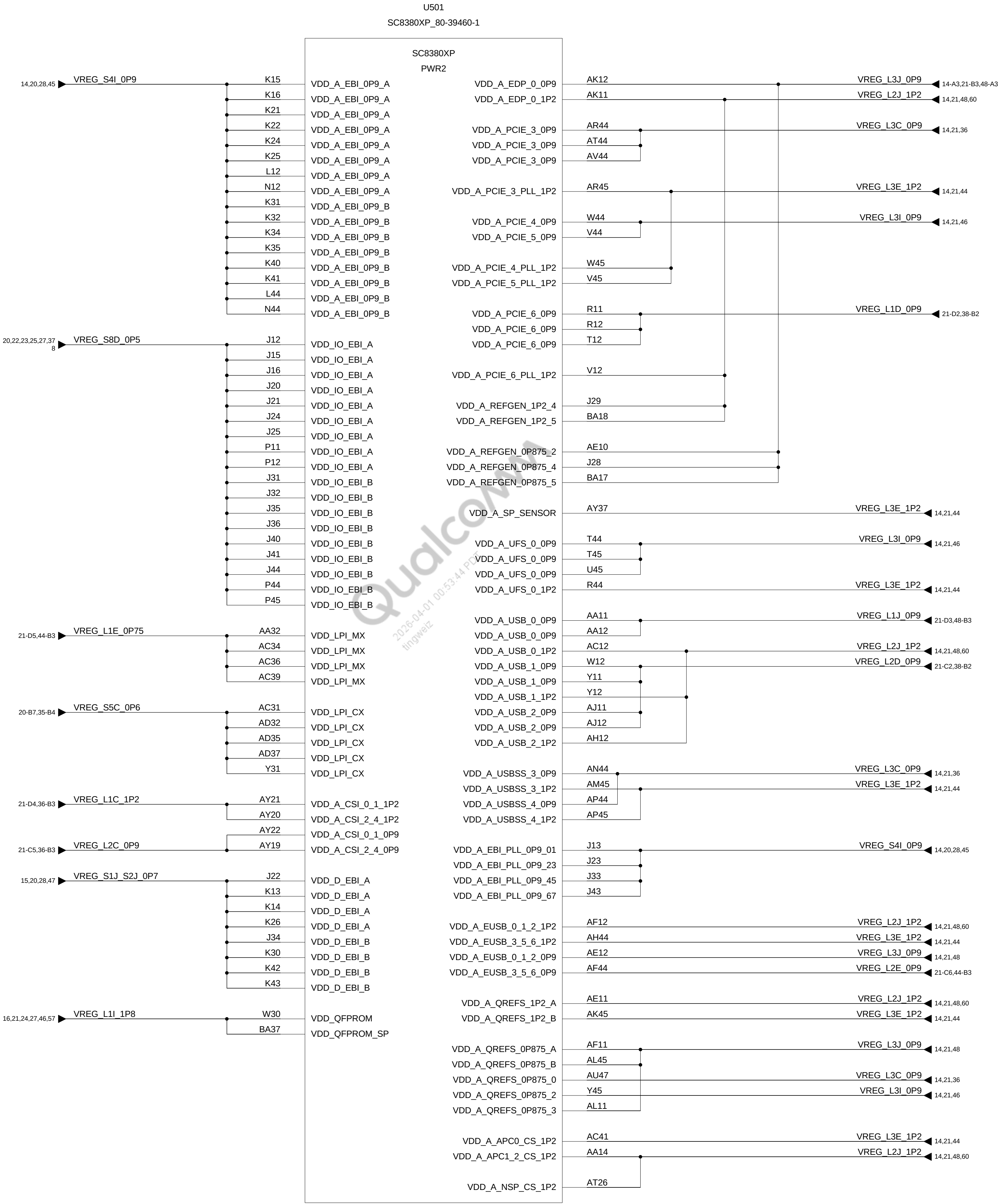
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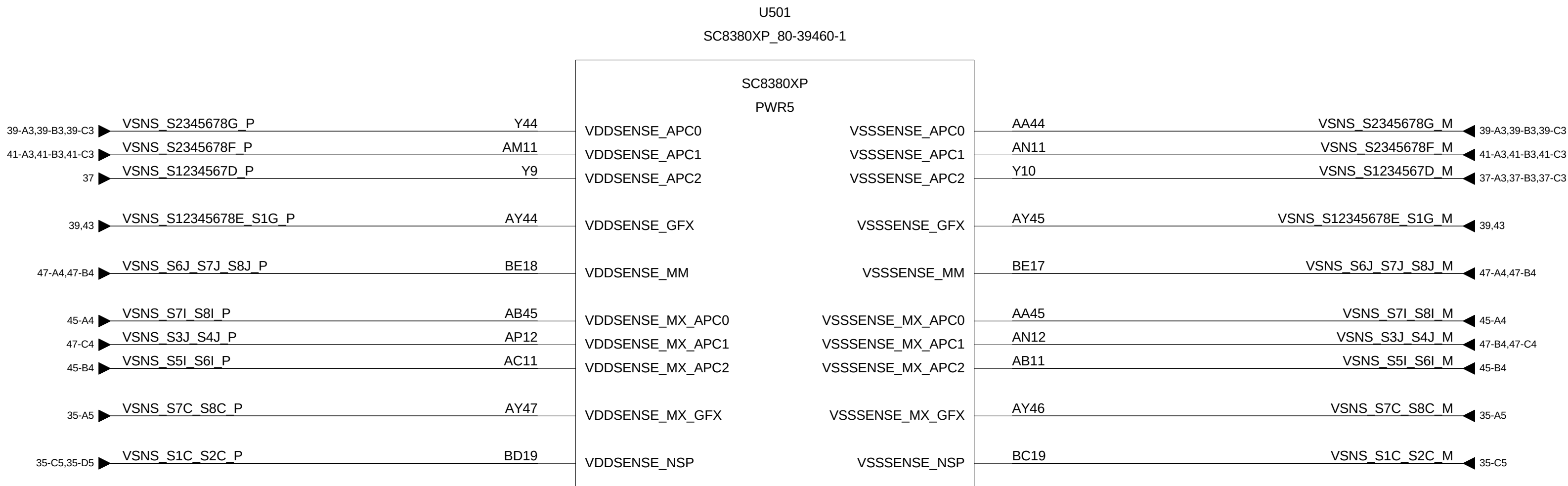
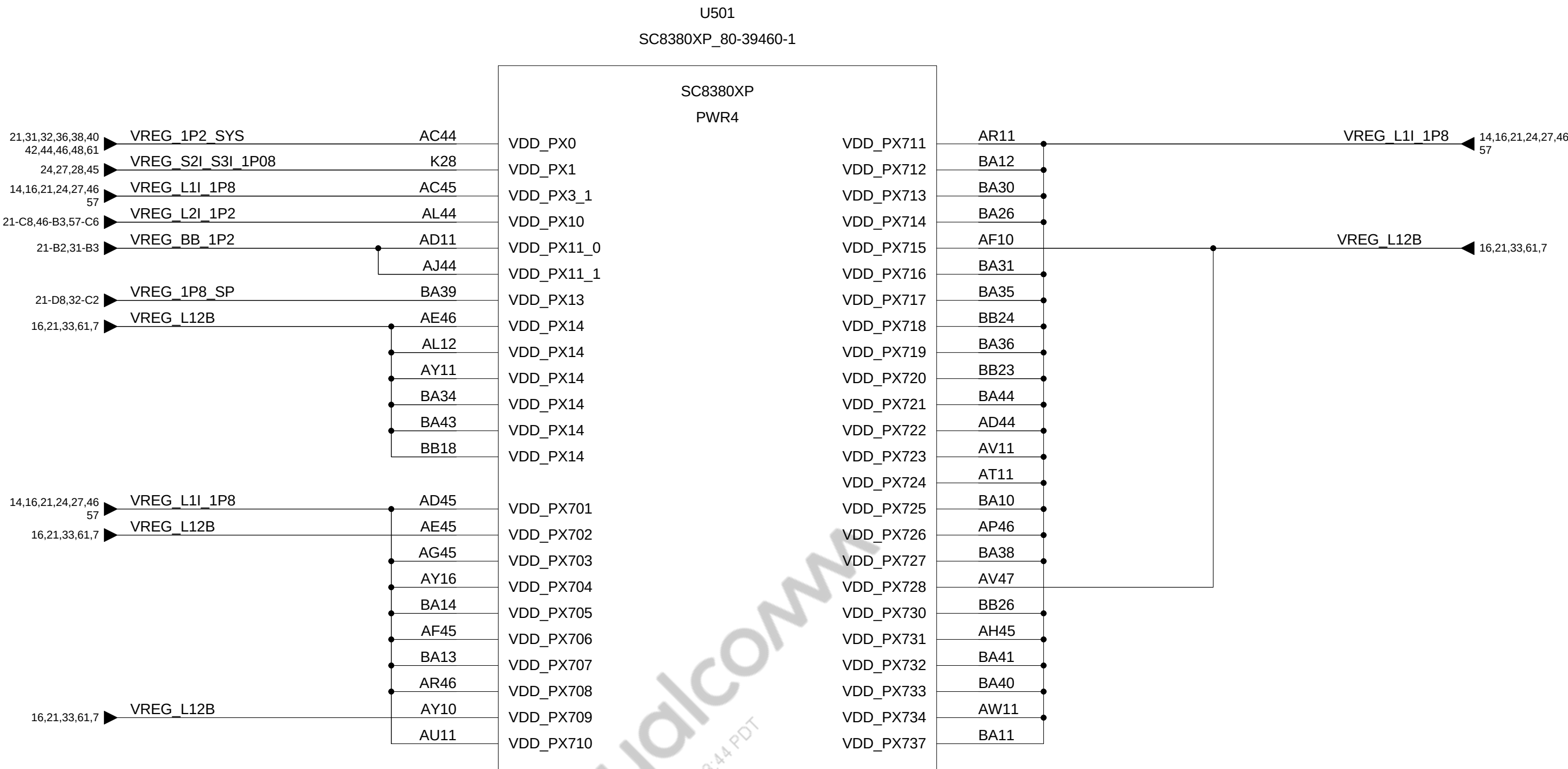
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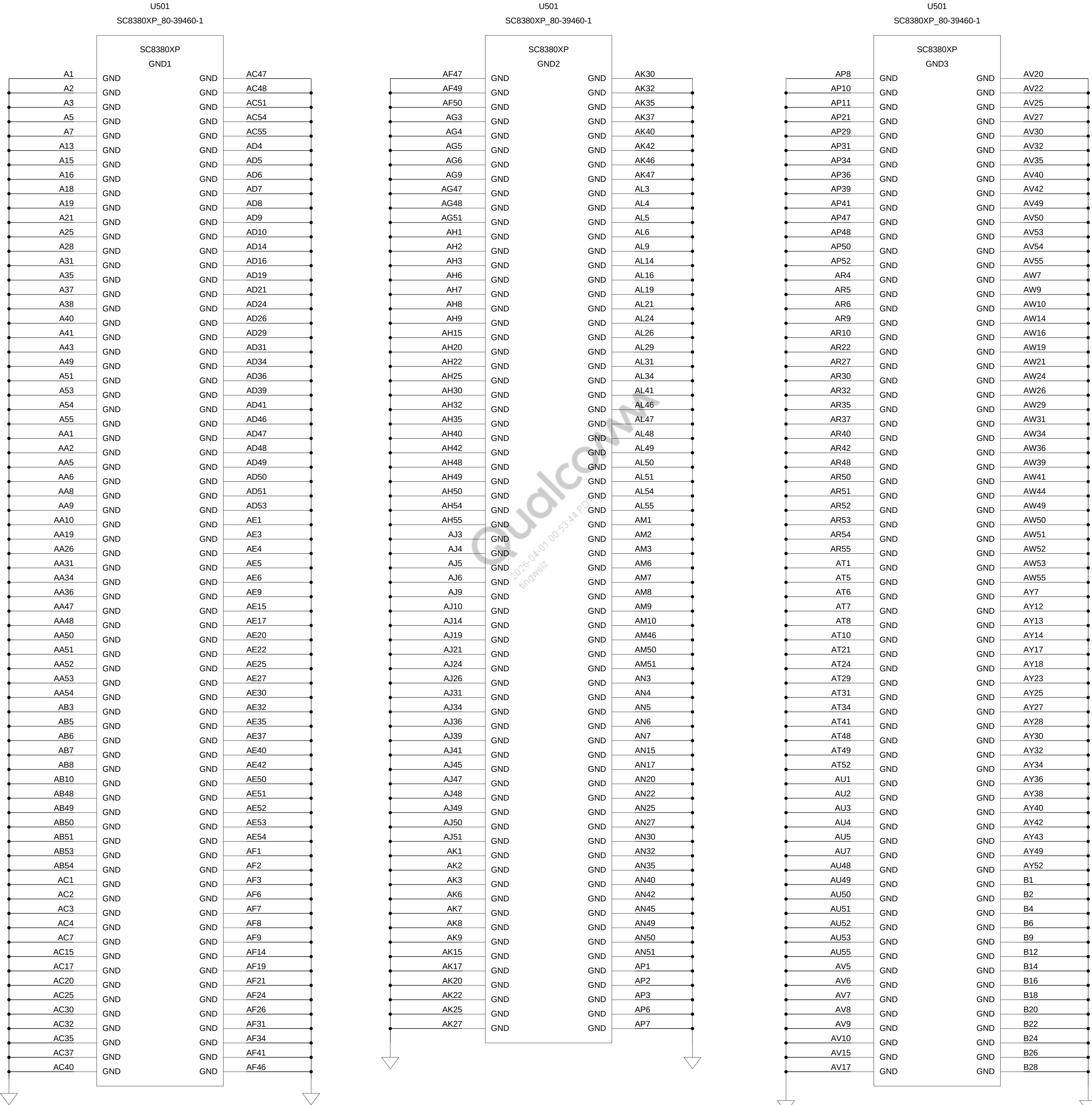
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A	INITIAL RELEASE		



REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		



SC8380XP_GND_1

SIZE	DRAWING NO		REV
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SCALE NONE		SHEET 17	OF 62

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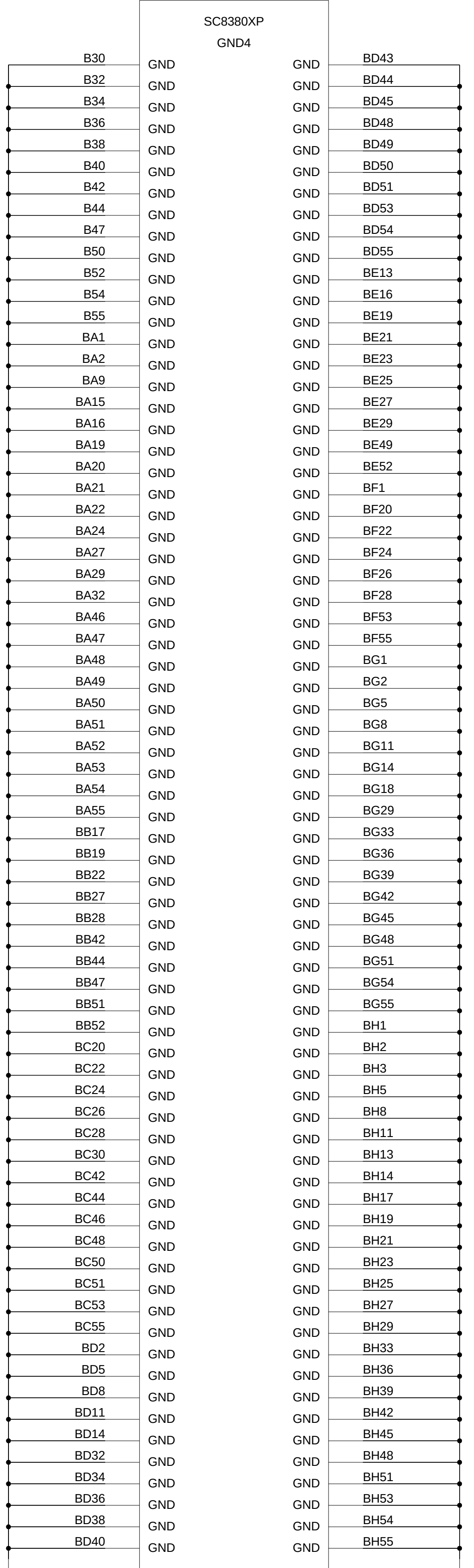
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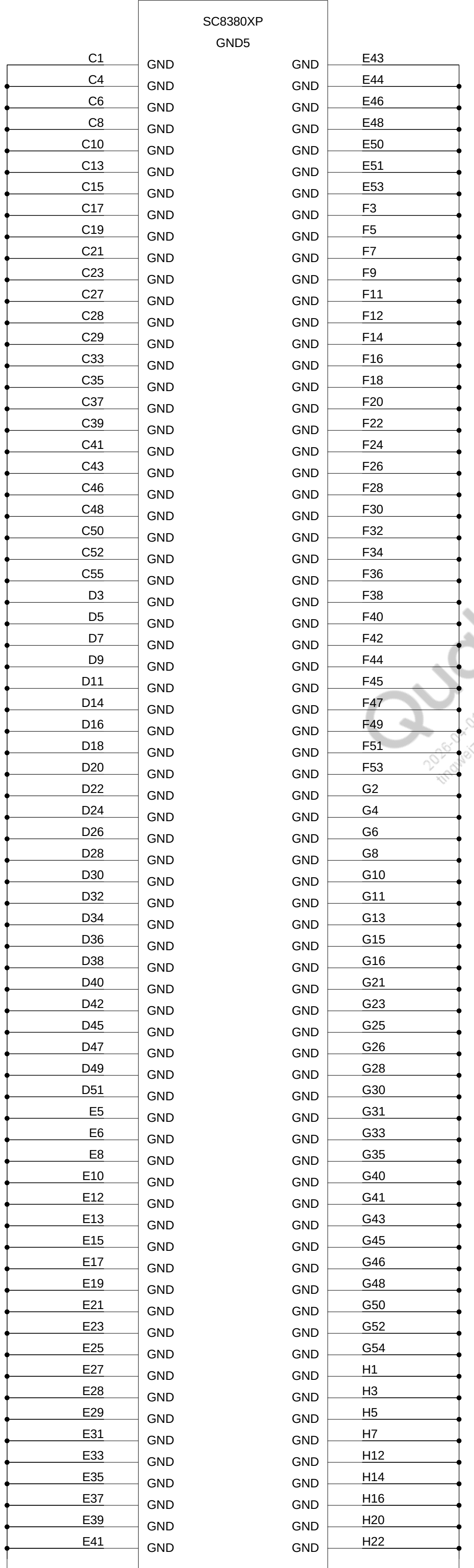
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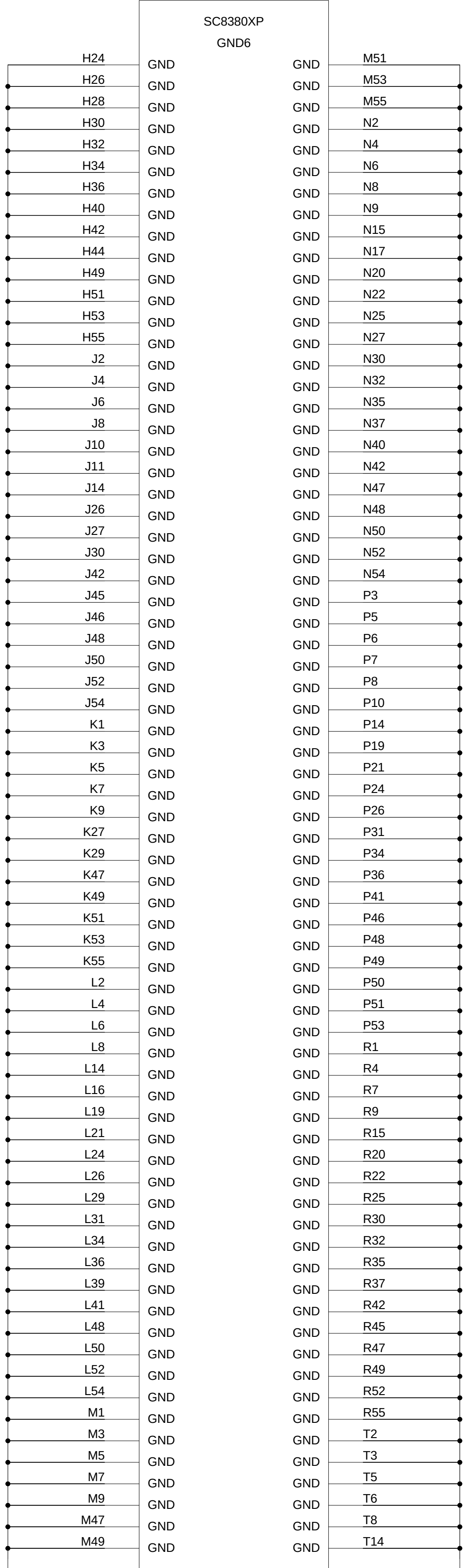
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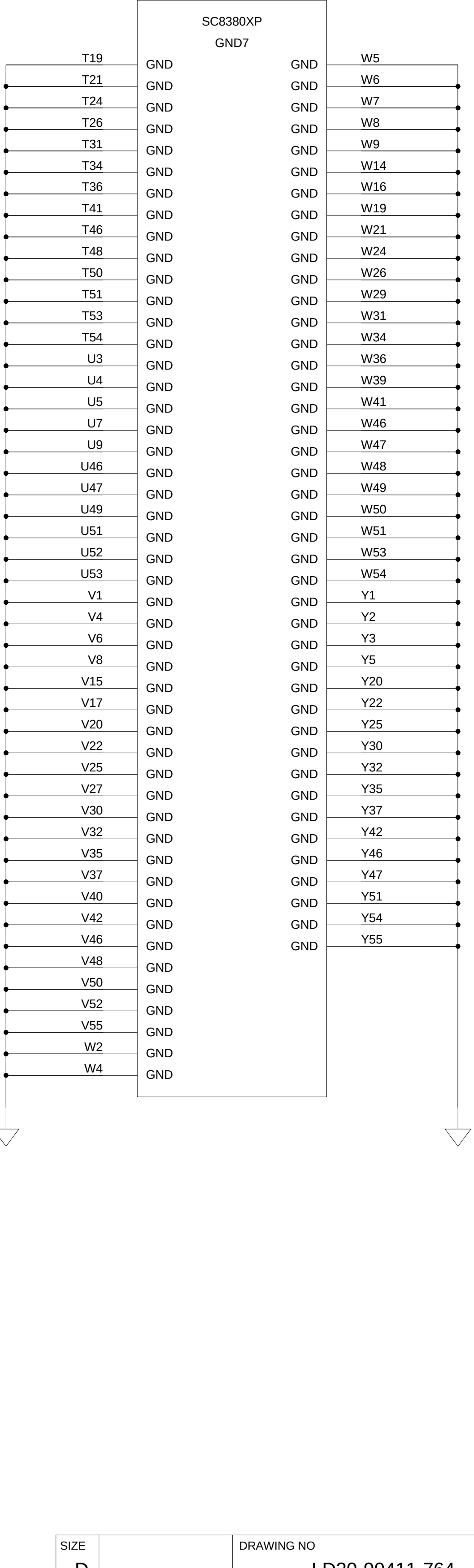
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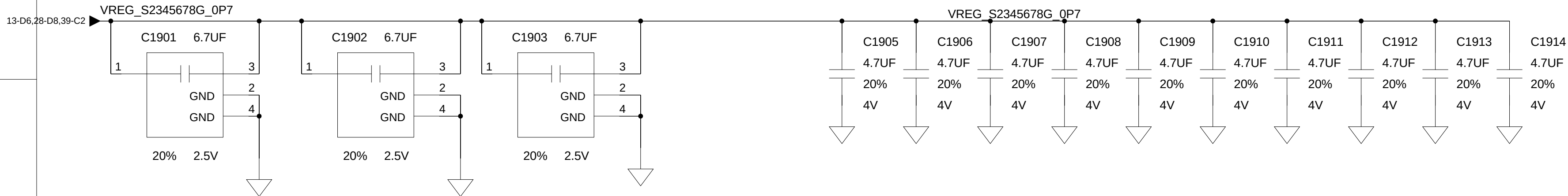


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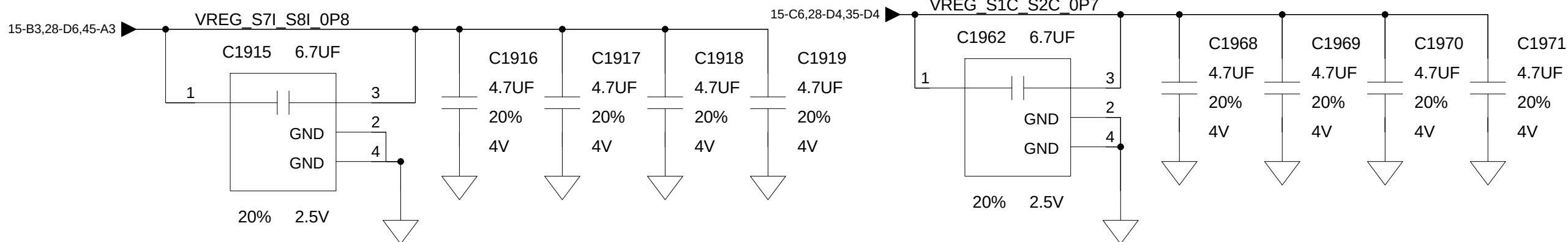
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A	INITIAL RELEASE		

SMPS Bulk Caps

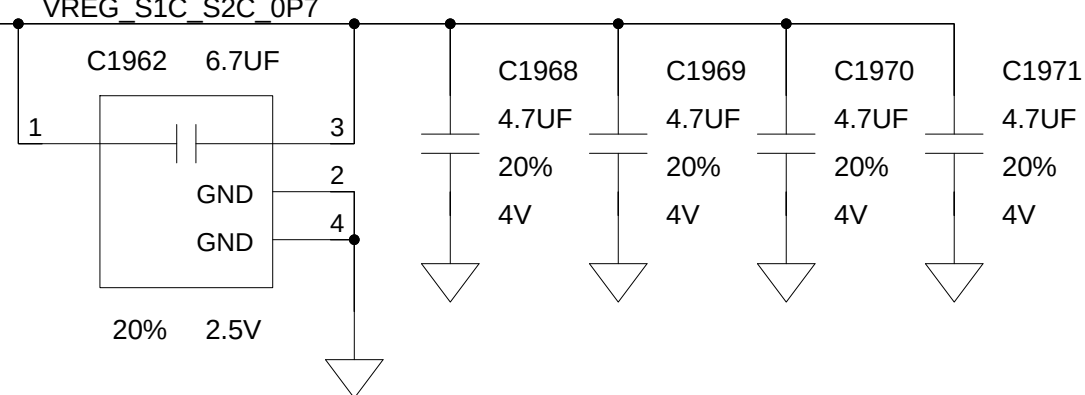
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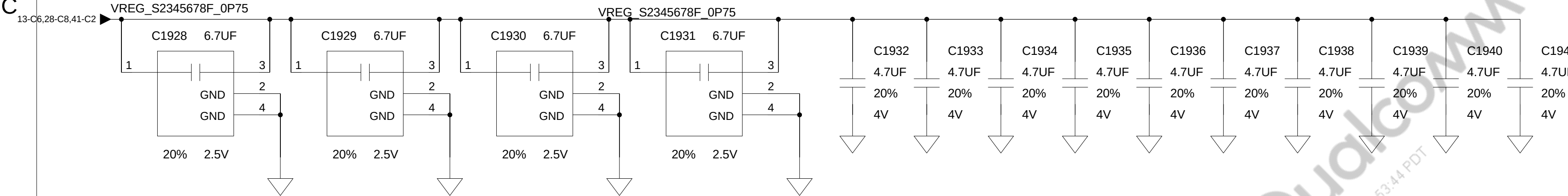
VDD_MX_APC0
Dedicated sense pins on package



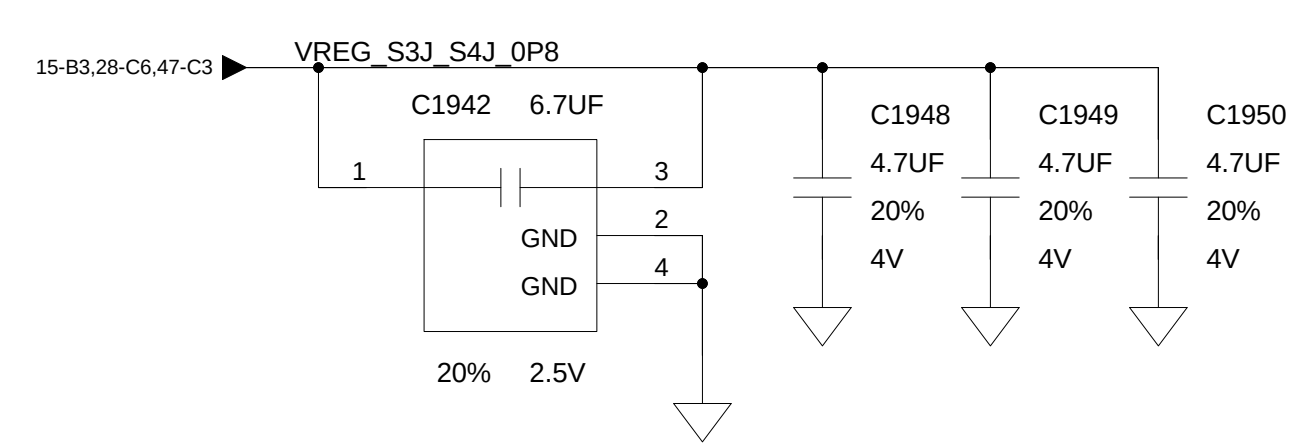
VDD_NSP
Dedicated sense pins on package



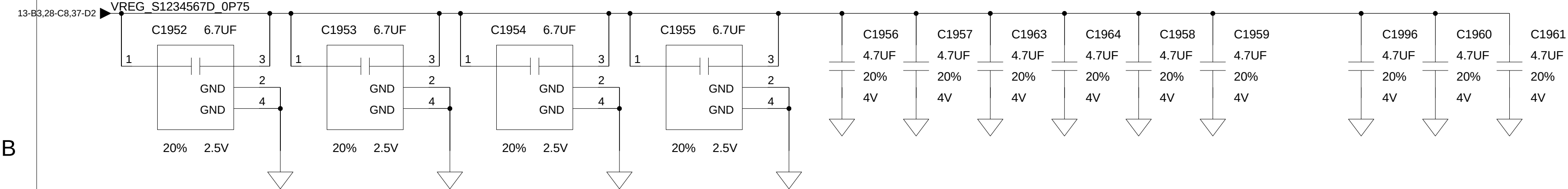
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Dedicated sense pins on package



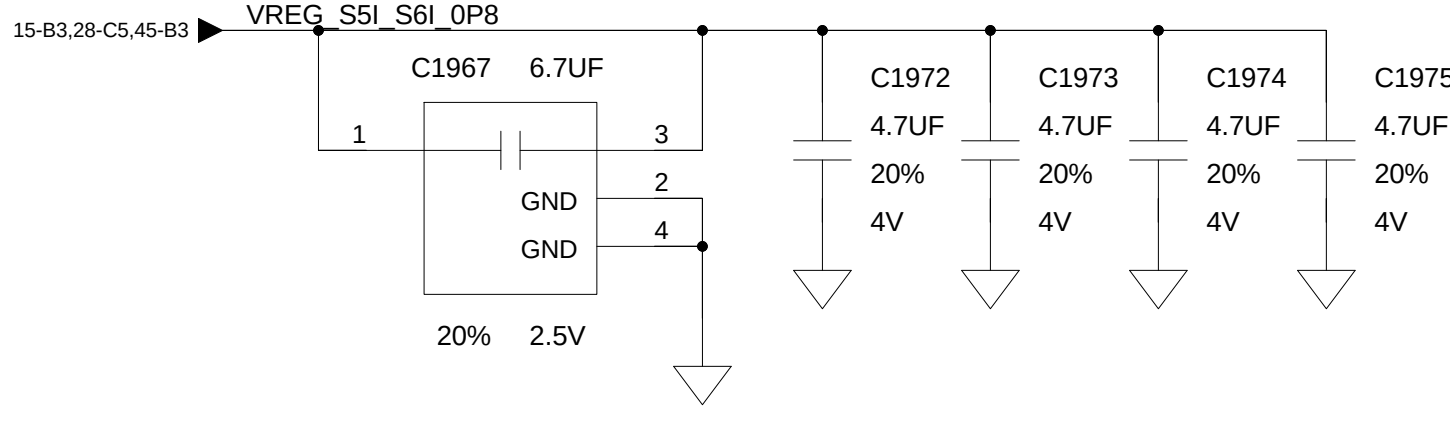
VDD_MX_APC1
Dedicated sense pins on package



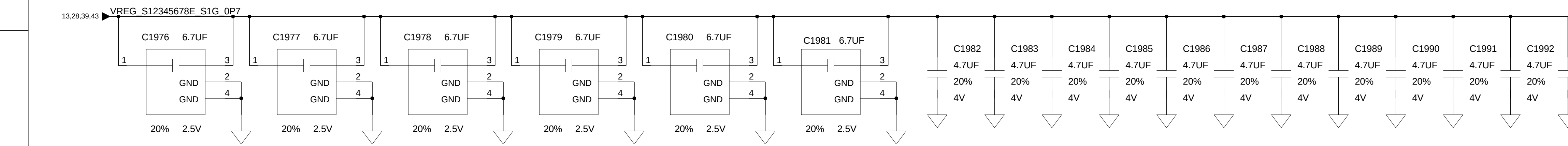
APC2 - PWR
Dedicated sense pins on package



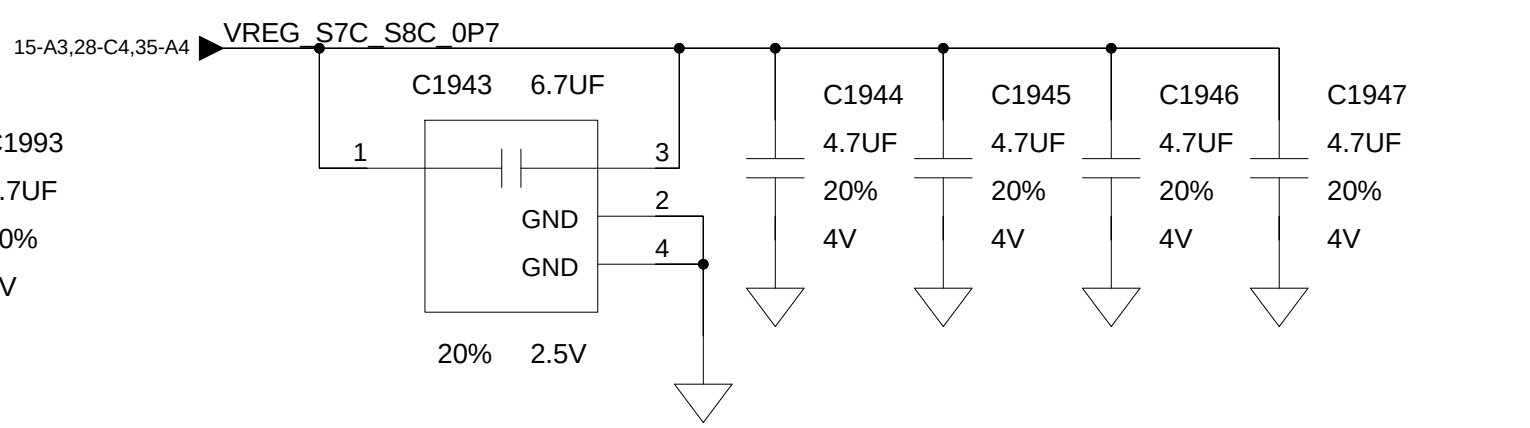
VDD_MX_APC2
Dedicated sense pins on package



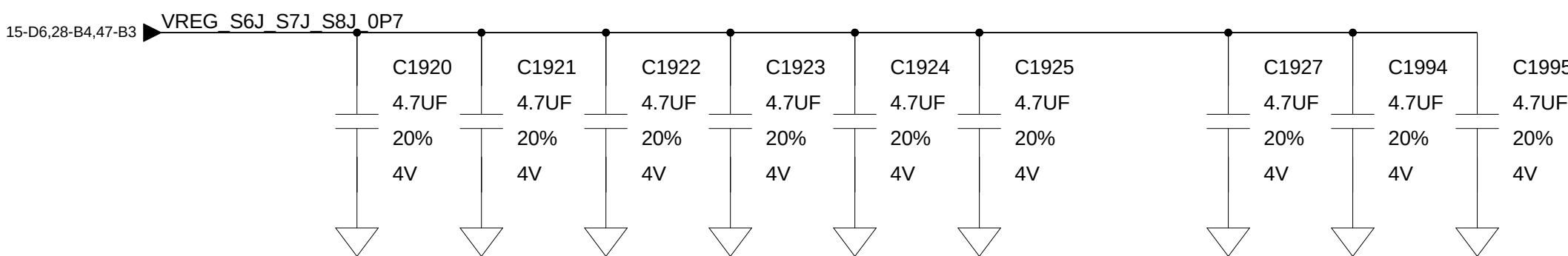
VDD_GFX
Dedicated sense pins on package



MX_GFX
Dedicated sense pins on package



VDD_MM
Dedicated sense pins on package

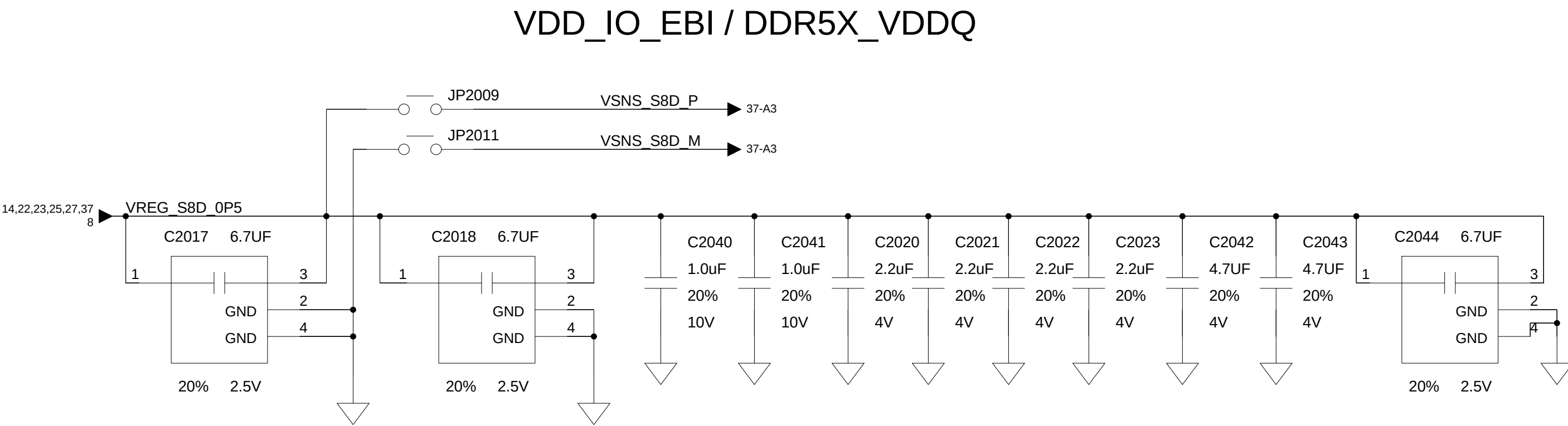
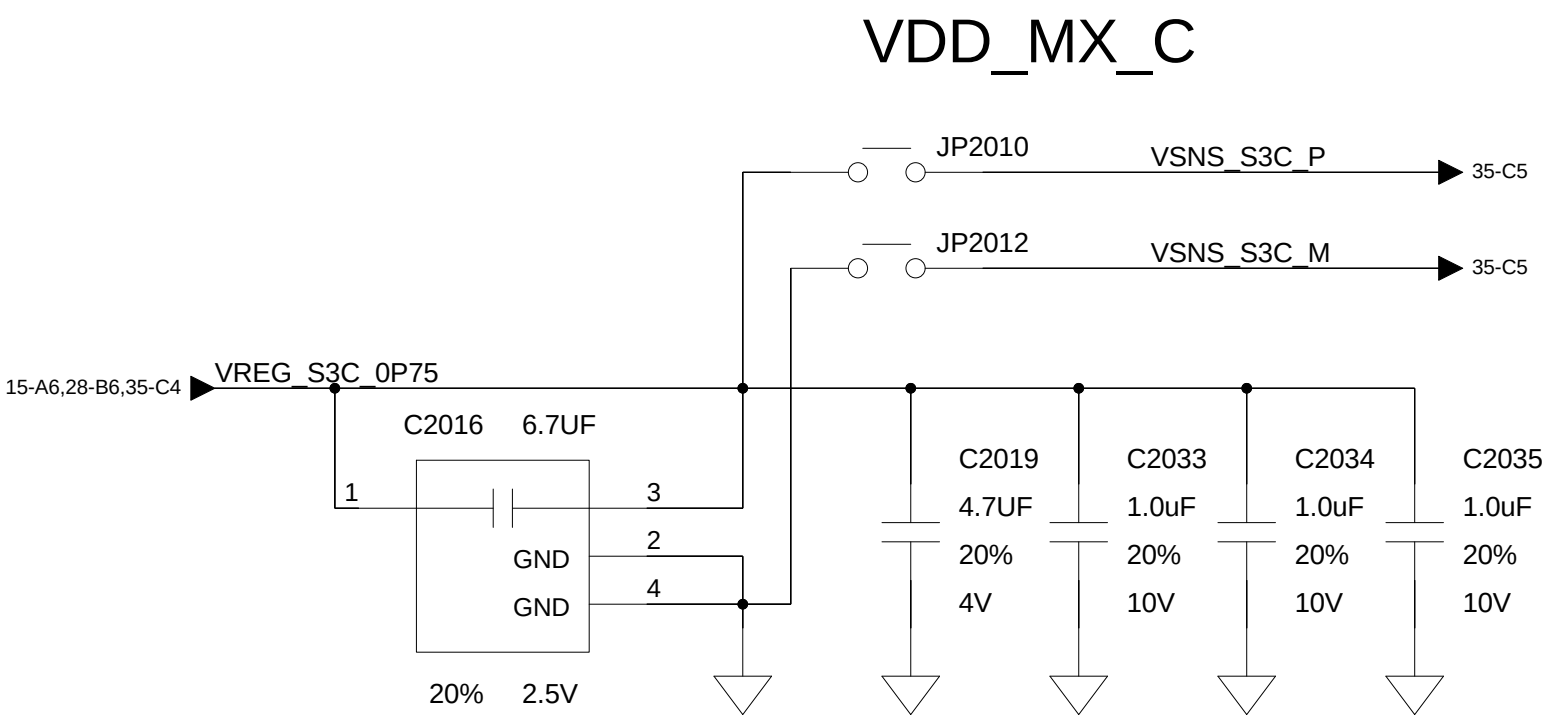
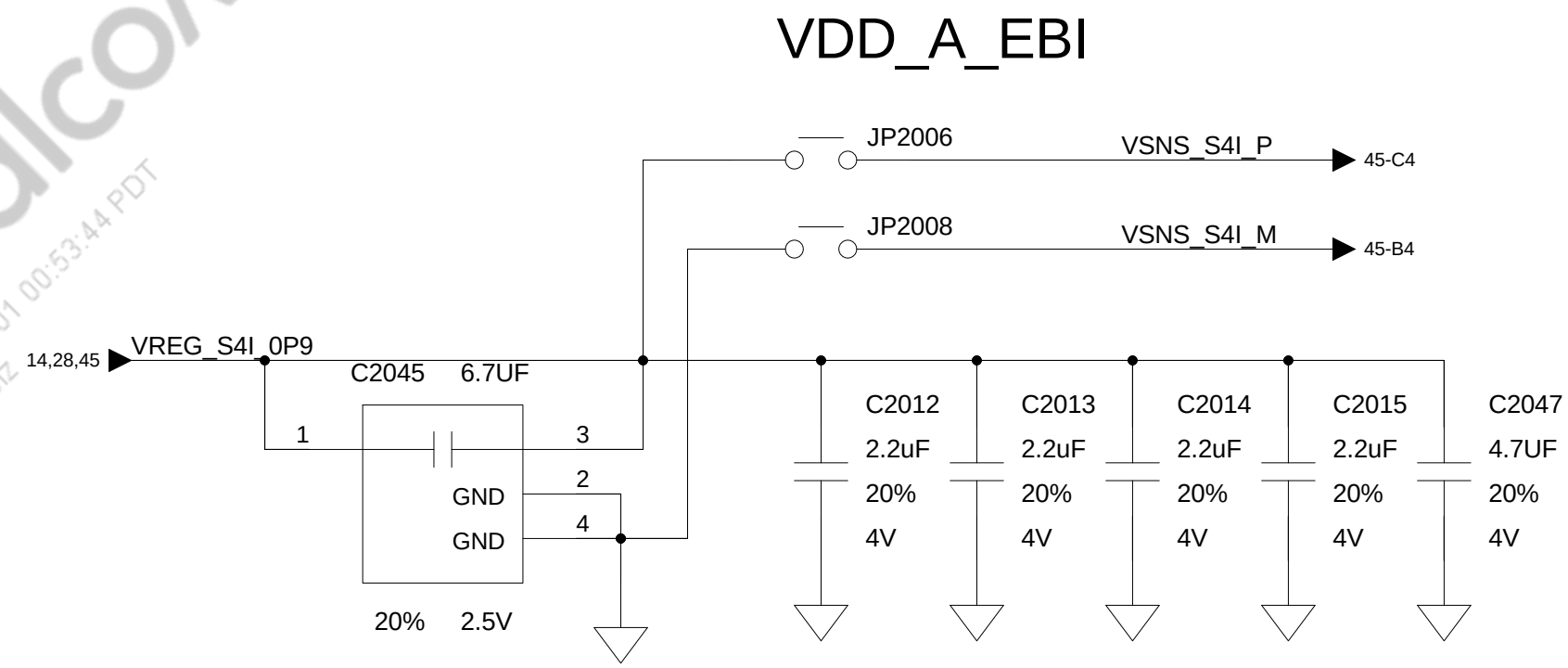
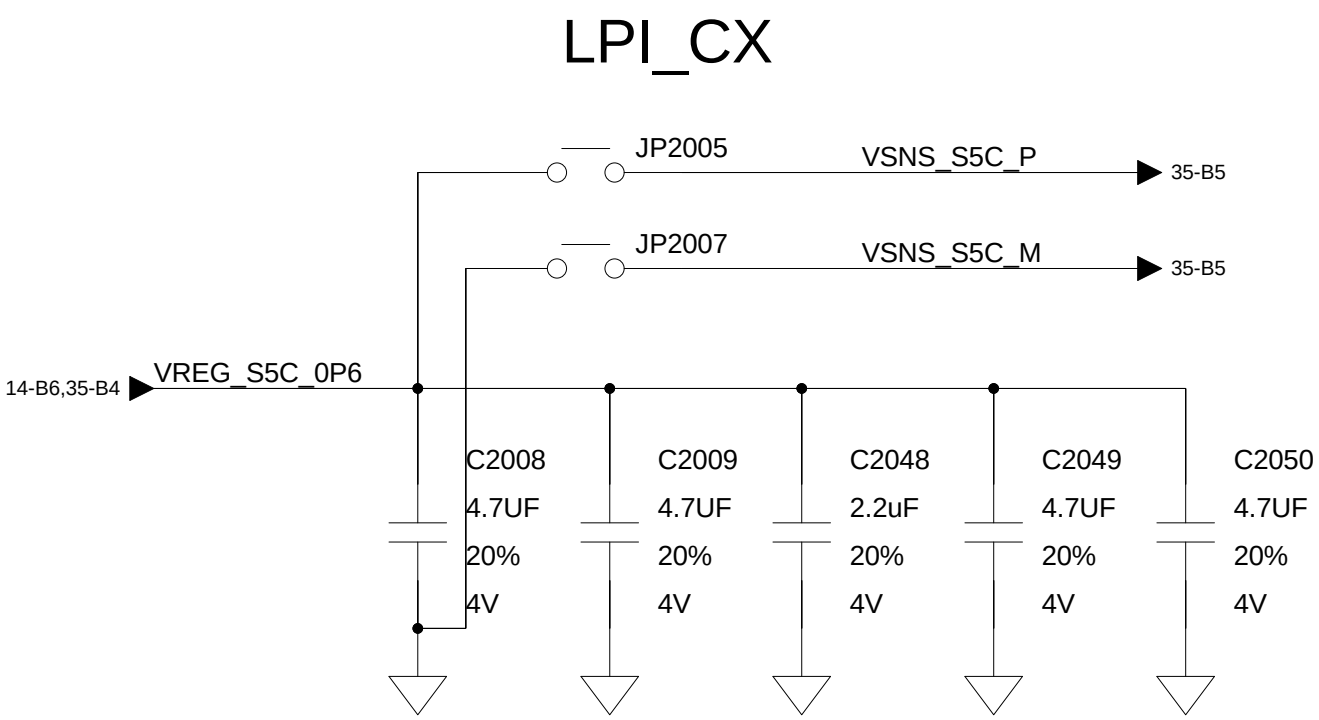
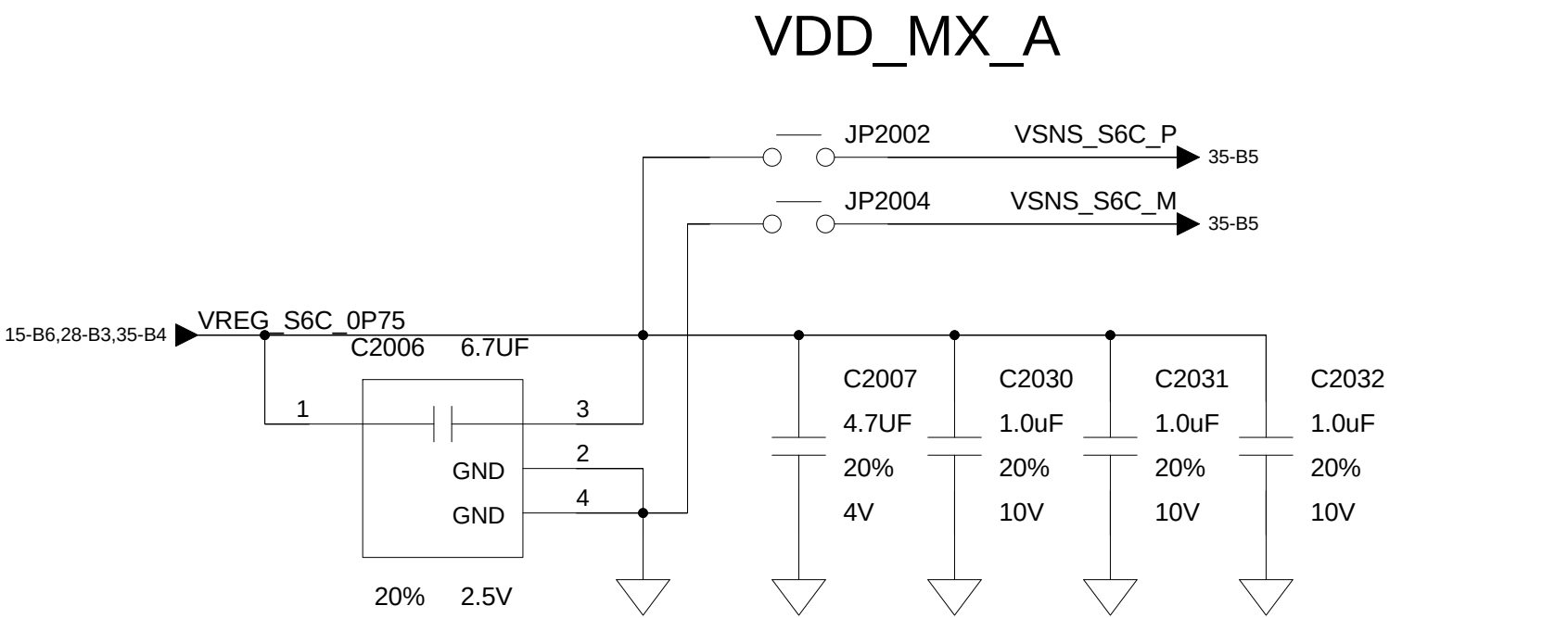
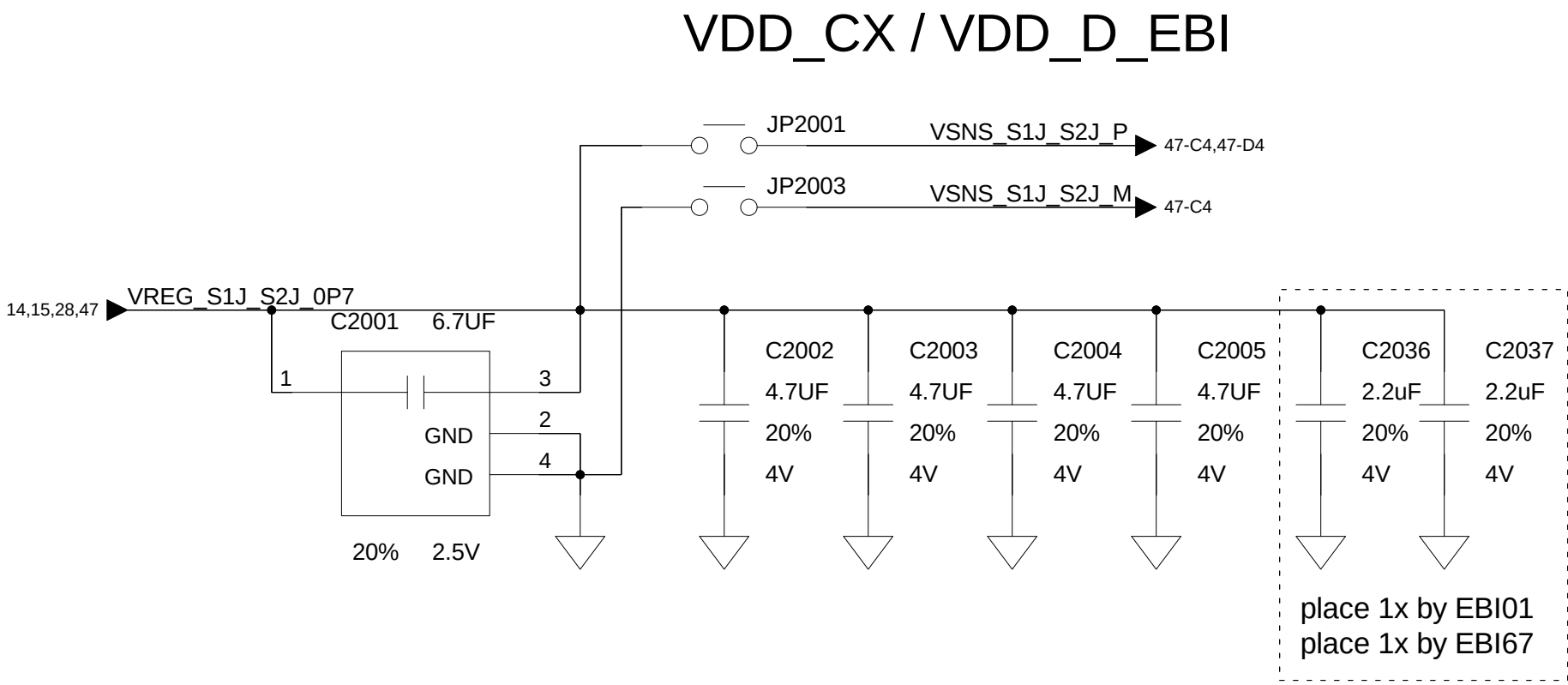


SC8380XP_DECAP1

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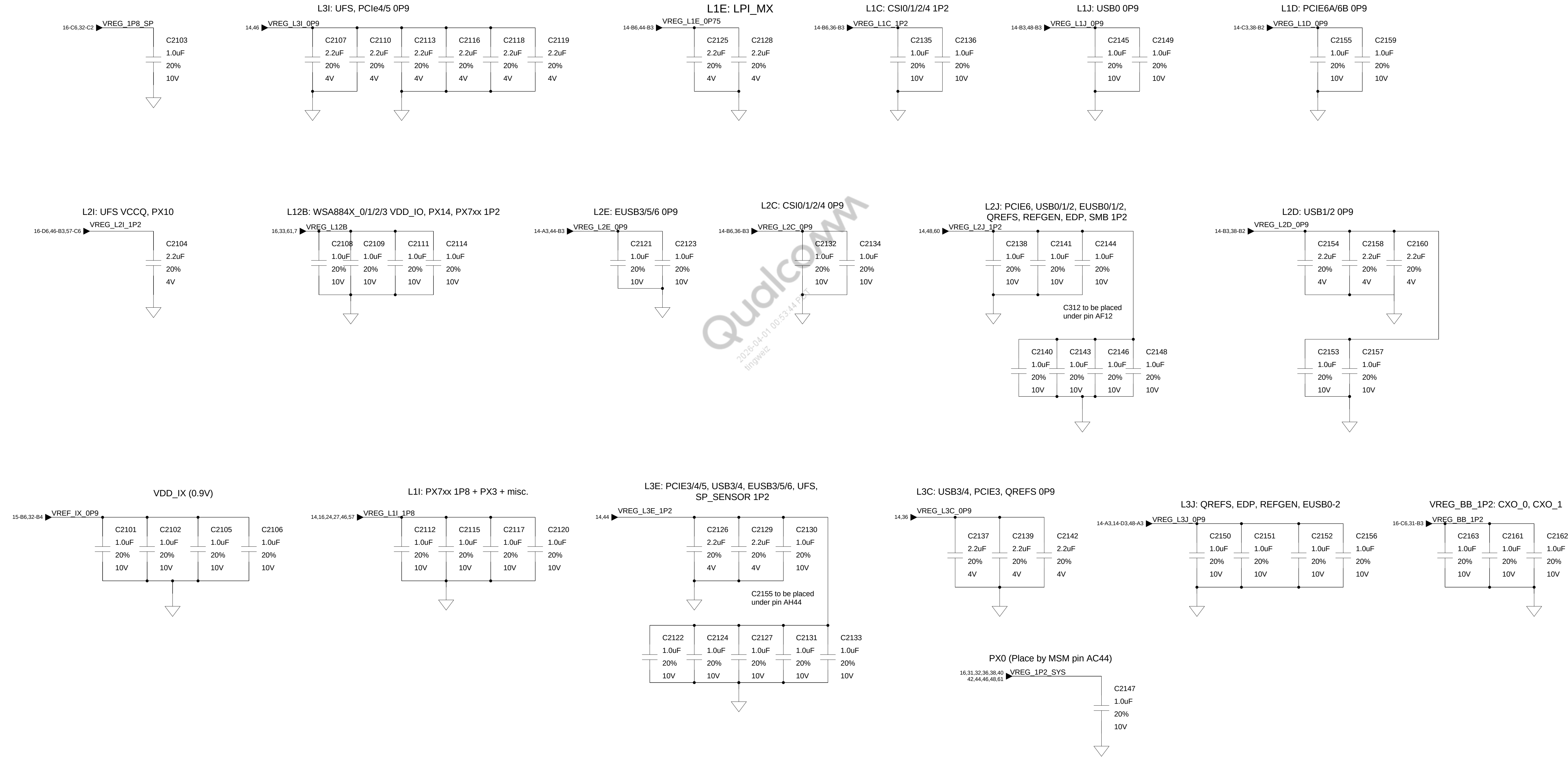
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SMPS Bulk Caps



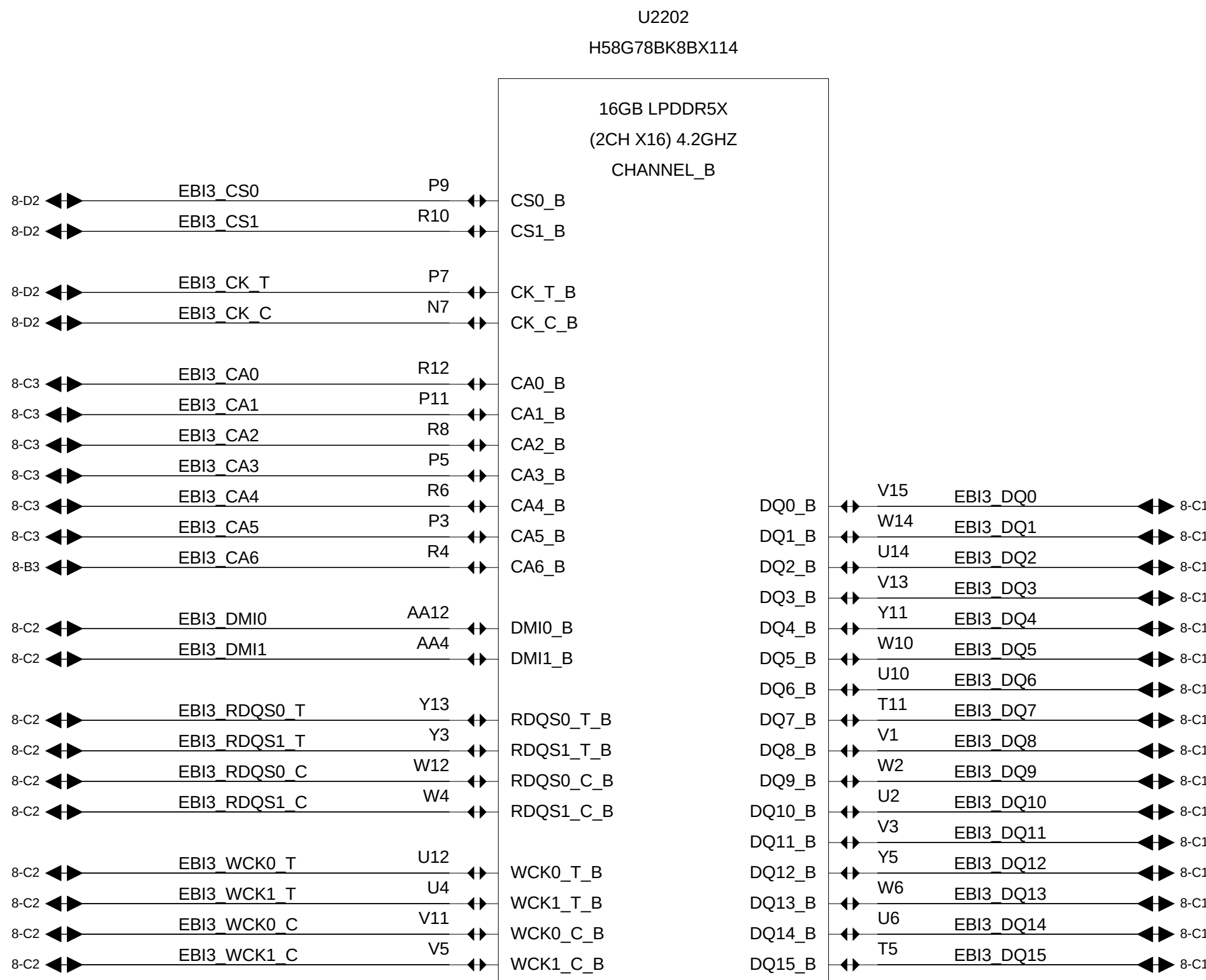
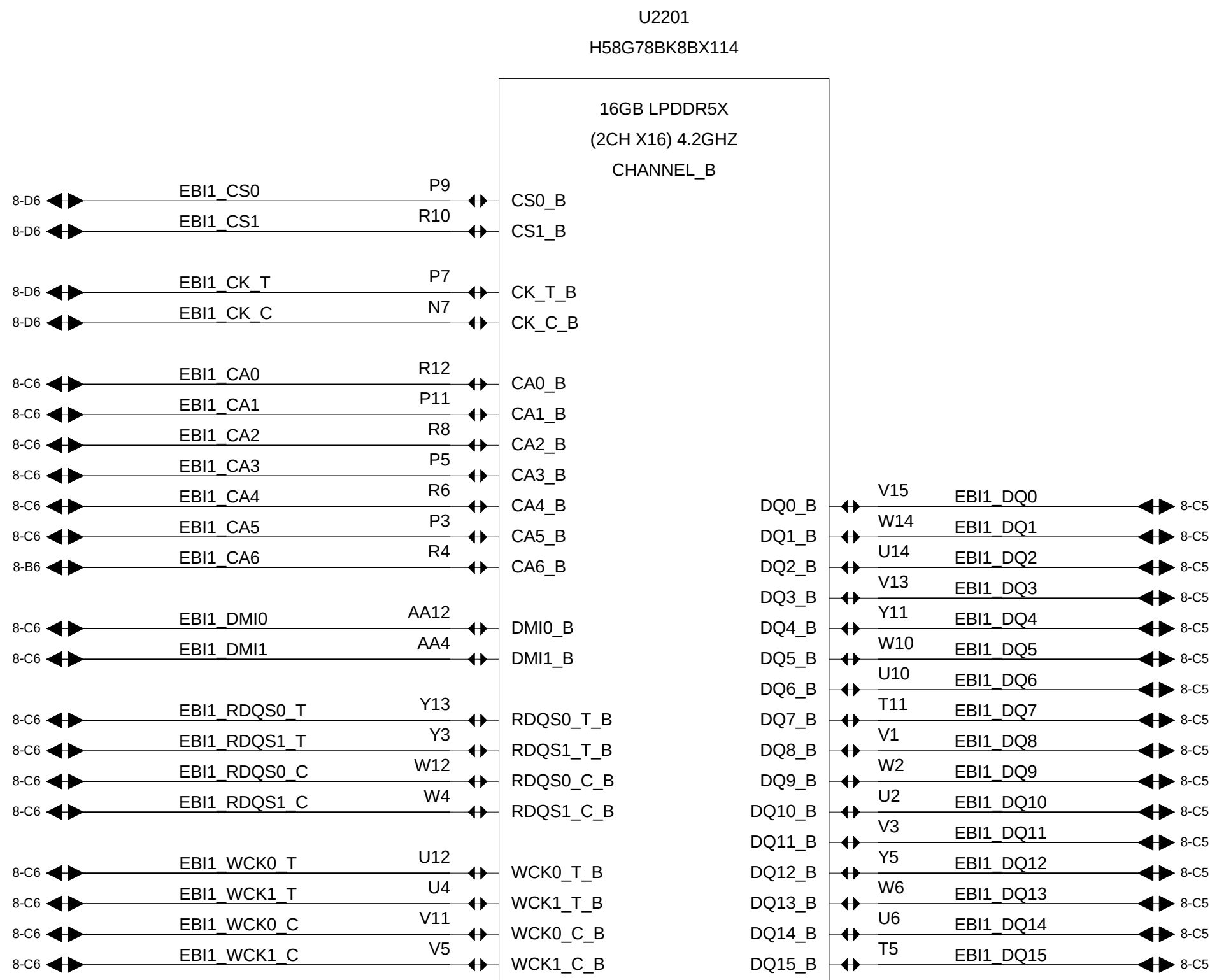
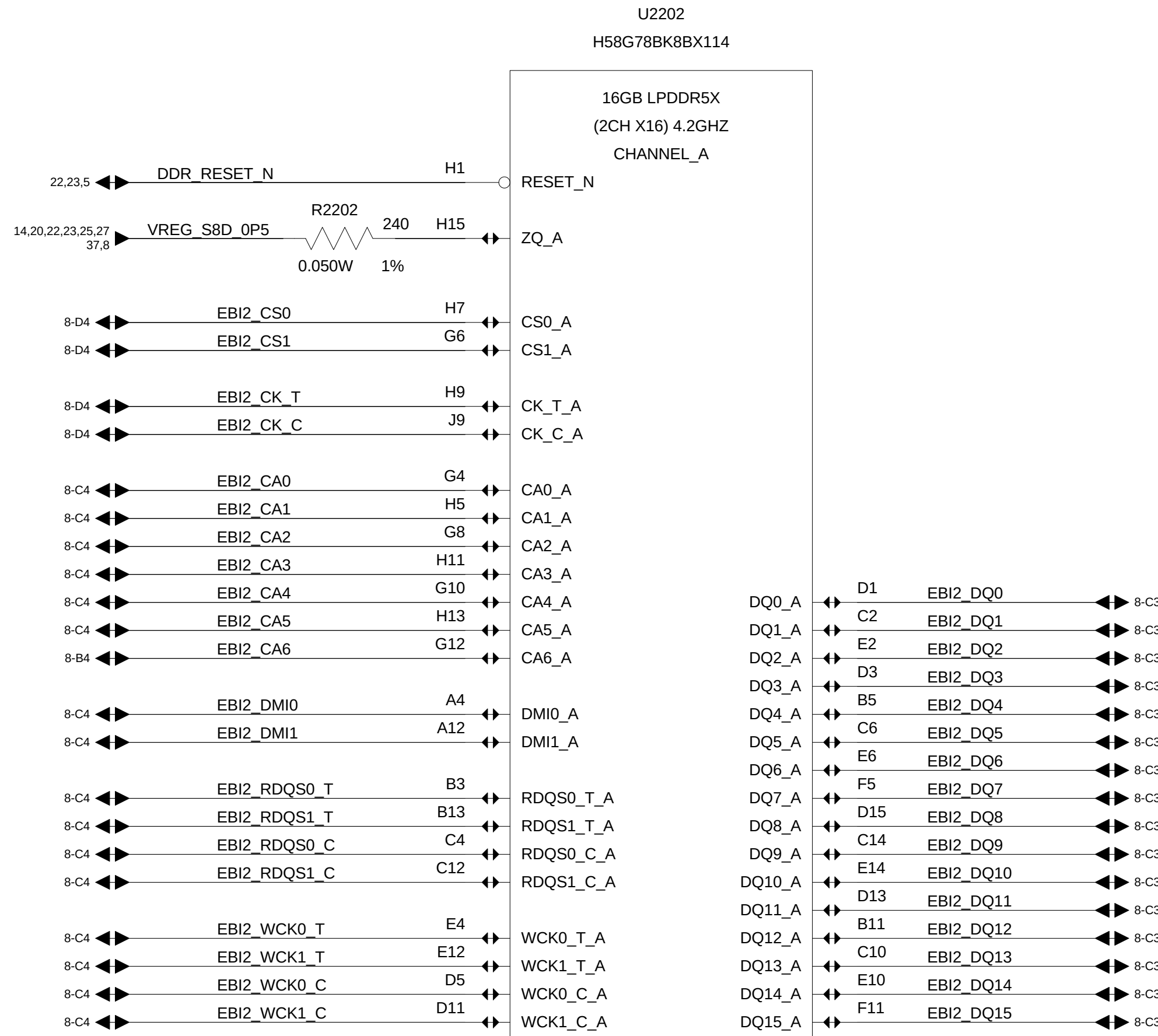
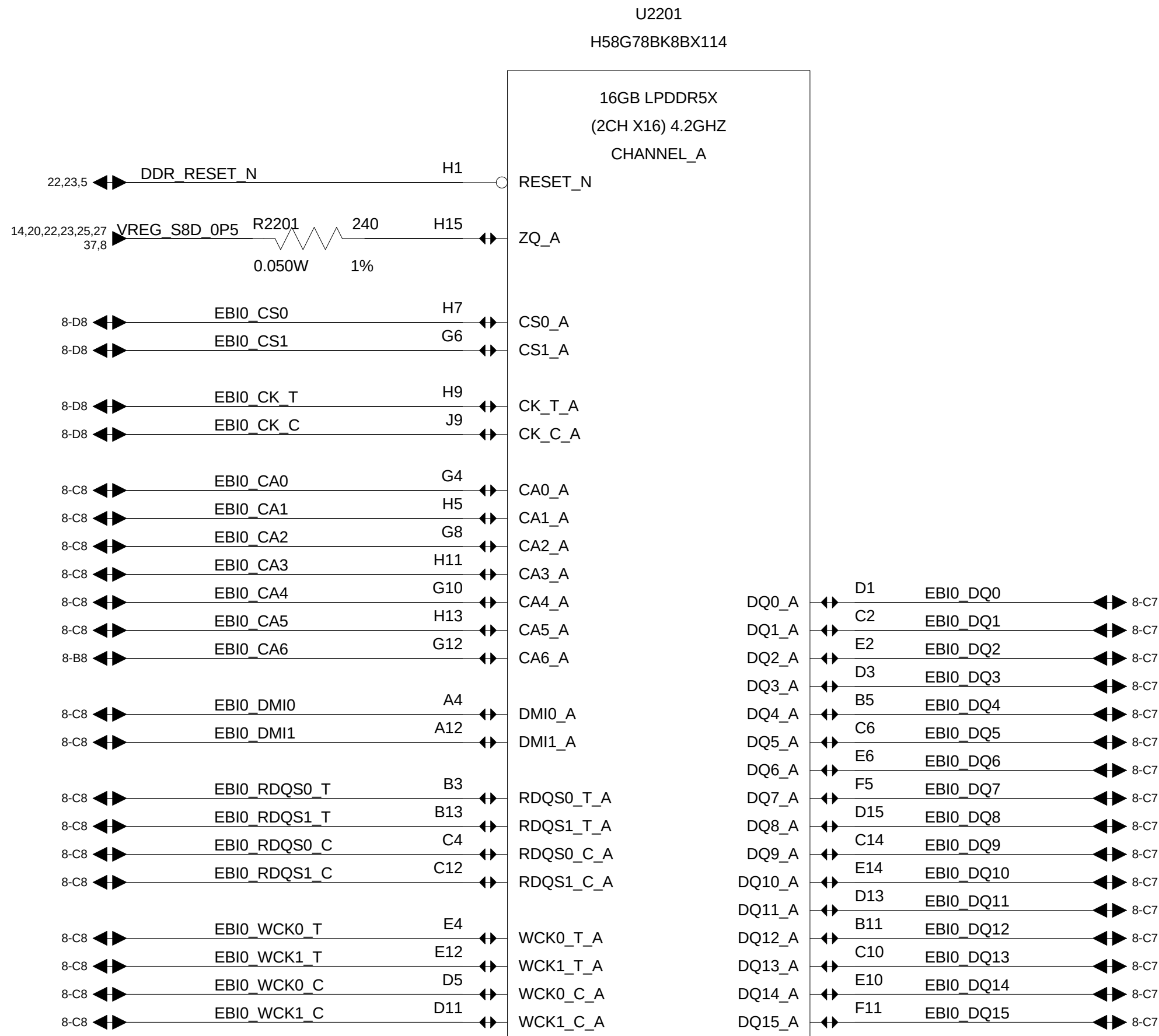
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LDO Output Caps

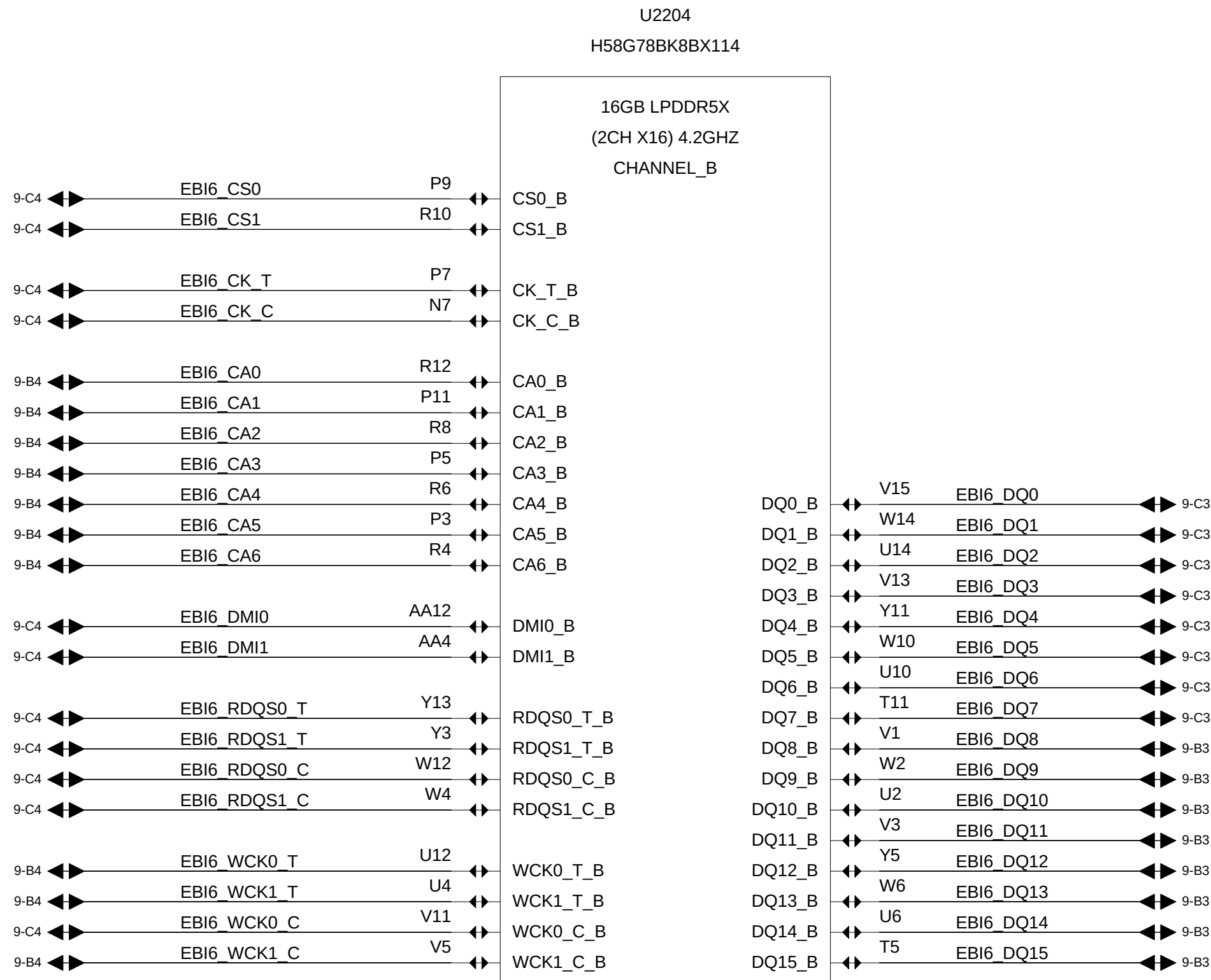
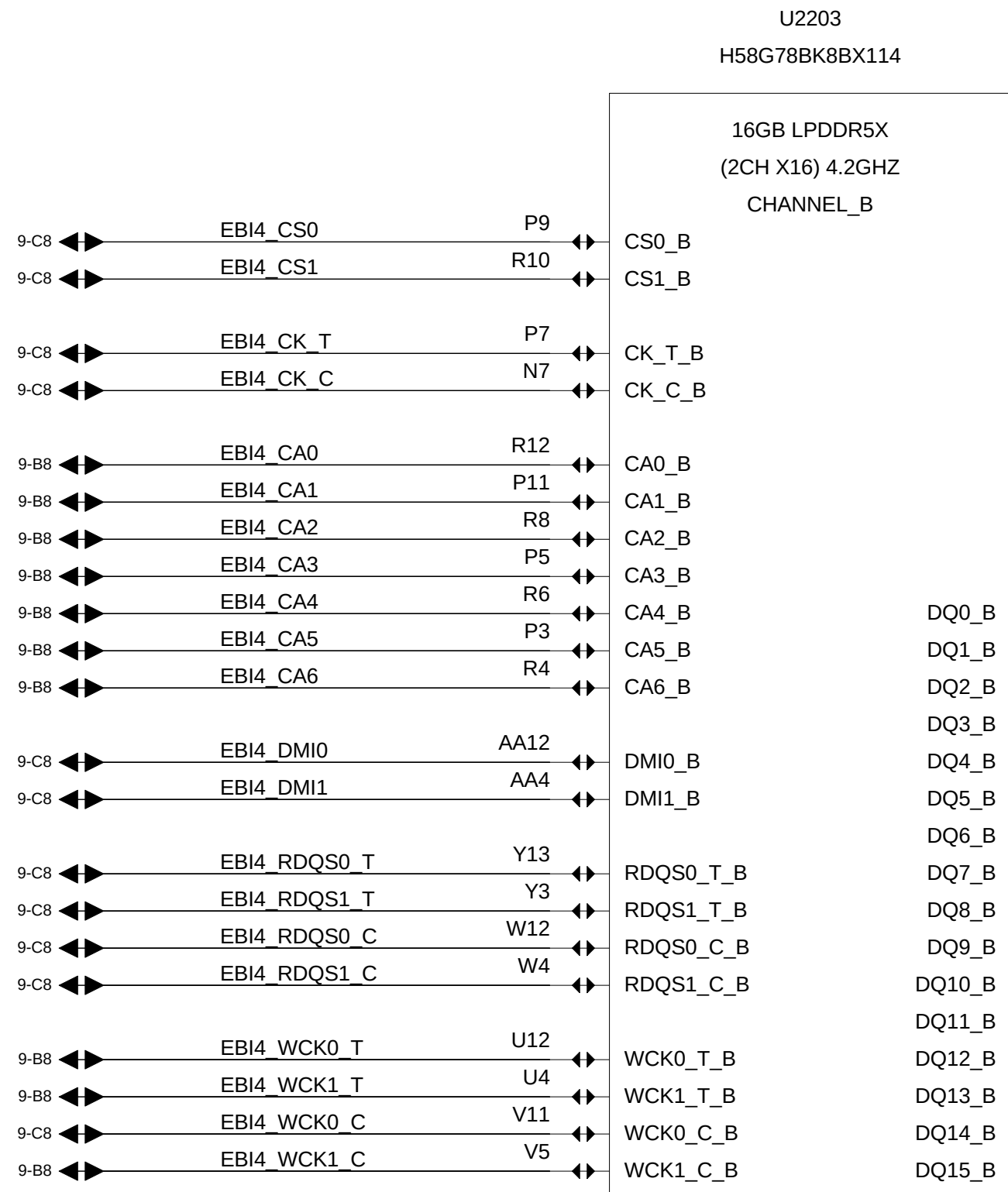
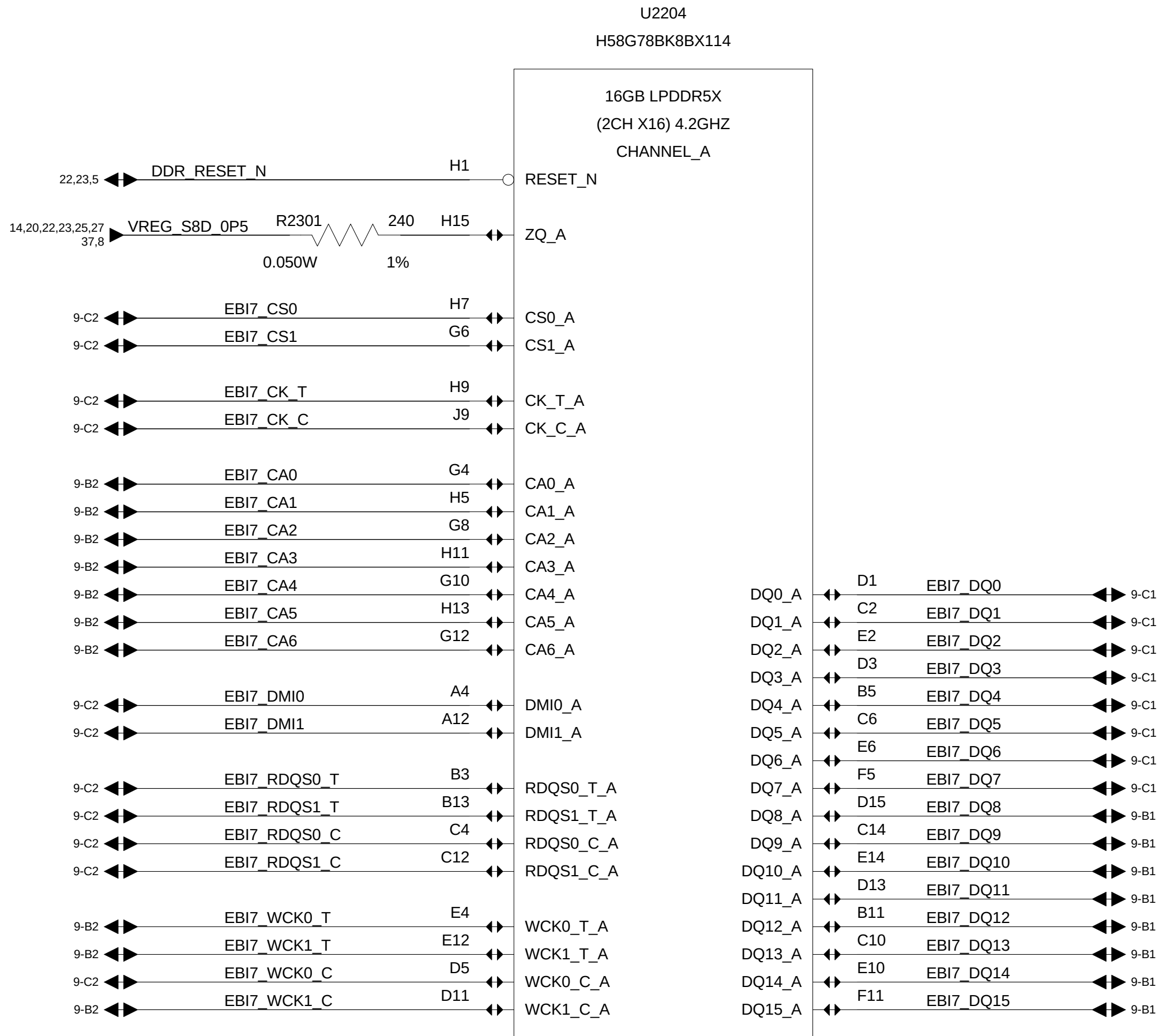
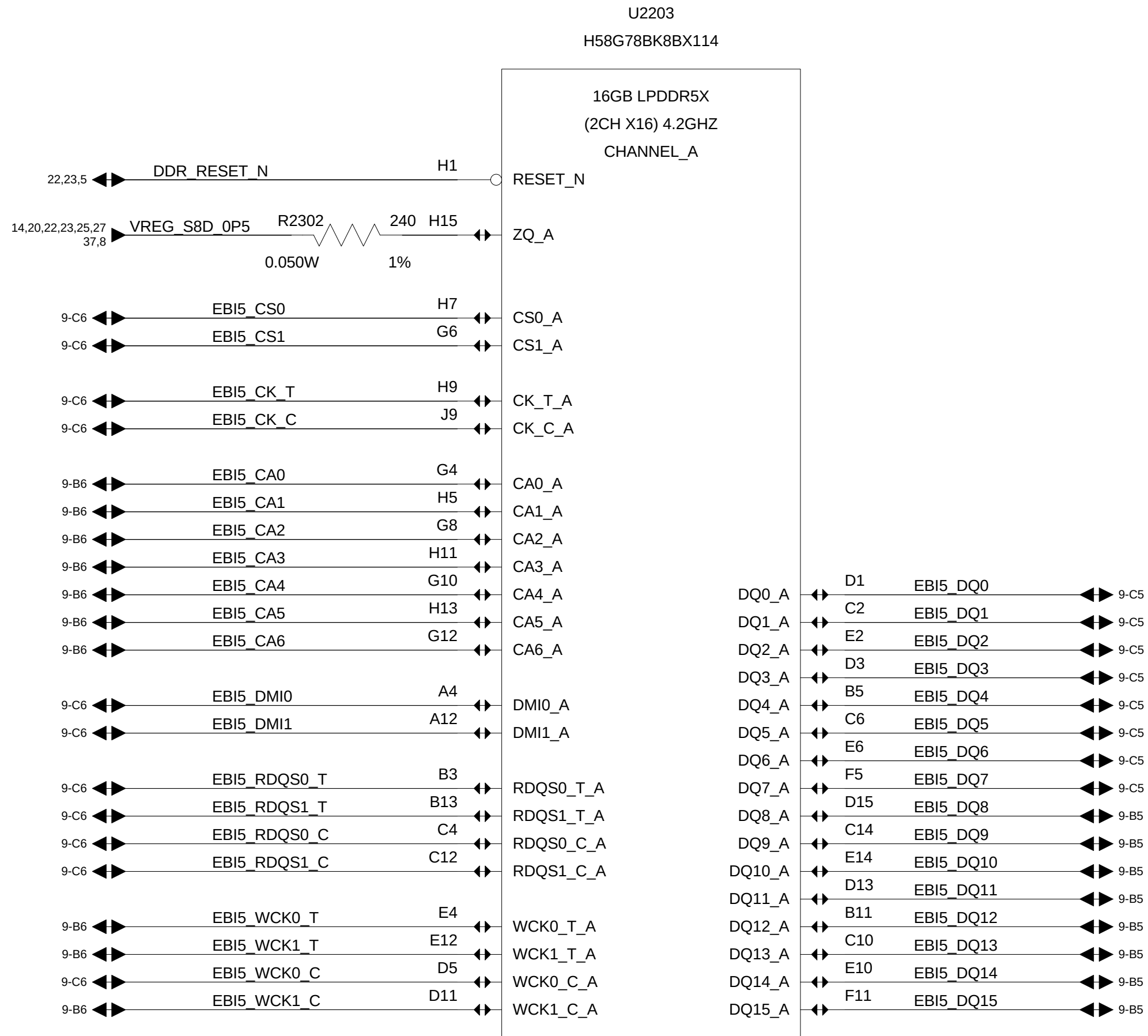


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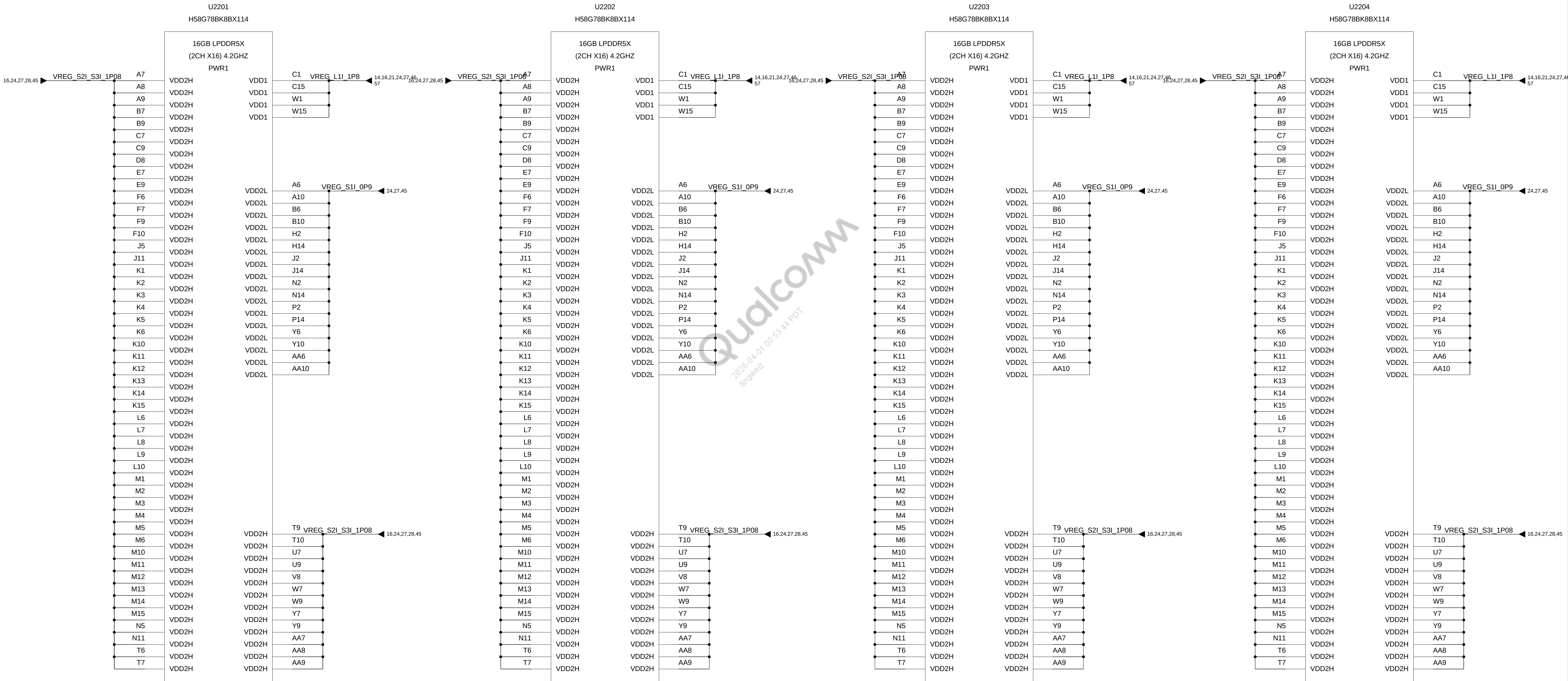
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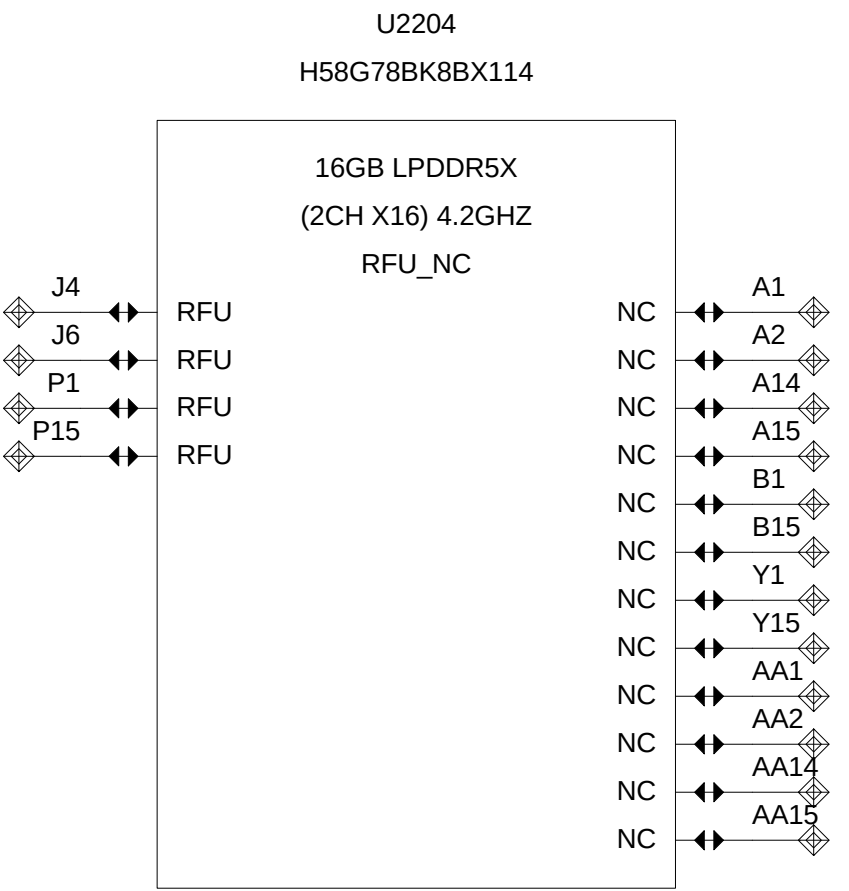
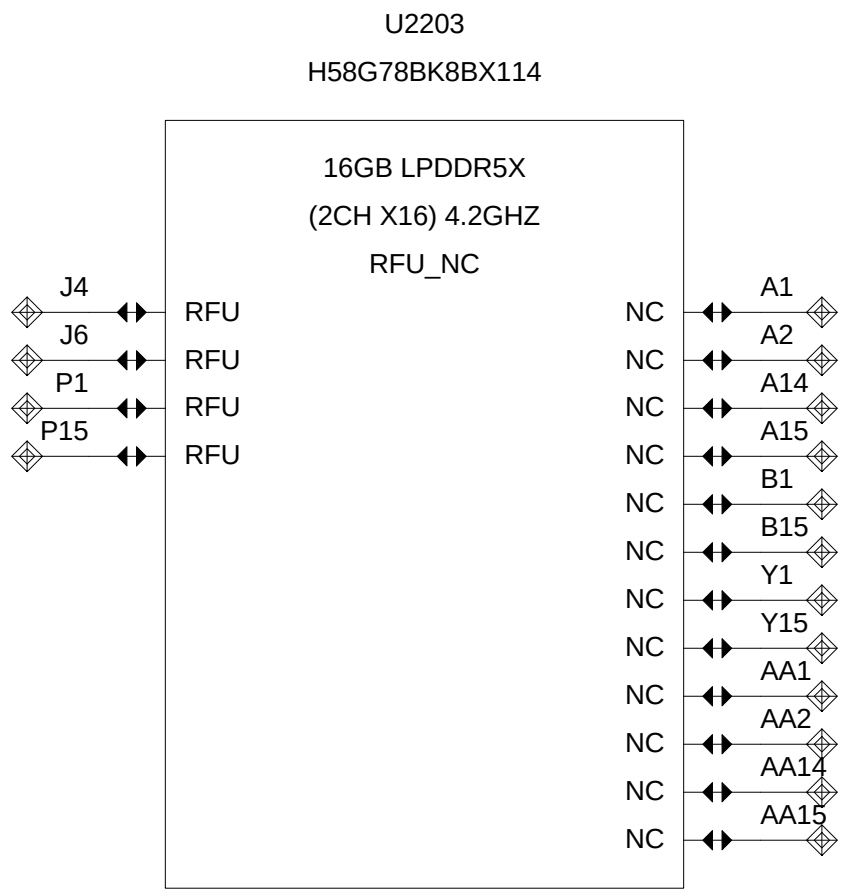
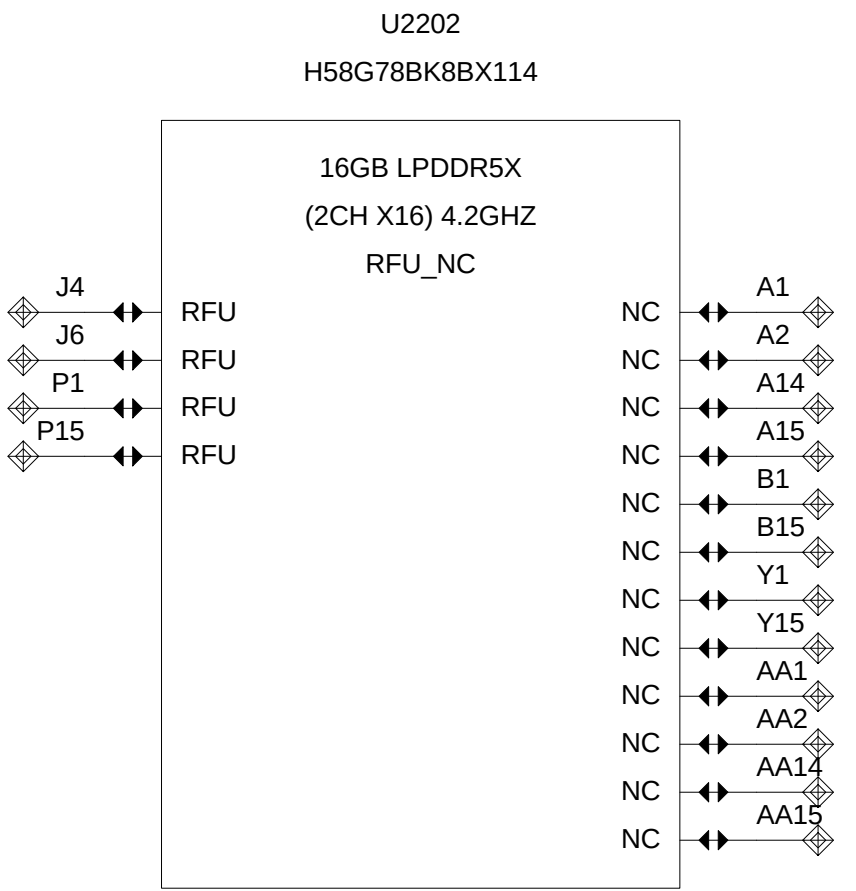
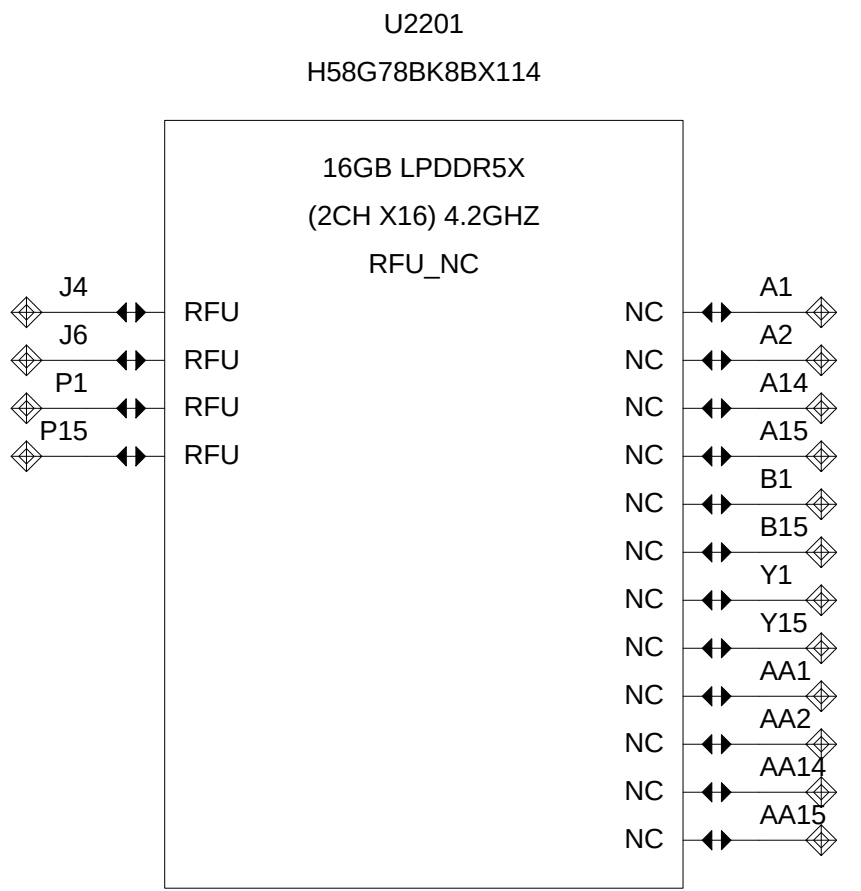
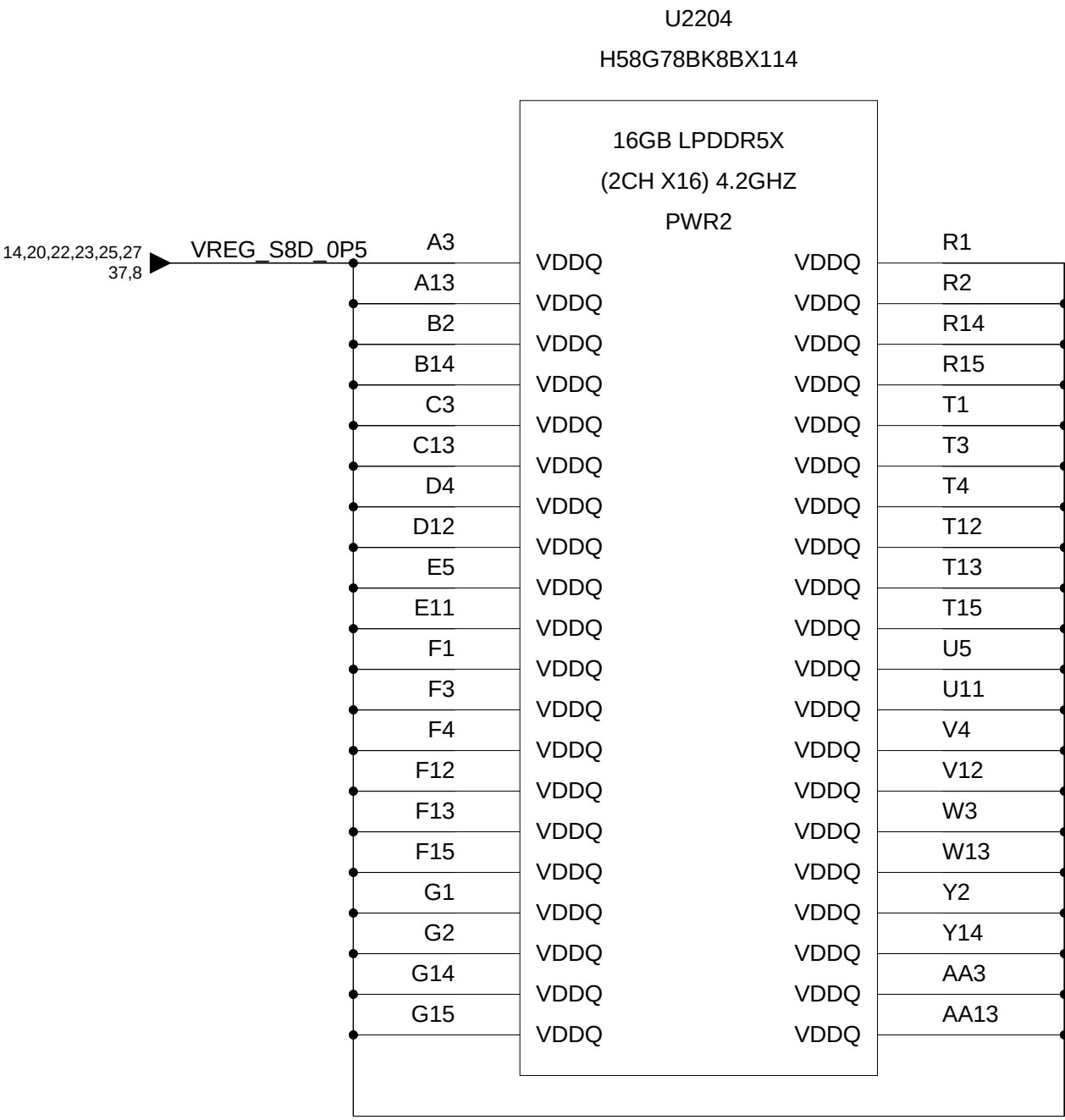
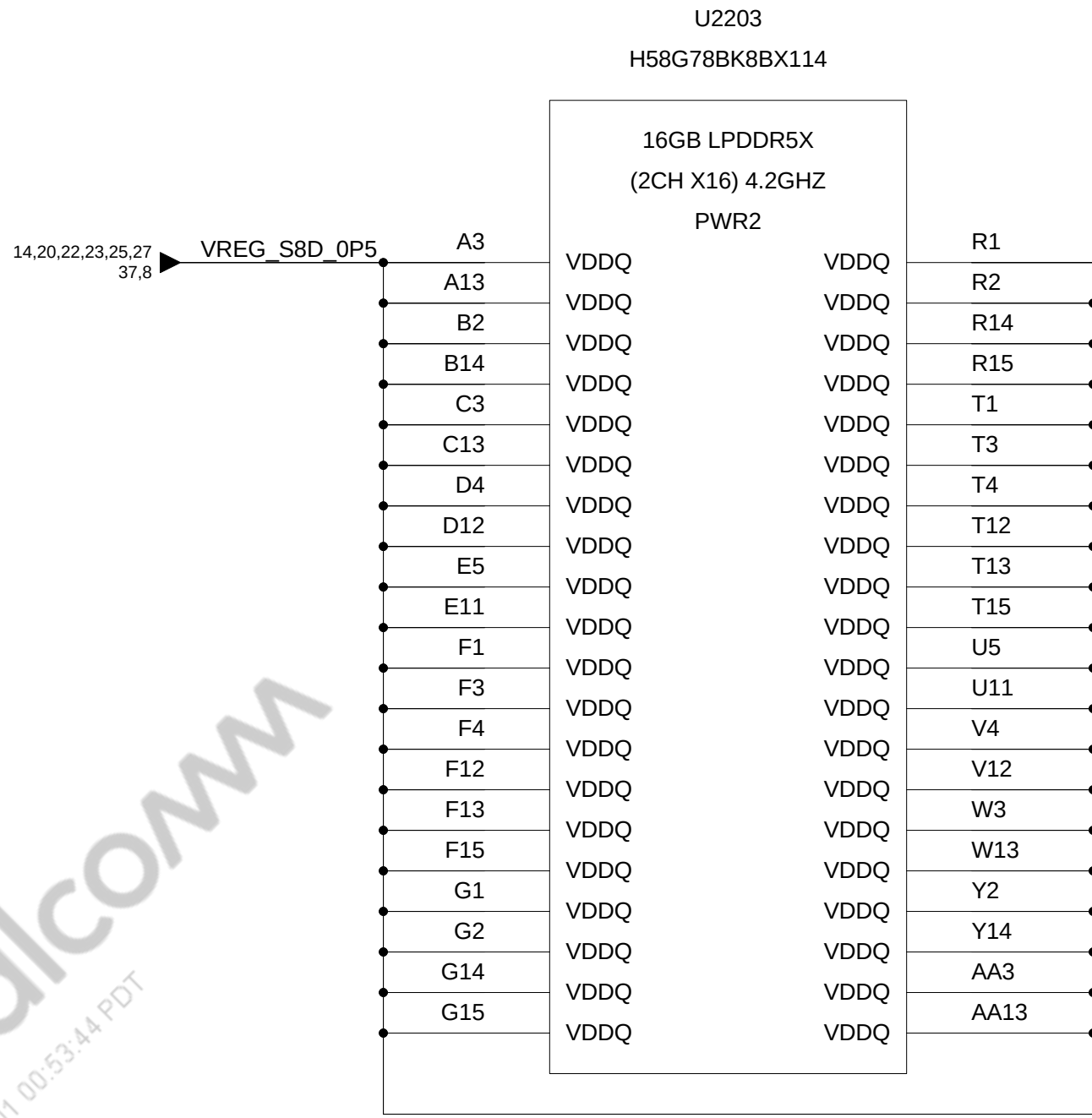
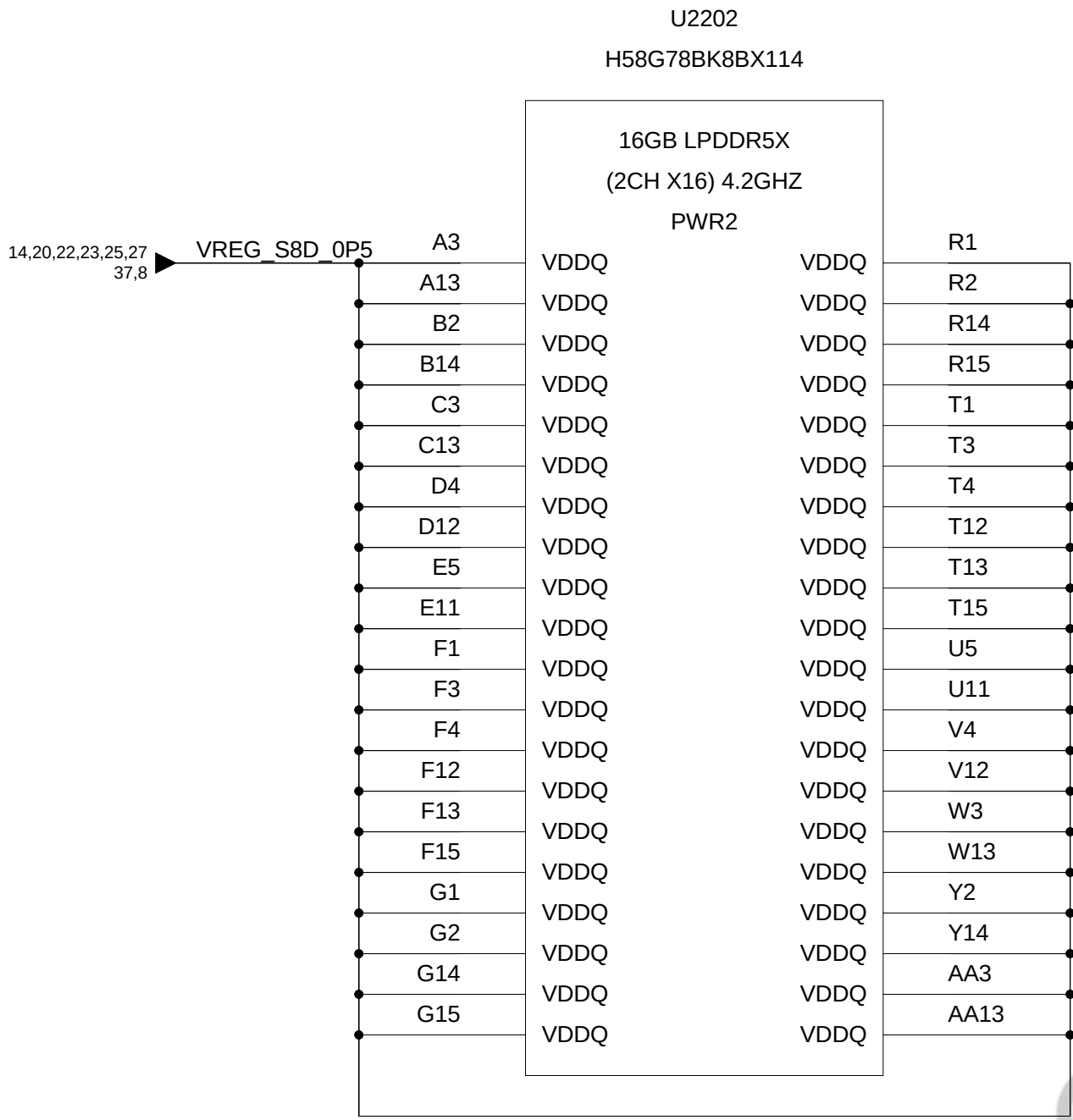
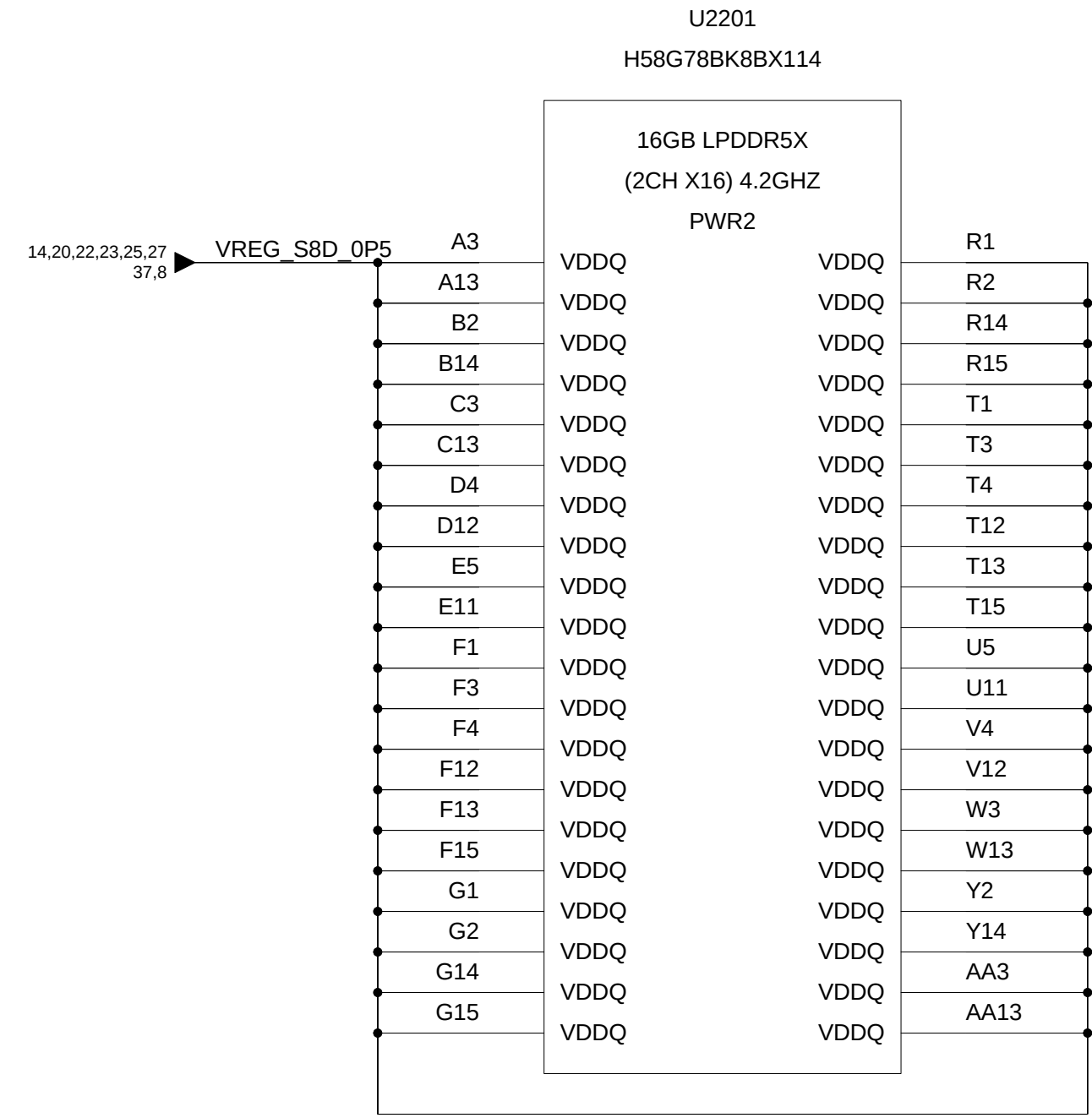
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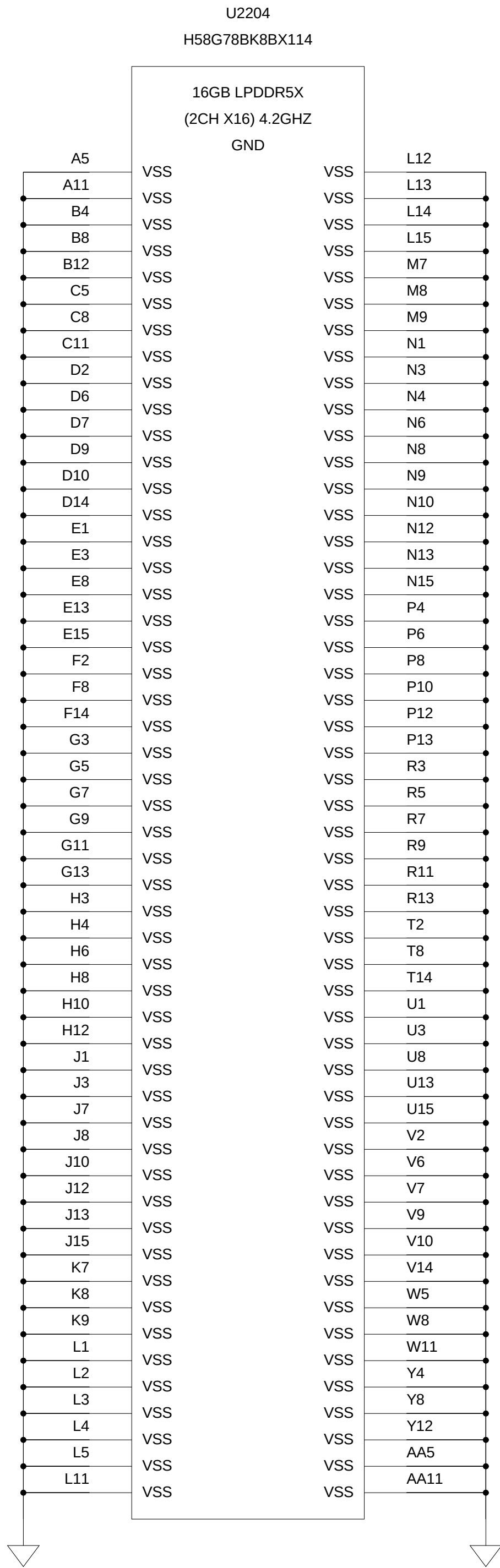
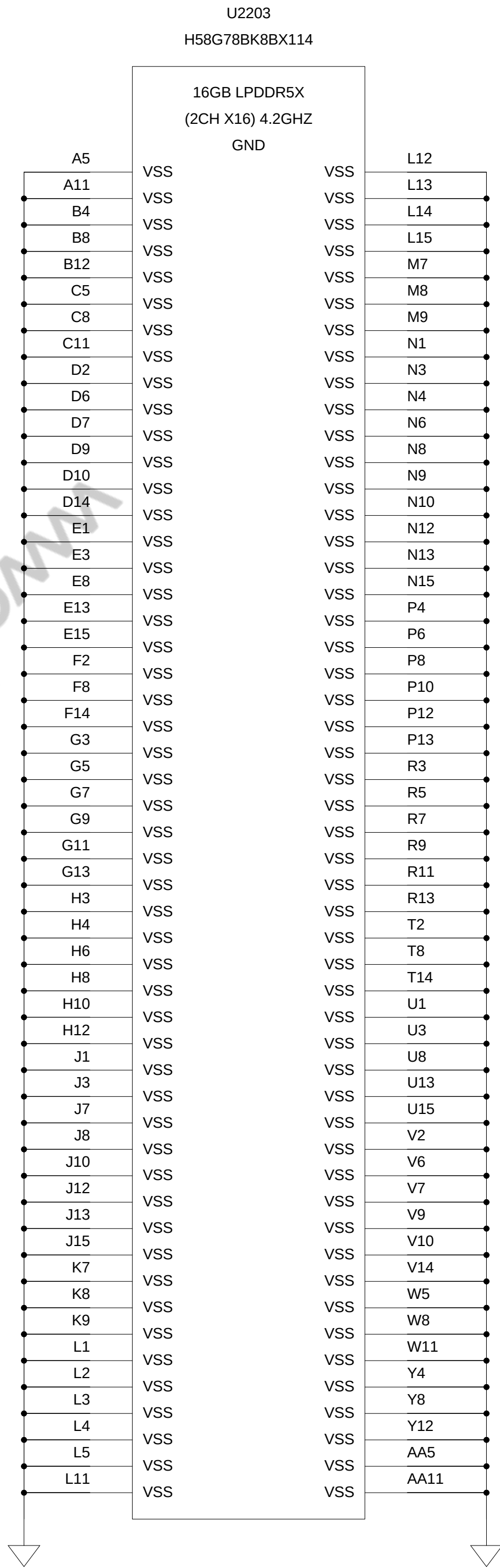
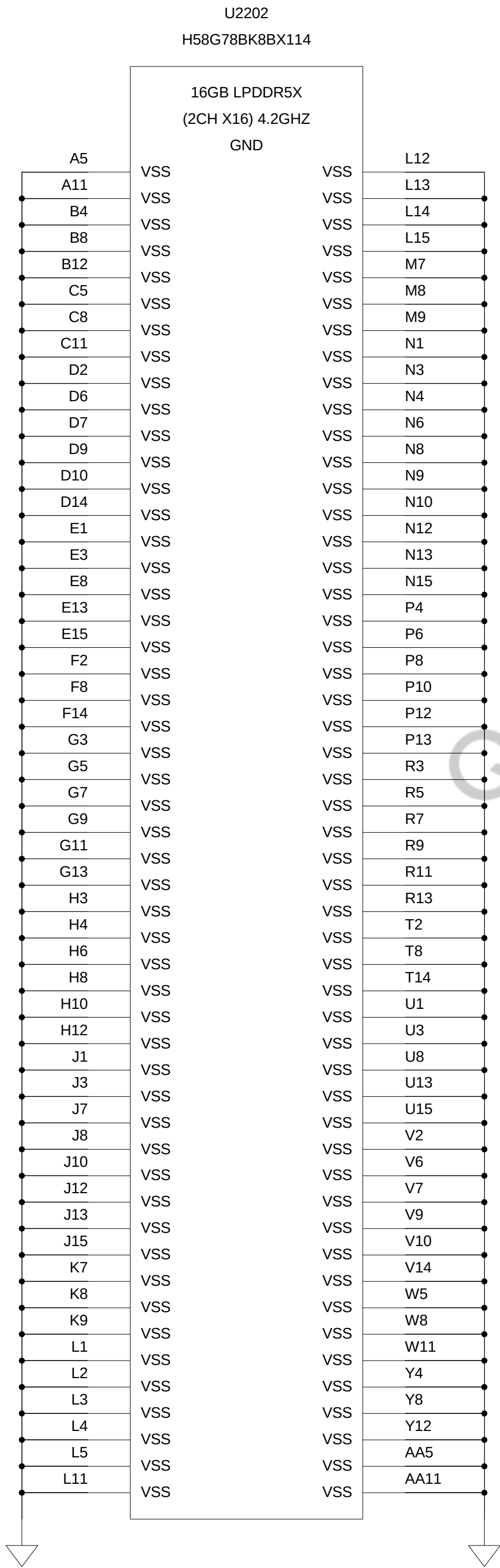
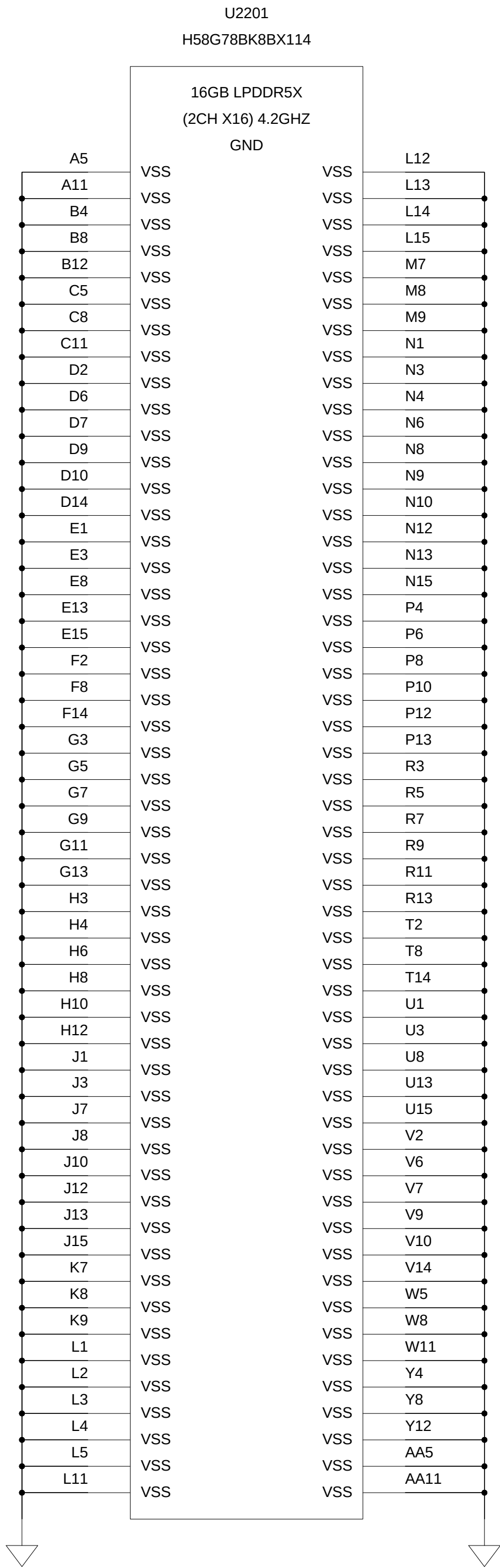
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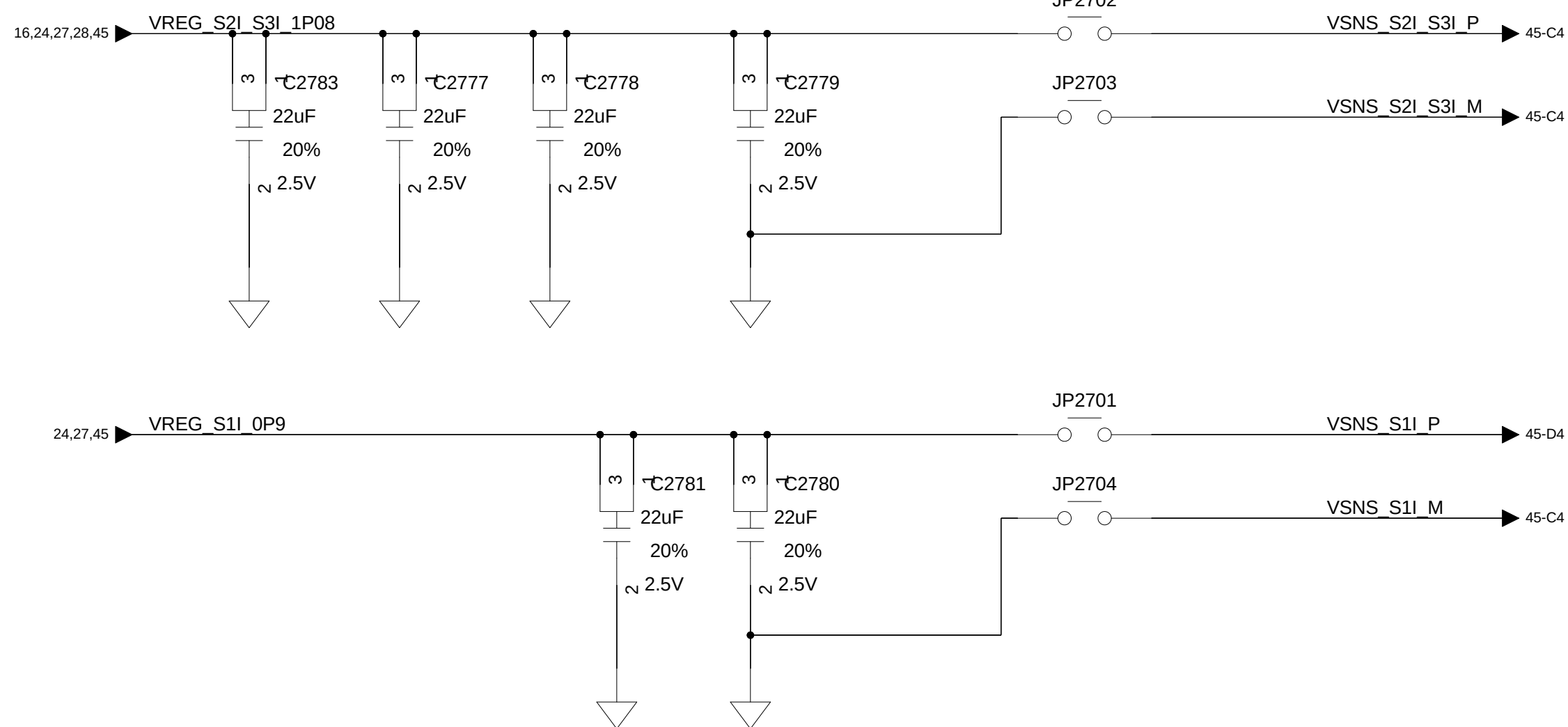


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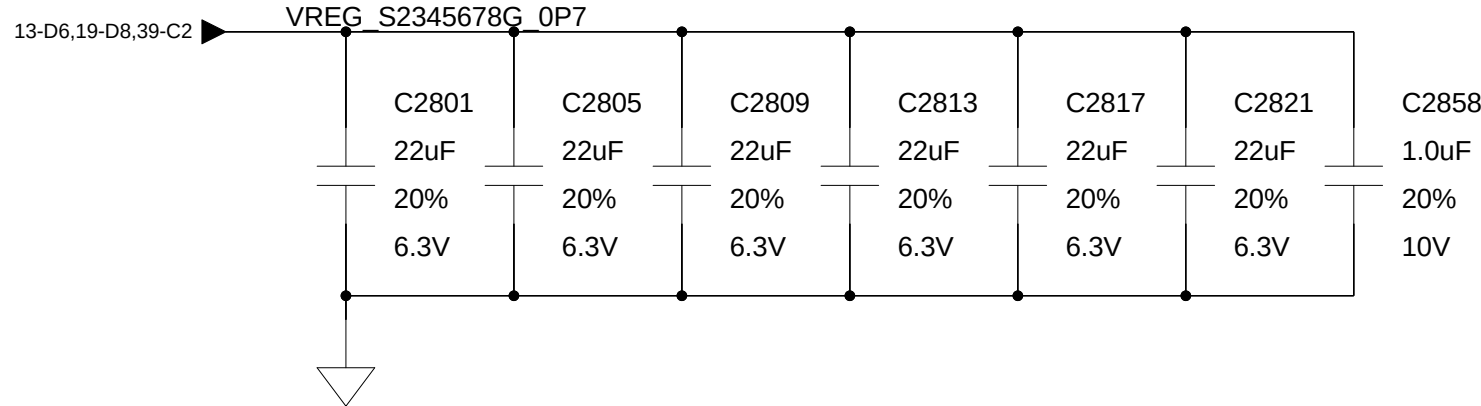




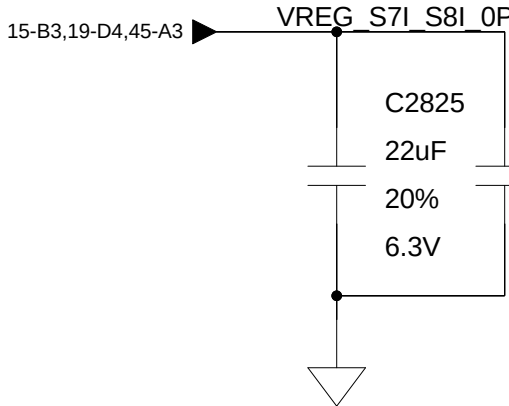
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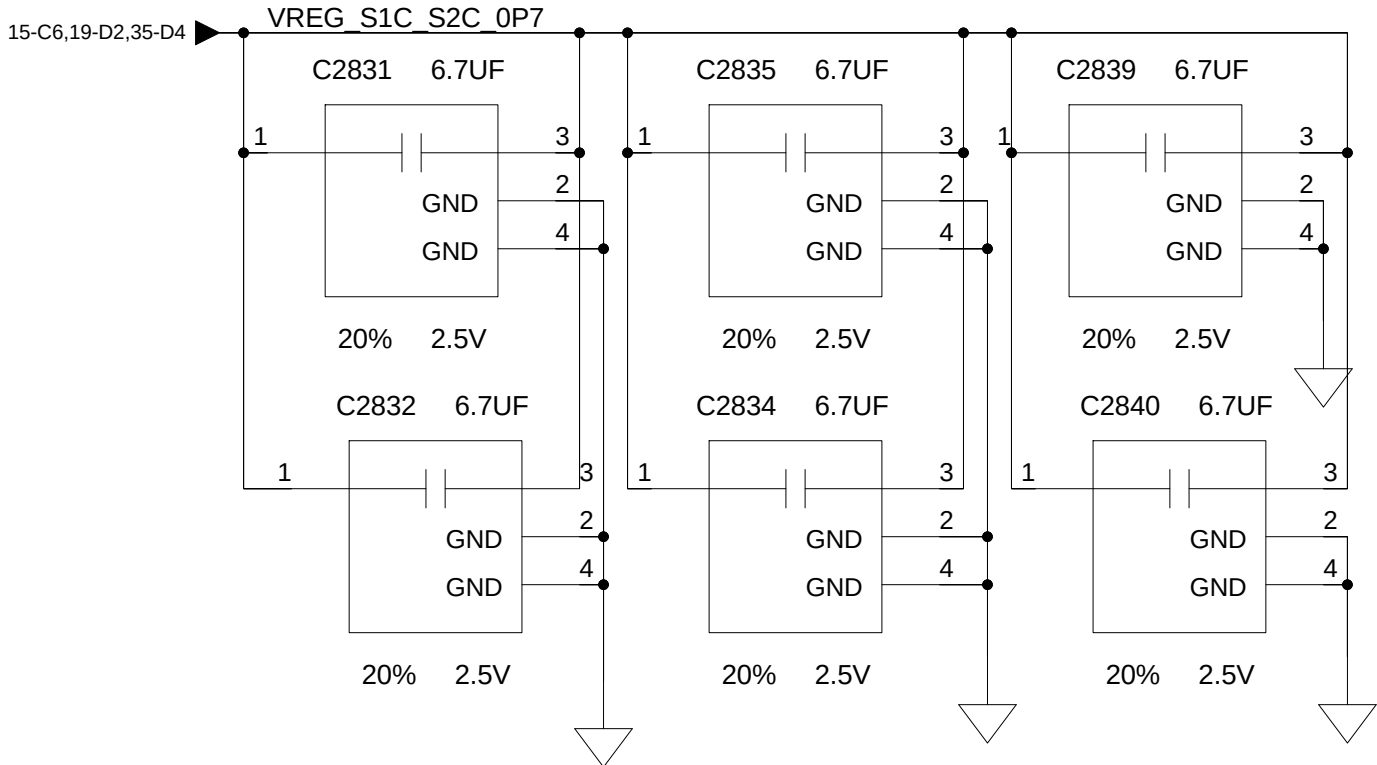
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Dedicated sense pins on package



VDD_MX_APC0
Dedicated sense pins on package

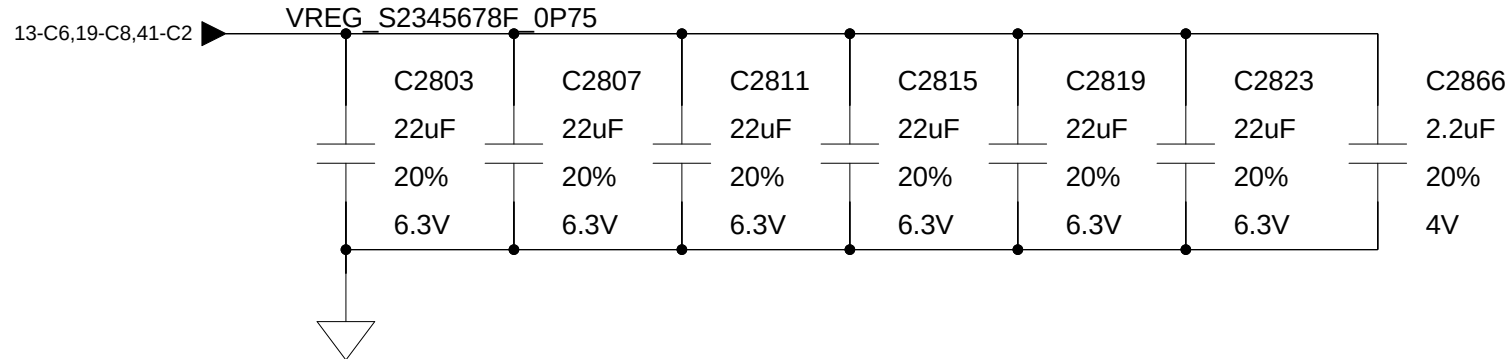


VDD_NSP
Dedicated sense pins on package

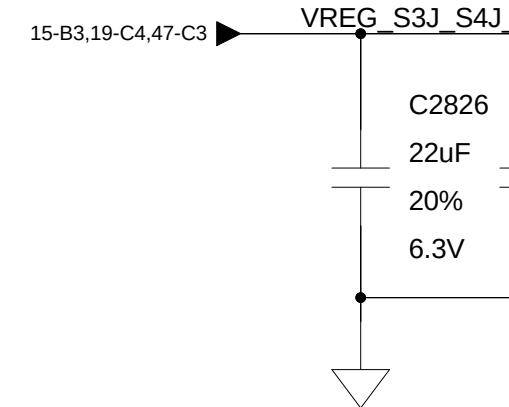


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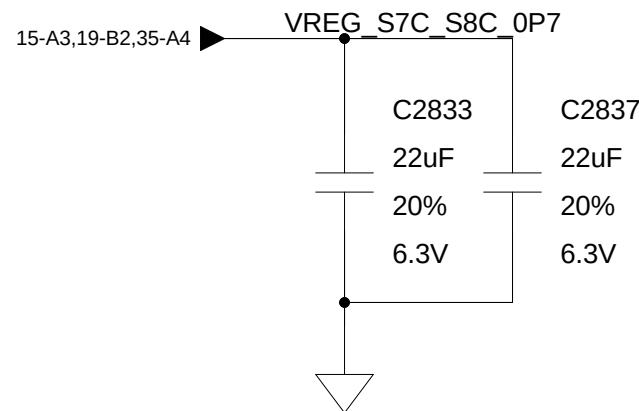
APC1 - PWR
Dedicated sense pins on package



VDD_MX_APC1
Dedicated sense pins on package

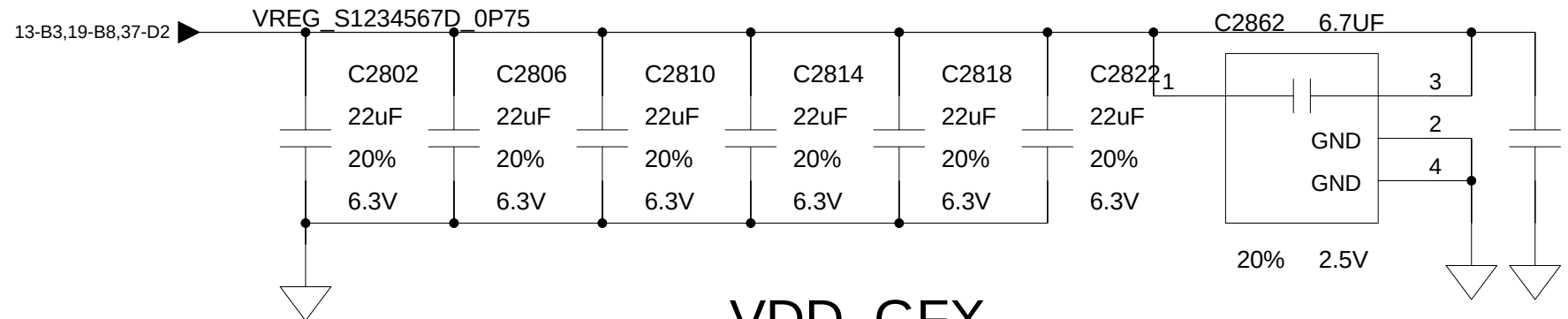


MX_GFX
Dedicated sense pins on package

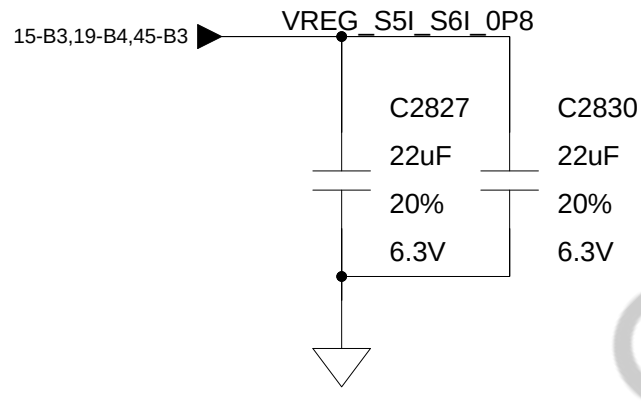


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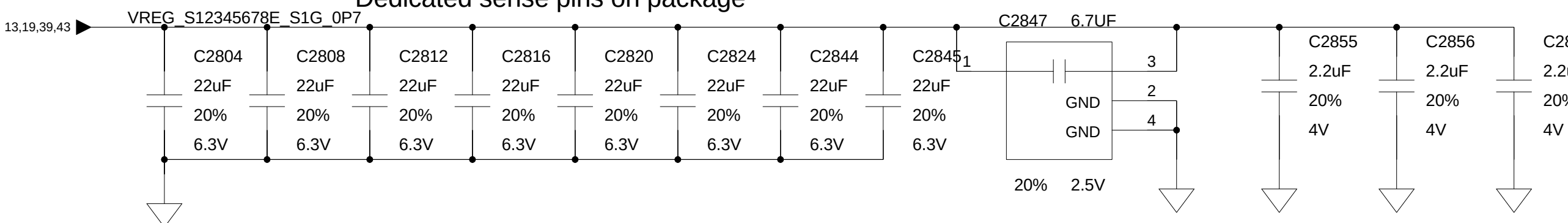
APC2 - PWR
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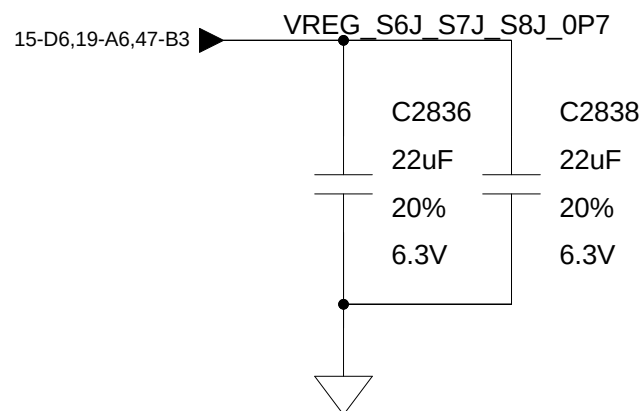
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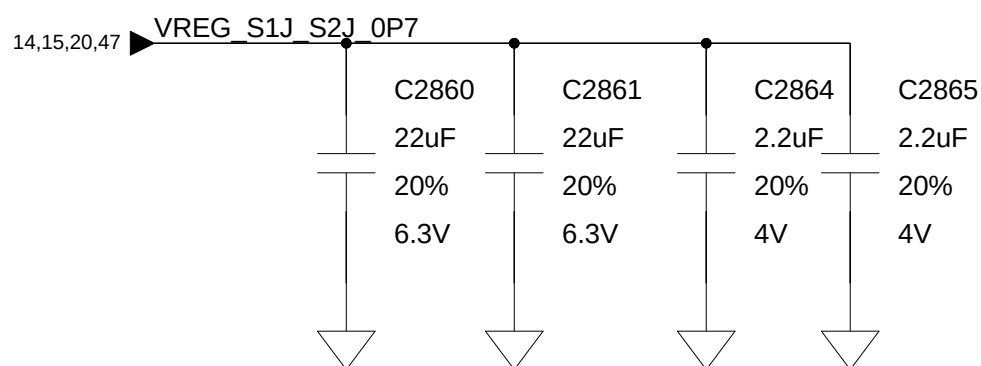
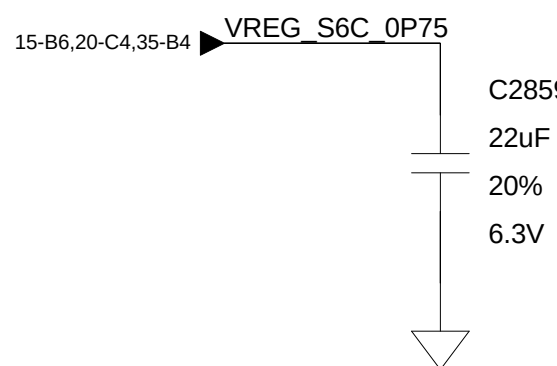
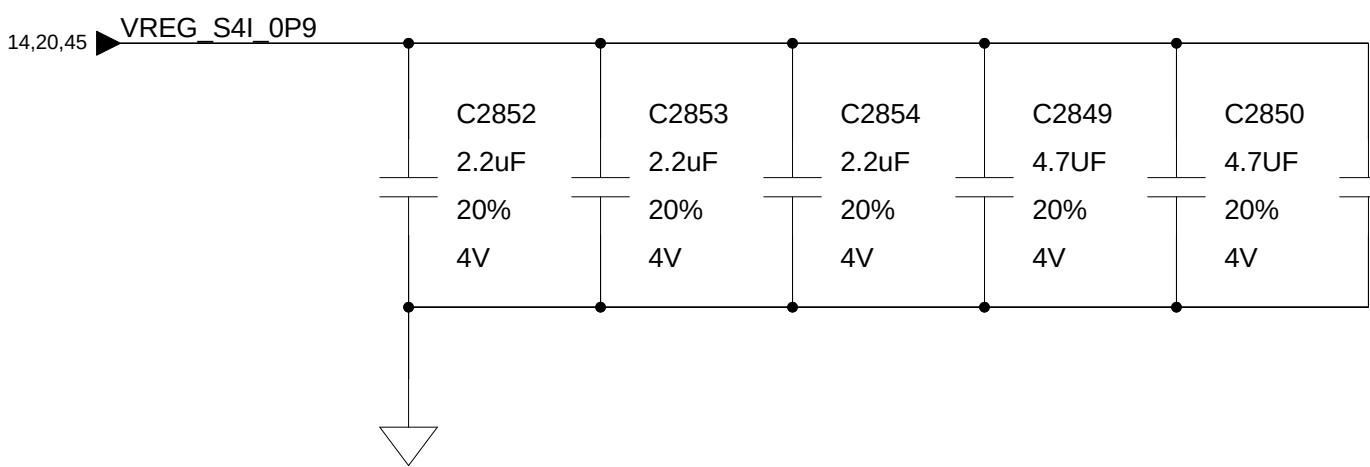
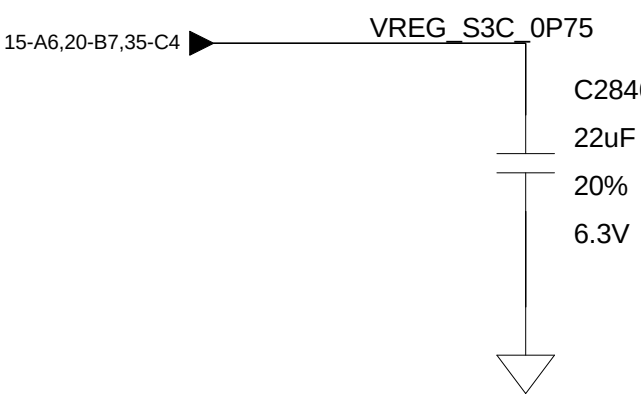
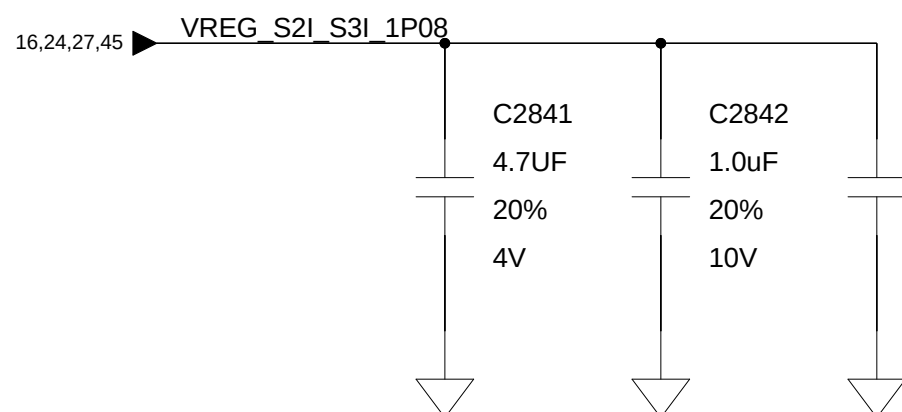
VDD_GFX
Dedicated sense pins on package



VDD_MM
Dedicated sense pins on package



A



SIZE	DRAWING NO	REV
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SCALE NONE	SHEET 28 OF 62	

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RESERVED_03

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SCALE	NONE	SHEET 29	OF 62

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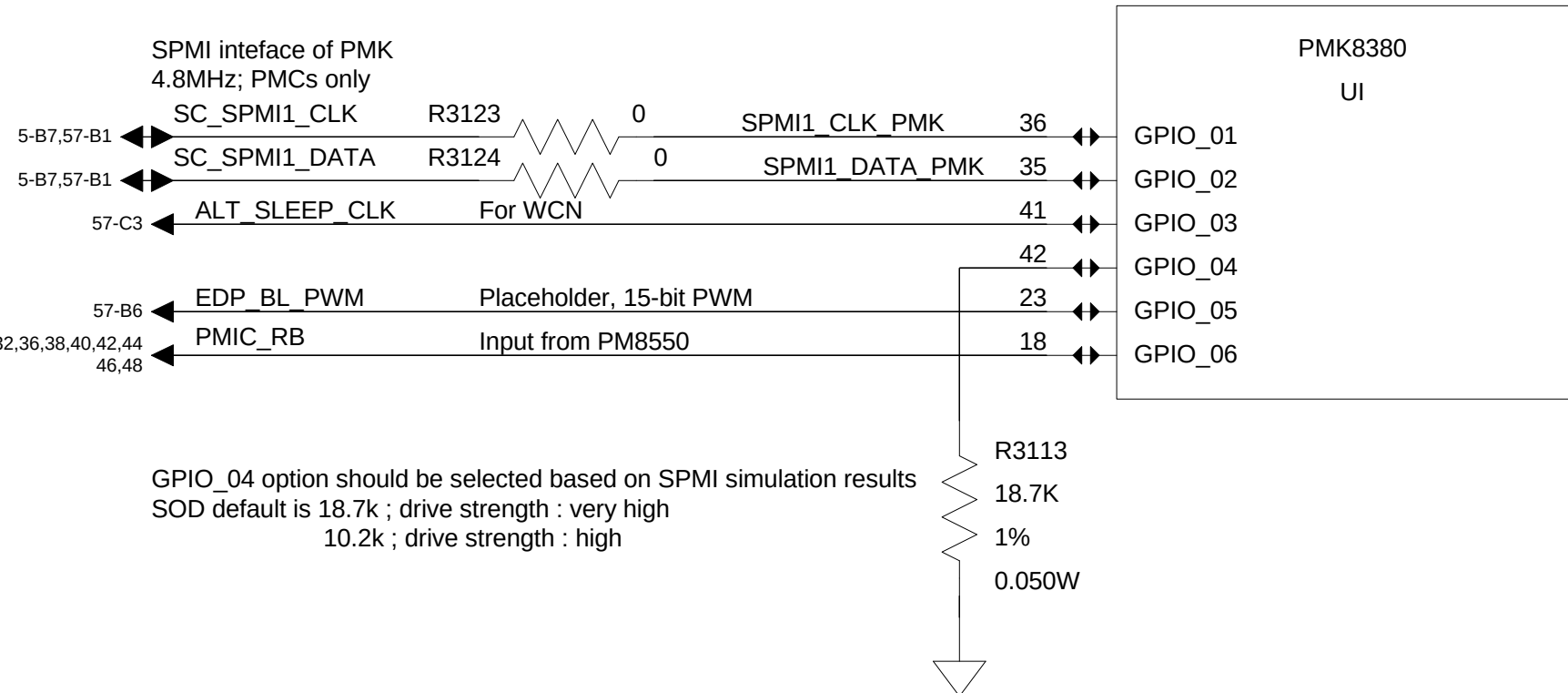
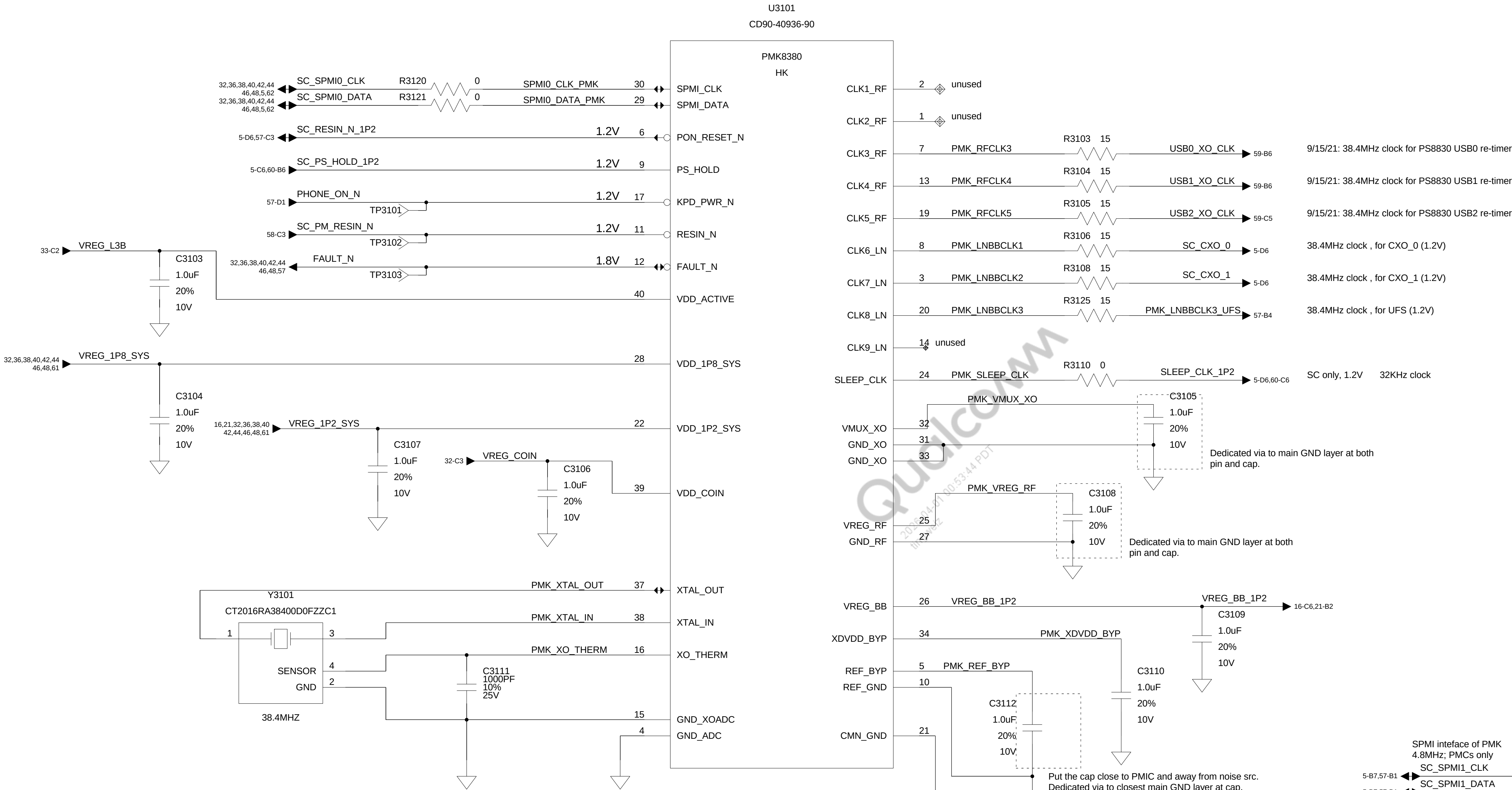


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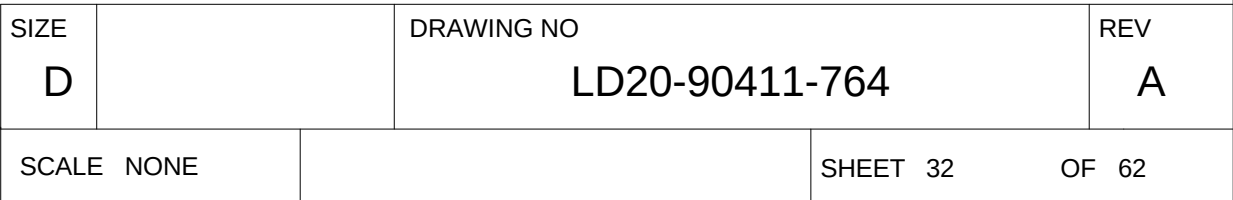
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SCALE	NONE	SHEET 30 OF 62	

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

PMK8380_A



PMIC8380C_B_HK_UI_GPIO



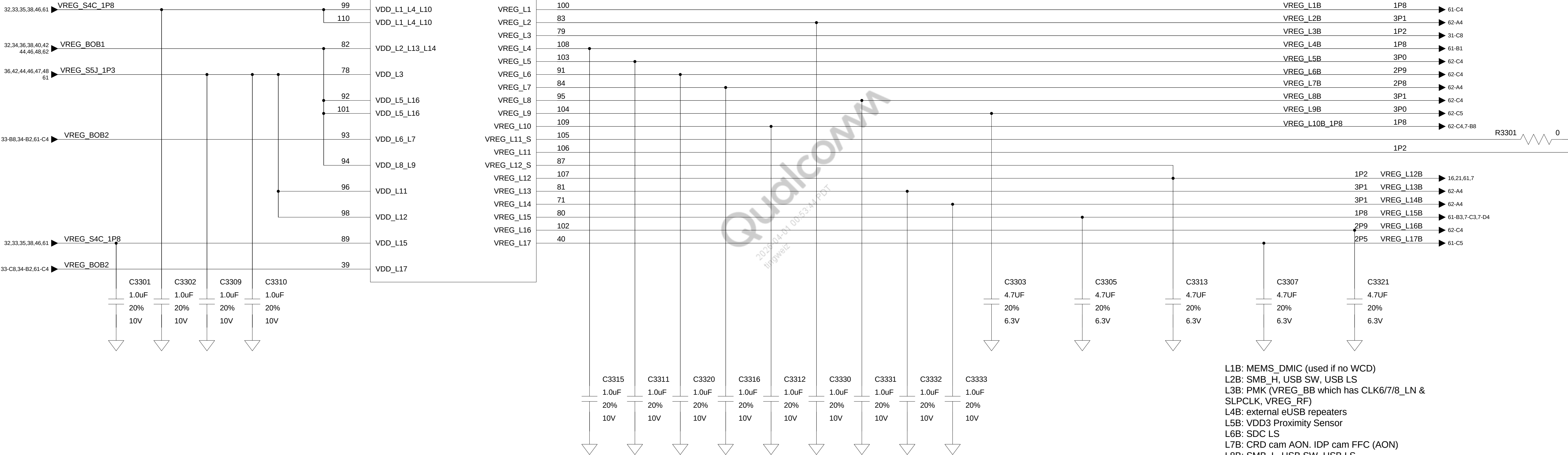
REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

PMC8380C_B

BOB2 sources: L17, L6, L7
BOB1 sources: L8, L9, L2, L13, L14, L5, L16
MV LDOs sources: L3, L11, L12
HV LDOs sources: L1, L4, L10, L15

U3201
CD90-40927-90

PMC8380C
VREG2

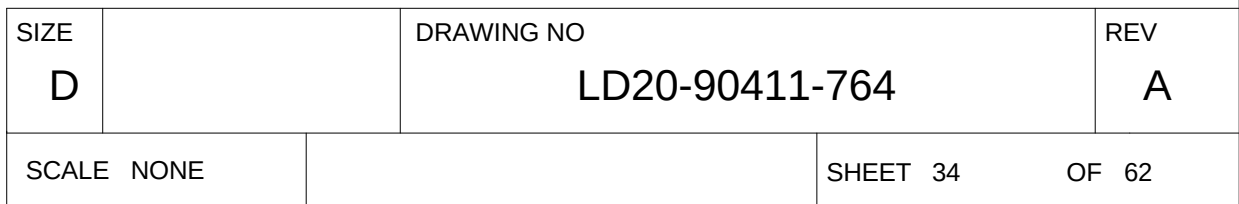


L1B: MEMS_DMIC (used if no WCD)
L2B: SMB_H, USB SW, USB LS
L3B: PMK (VREG_BB which has CLK6/7/8_LN & SLPCLK, VREG_RF)
L4B: external eUSB repeaters
L5B: VDD3 Proximity Sensor
L6B: SDC LS
L7B: CRD cam AON, IDP cam FFC (AON)
L8B: SMB_L, USB SW, USB LS
L9B: SD_MMC_VDD
L10B: Sensors
L11B: unused
L12B: WSA884X_0/1/2/3 VDD_IO, PX14, PX7xx 1P2, SDC LS, misc. LS
L13B: external eUSB repeaters
L14B: SMB_K, USB SW, USB LS
L15B: WSA884X_0/1/2/3 VDD/VDD_IO_OPT, WCD
VDD_TX_RX/PX/BUCK, TouchS VDD, Misc., PX3, QSPI VCC, USB4.0 re-timer SWs, MAX34417 VDDIO, WCN VDD18_IO and VDD18_AON
L16B: IDP RFC wide cam
L17B: UFS VCC

PMC8380C_B_LDO

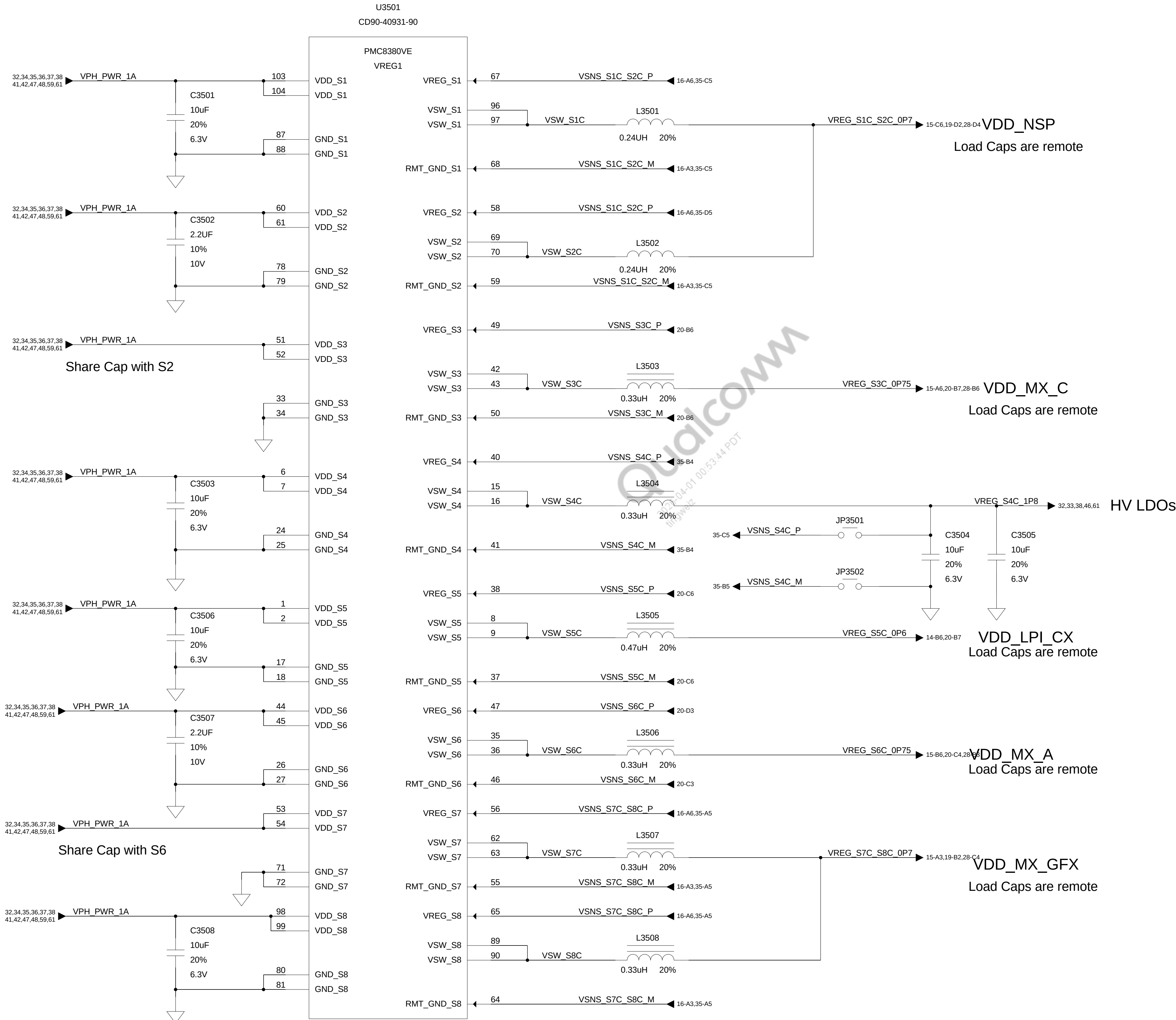
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SCALE NONE	SHEET 33 OF 62	

PMC8380C_B_BOB



REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
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PMC8380VE_C

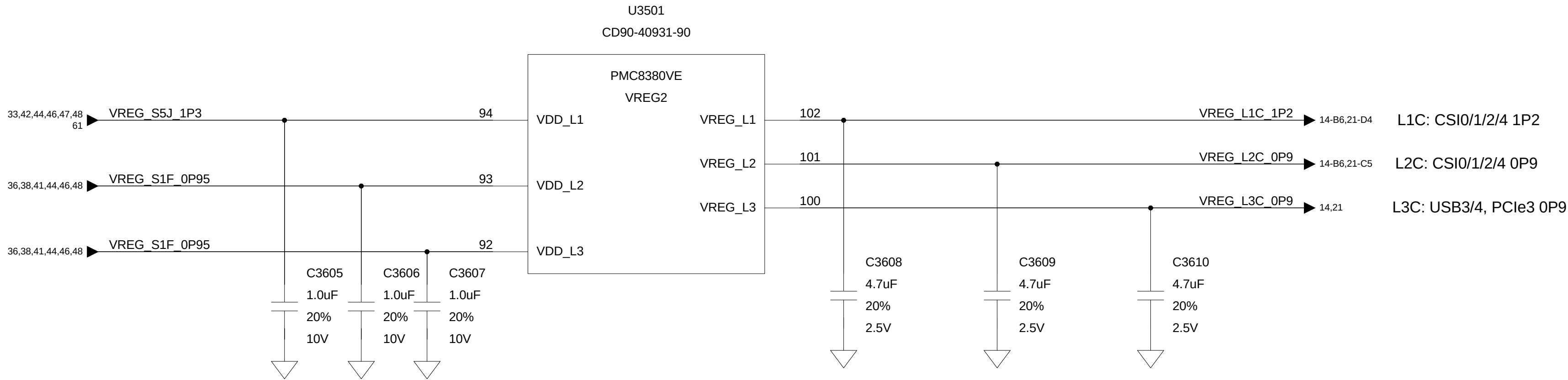
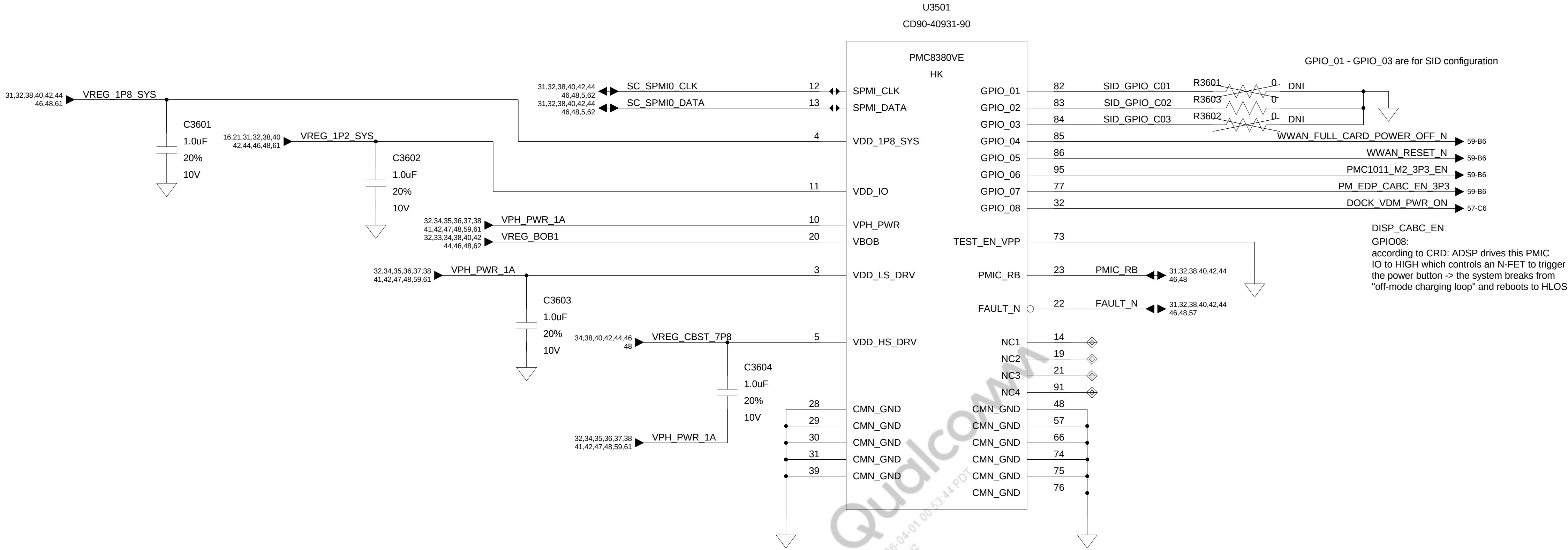


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PMC8380VE_C

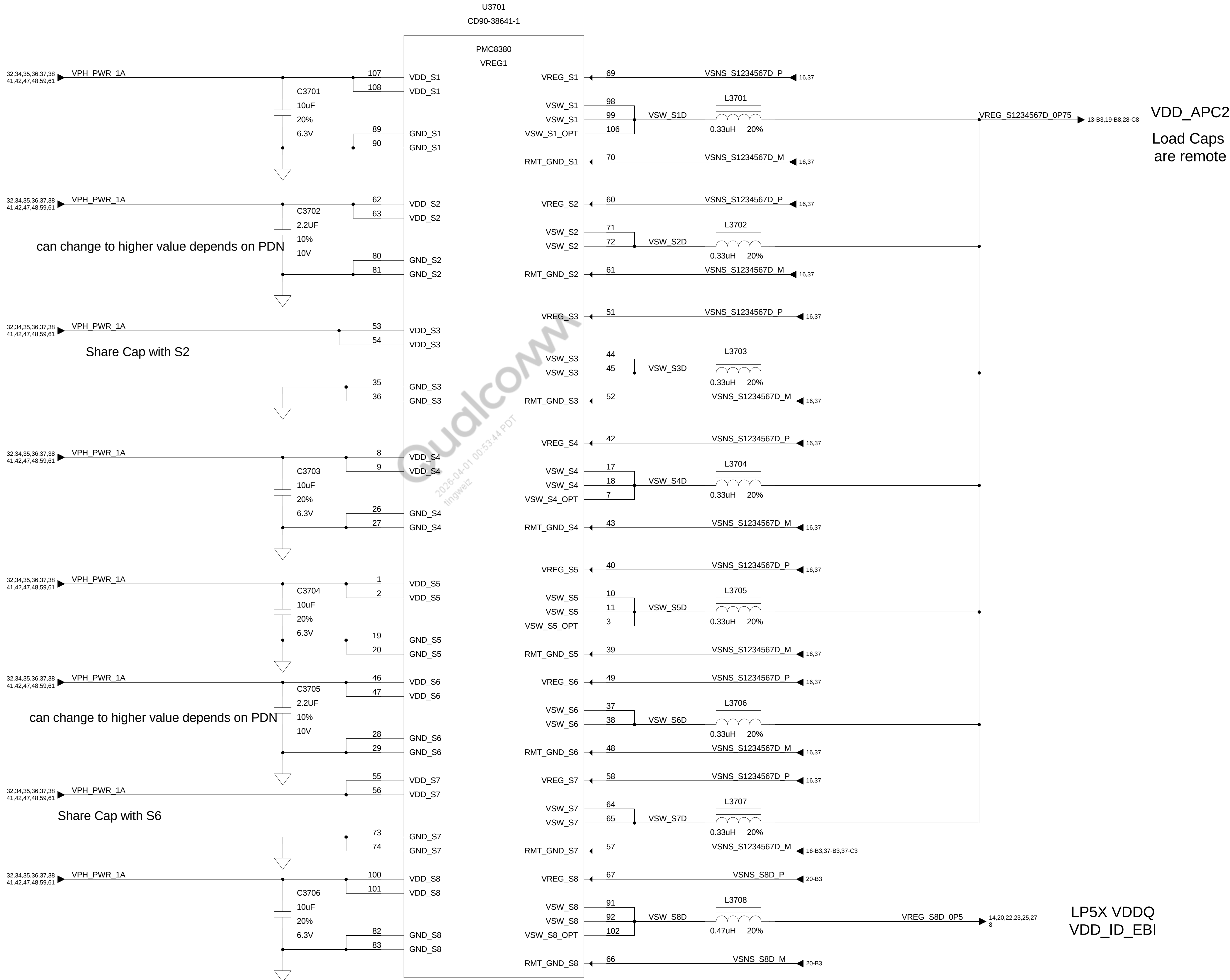
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SID	PMIC	GPIO3	GPIO2	GPIO1
2	V8C	open	GND	open
3	C8D	GND	GND	open
4	C8E	GND	GND	GND
5	C8F	open	GND	GND
6	C8G	GND	open	open
7	C8H	GND	open	GND
8	V8I	GND	open	GND
9	V8J	GND	open	open
A	C8K	open	open	open
B	C8L	open	open	GND
D	C8N	open	GND	open



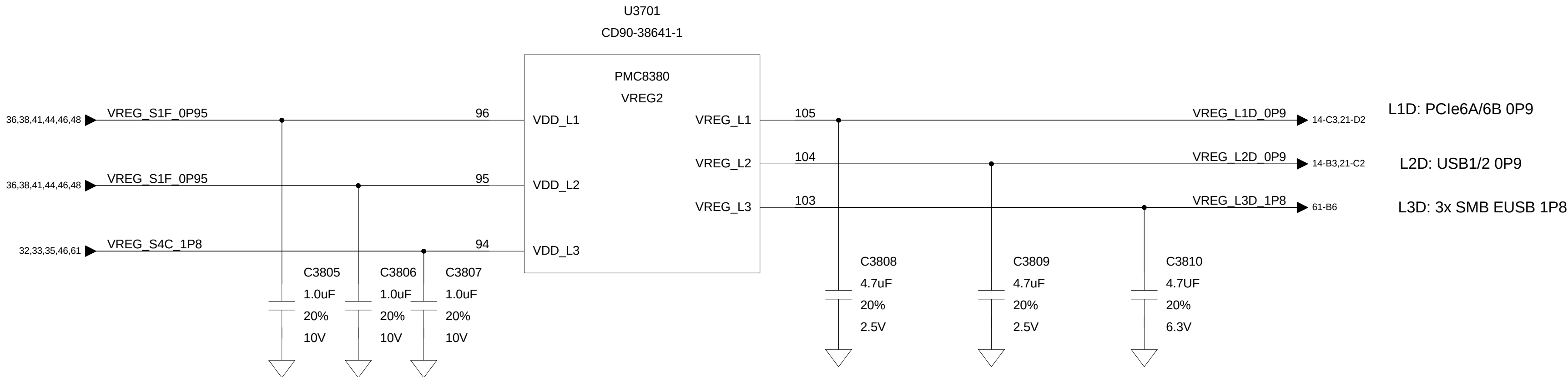
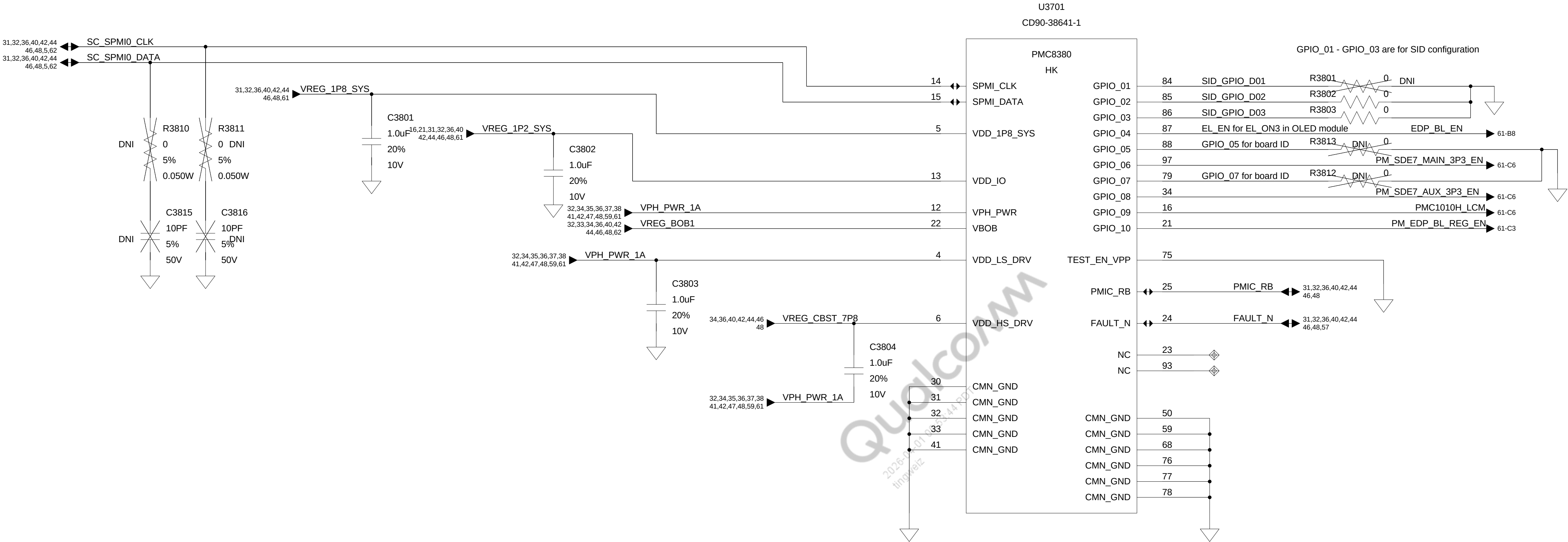
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PMC8380_D



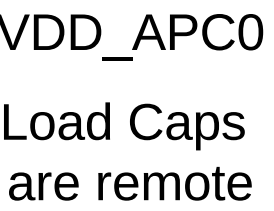
SIZE	DRAWING NO	REV
D	LD20-90411-764	A
SCALE NONE	SHEET 37 OF 62	

PMC8380_D



SID	PMIC	GPIO3	GPIO2	GPIO1
2	V8C	open	GND	open
3	C8D	GND	GND	open
4	C8E	GND	GND	GND
5	C8F	open	GND	GND
6	C8G	GND	open	open
7	C8H	GND	open	GND
9	V8J	GND	open	open
A	C8K	open	open	open
B	C8L	open	open	GND
D	C8N	open	GND	open

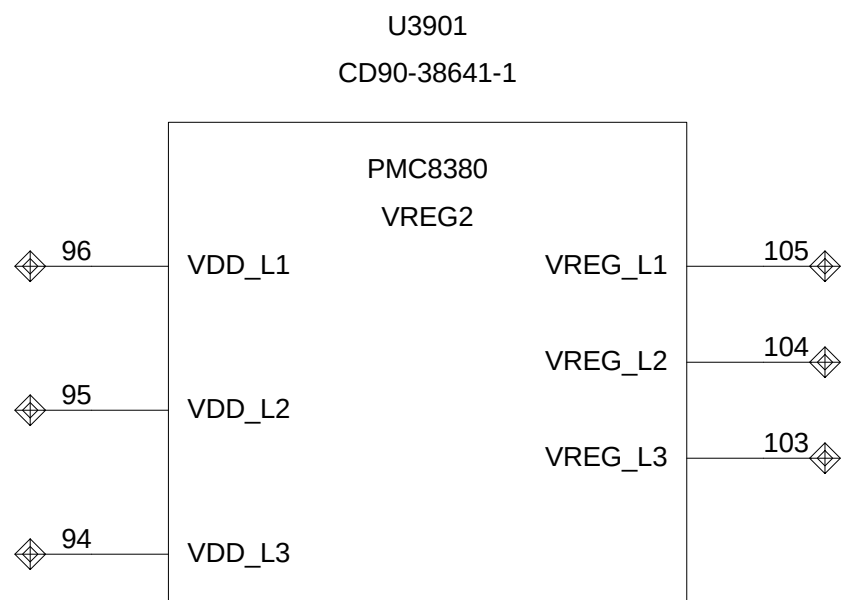
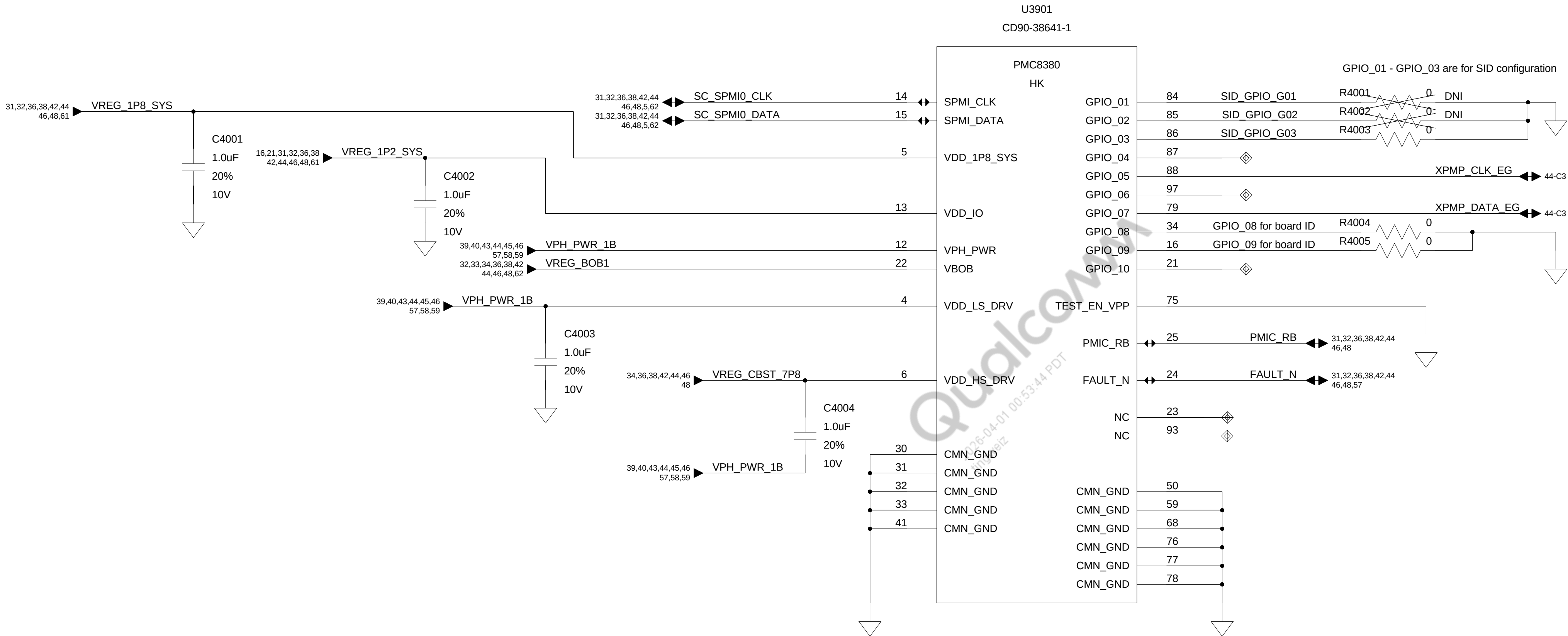
VDD_GFX
Load Caps
are remote



REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

PMC8380_G

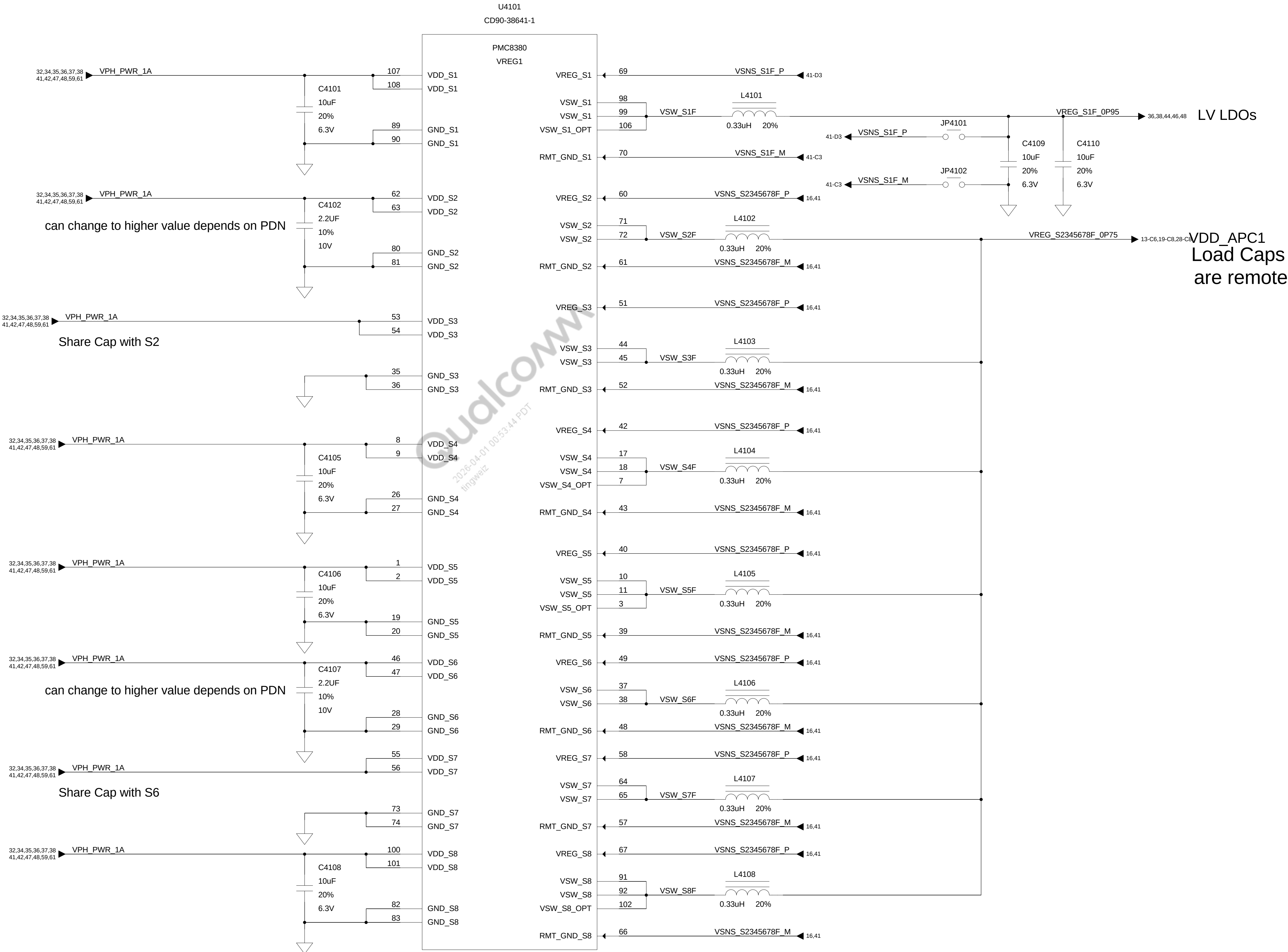
SID	PMIC	GPIO3	GPIO2	GPIO1
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3	C8D	GND	GND	open
4	C8E	GND	GND	GND
5	C8F	open	GND	GND
6	C8G	GND	open	open
7	C8H	GND	open	GND
8	V8I	GND	open	GND
9	V8J	GND	open	open
A	C8K	open	open	open
B	C8L	open	open	GND
D	C8N	open	GND	open



Not use

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

PMC8380_F

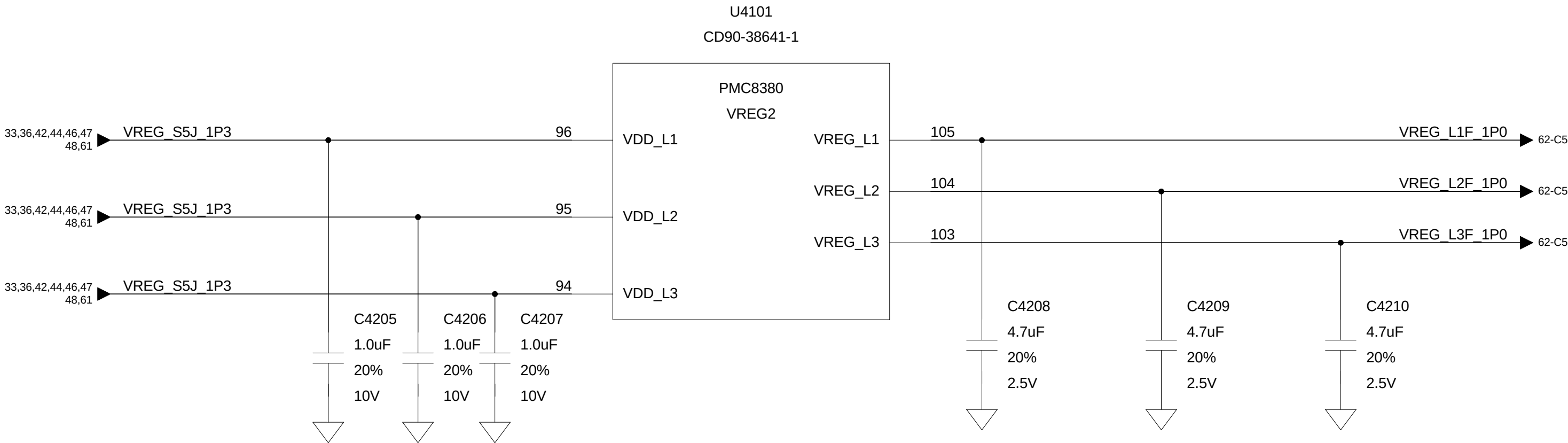
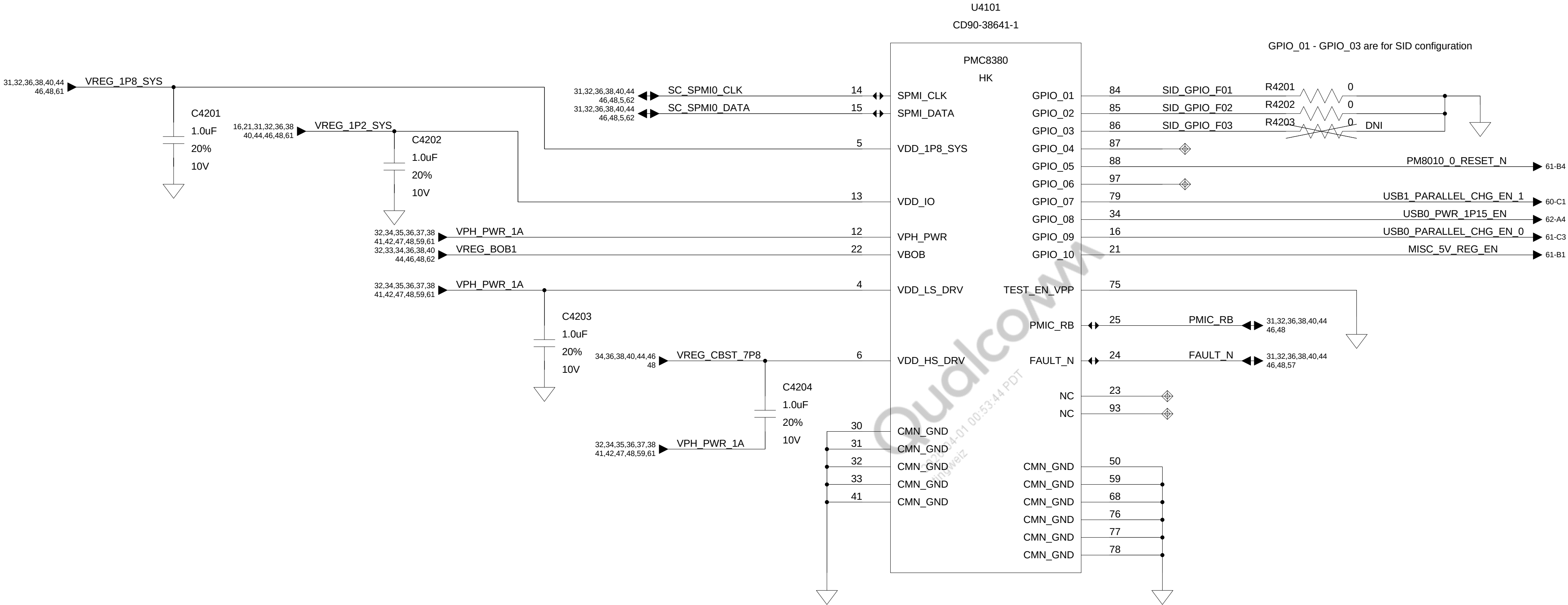


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PMC8380_F

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
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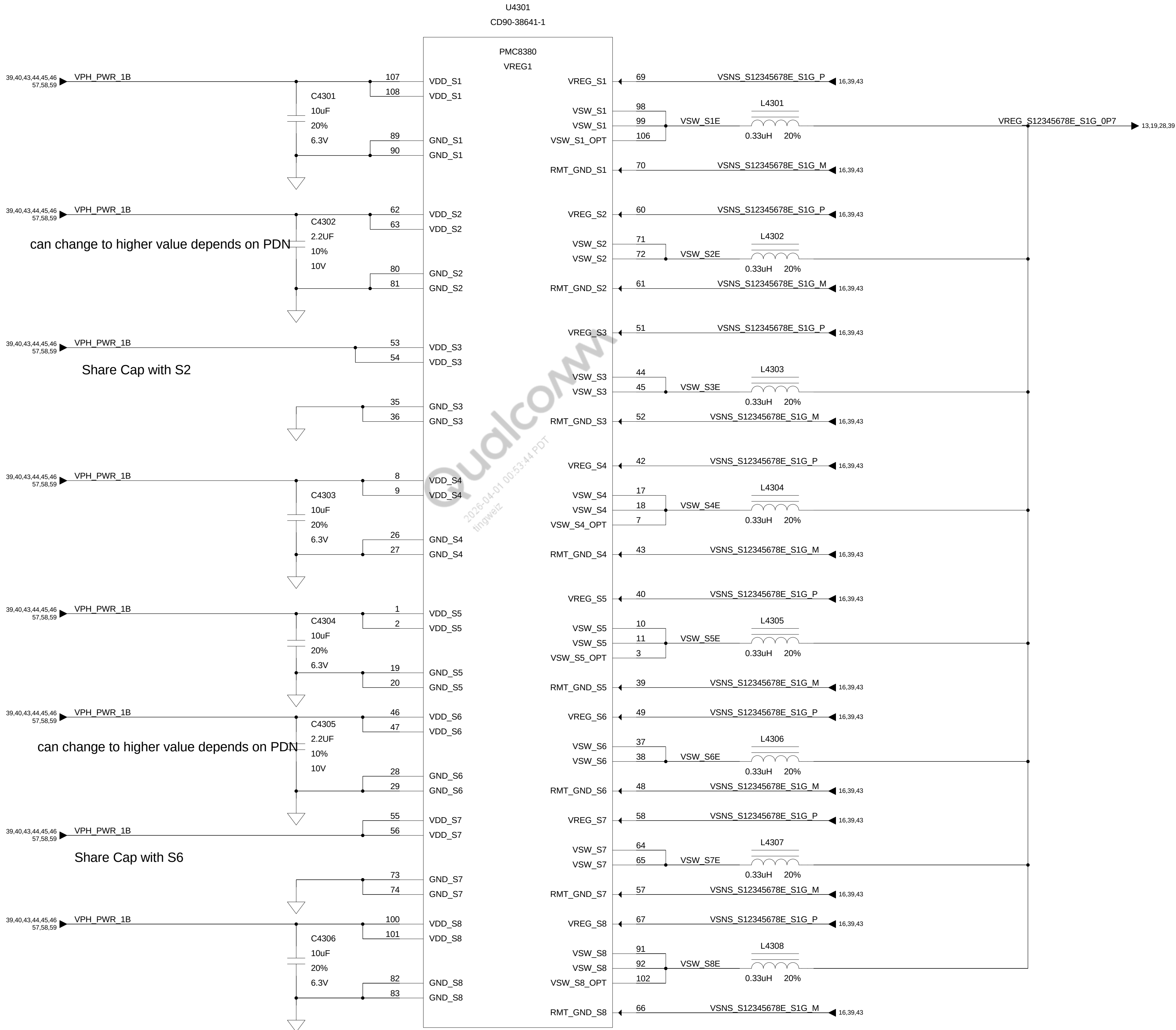
SID	PMIC	GPIO3	GPIO2	GPIO1
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3	C8D	GND	GND	open
4	C8E	GND	GND	GND
5	C8F	open	GND	GND
6	C8G	GND	open	open
7	C8H	GND	open	GND
8	V8I	GND	open	GND
9	V8J	GND	open	open
A	C8K	open	open	open
B	C8L	open	open	GND
D	C8N	open	GND	open



L1F: Option power for KB8010
L2F: Option power for KB8010
L3F: Option power for KB8010

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

PMC8380_E



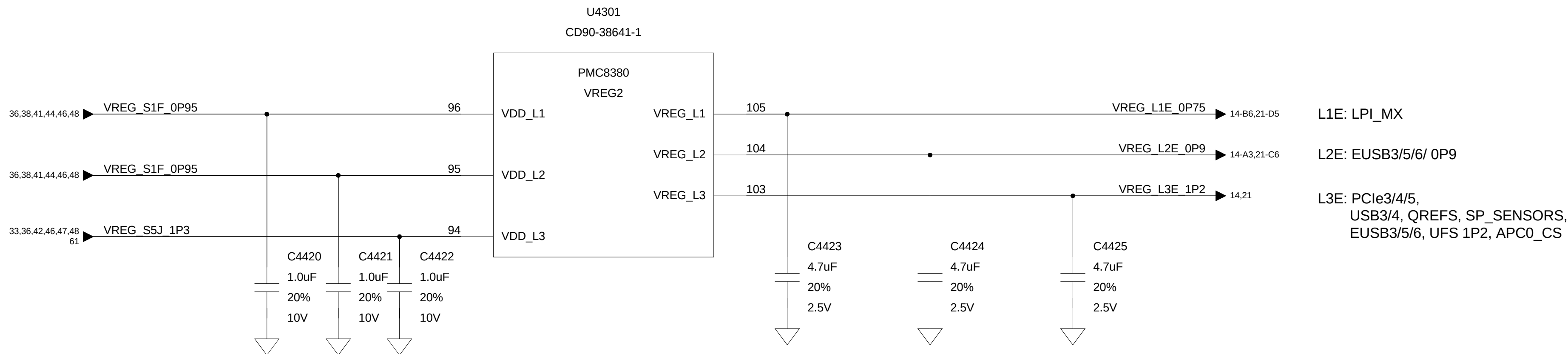
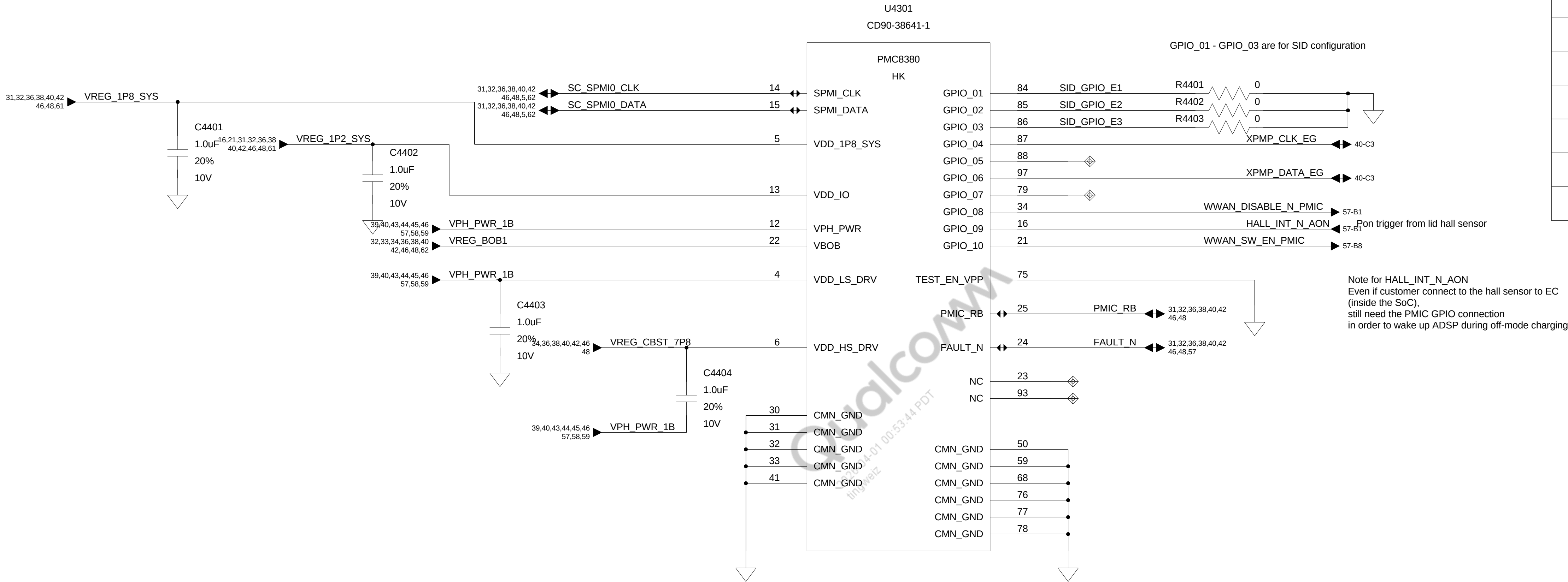
VDD_GFX
Load Caps
are remote

SIZE	DRAWING NO	REV
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SCALE NONE	SHEET 43 OF 62	

PMC8380_E

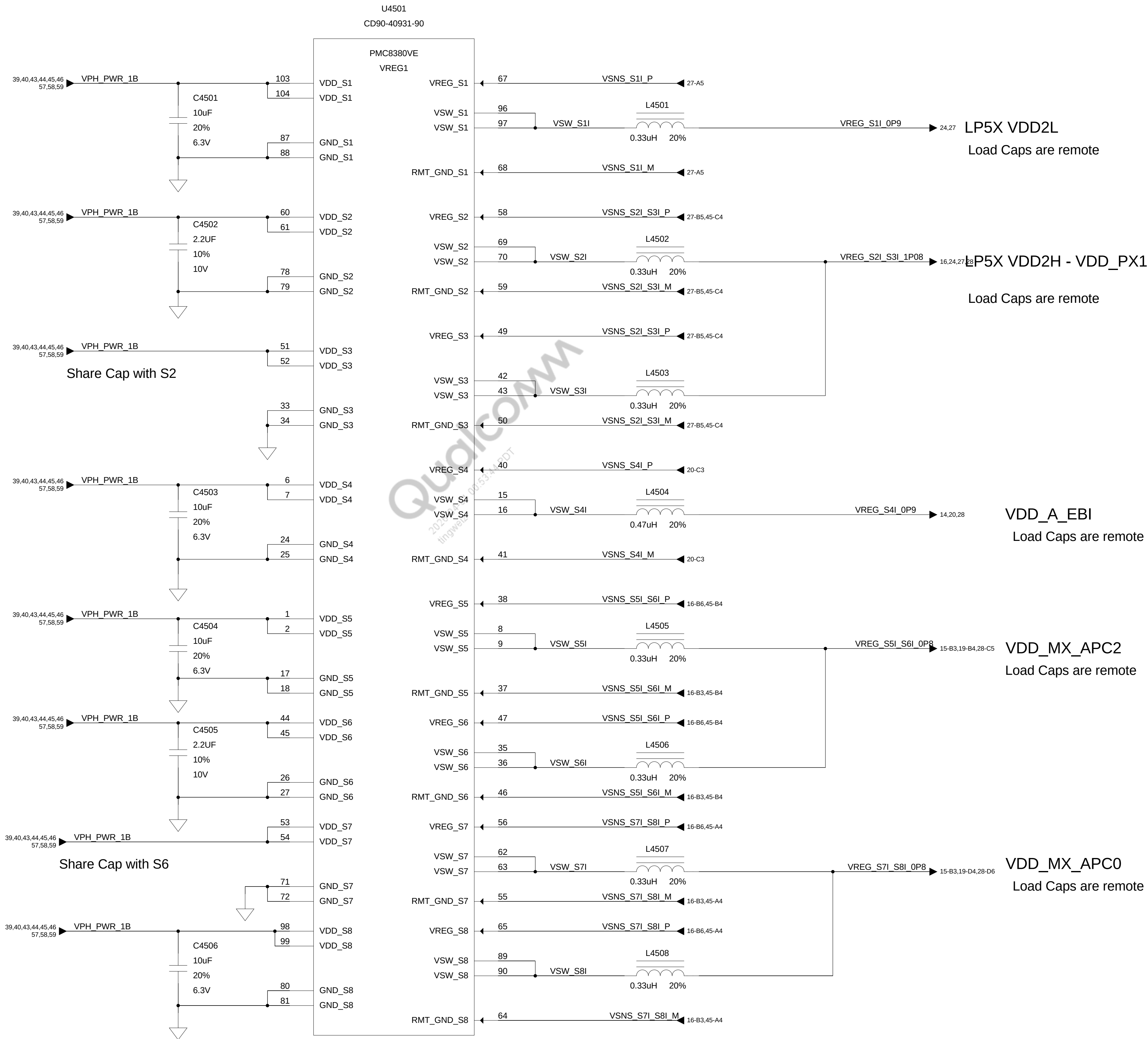
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LTR	DESCRIPTION	DATE	APPROVED
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SID	PMIC	GPIO3	GPIO2	GPIO1
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3	C8D	GND	GND	open
4	C8E	GND	GND	GND
5	C8F	open	GND	GND
6	C8G	GND	open	open
7	C8H	GND	open	GND
8	V8I	GND	open	GND
9	V8J	GND	open	open
A	C8K	open	open	open
B	C8L	open	open	GND
D	C8N	open	GND	open



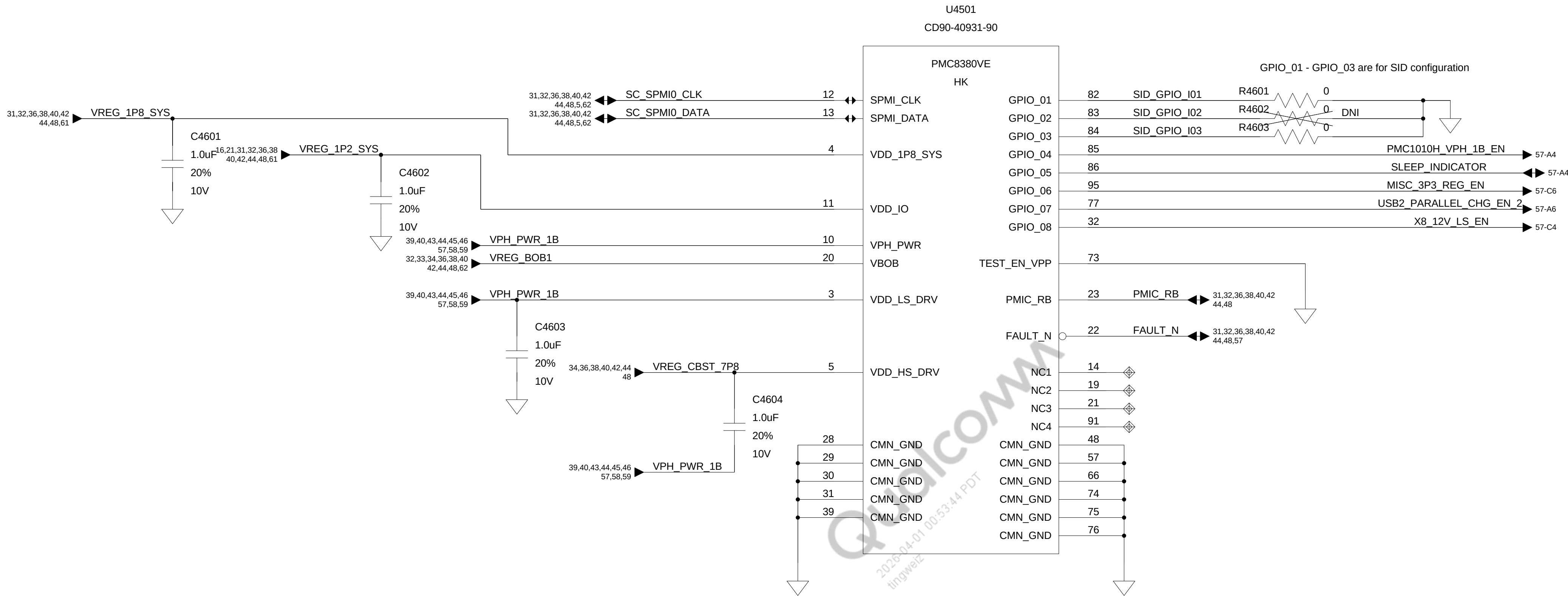
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PMC8380VE_I



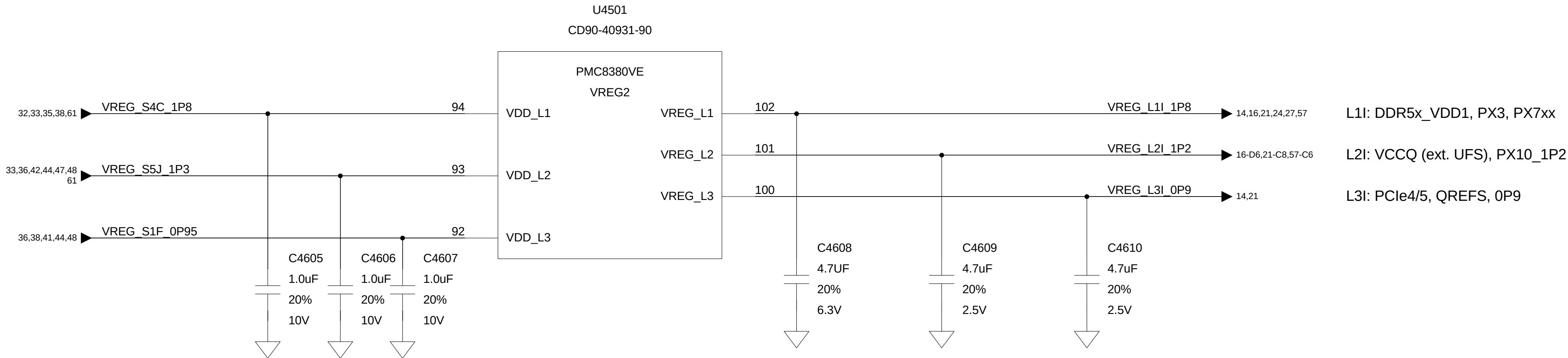
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SCALE NONE	SHEET 45 OF 62	

PMC8380VE_I



REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

SID	PMIC	GPIO3	GPIO2	GPIO1
2	V8C	open	GND	open
3	C8D	GND	GND	open
4	C8E	GND	GND	GND
5	C8F	open	GND	GND
6	C8G	GND	open	open
7	C8H	GND	open	GND
8	V8I	GND	open	GND
9	V8J	GND	open	open
A	C8K	open	open	open
B	C8L	open	open	GND
D	C8N	open	GND	open

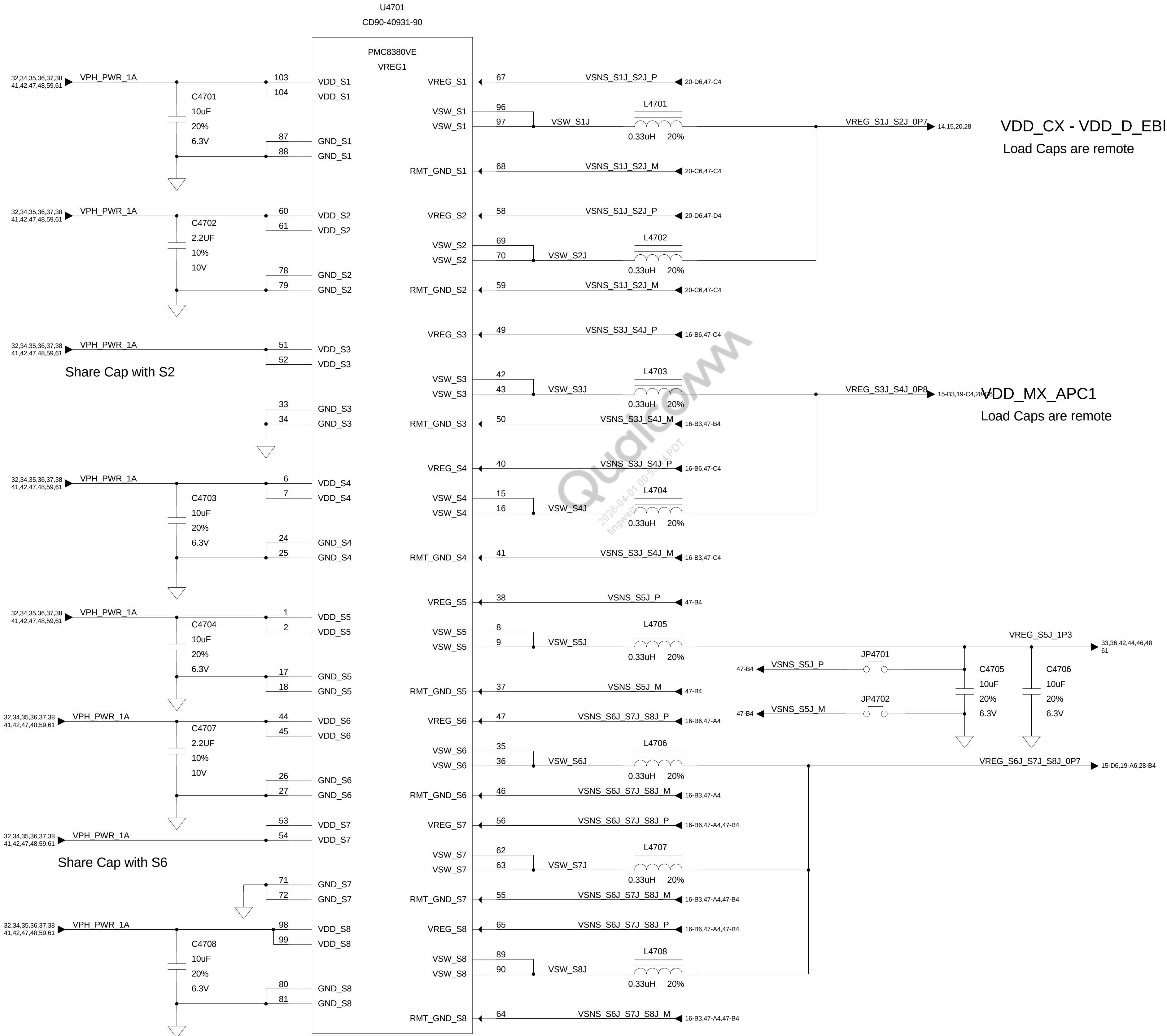


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REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
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PMC8380VE_J

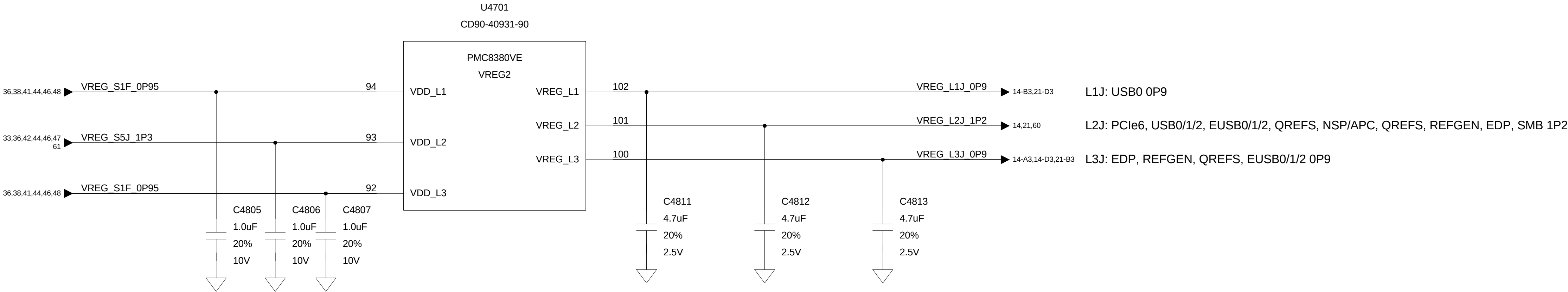
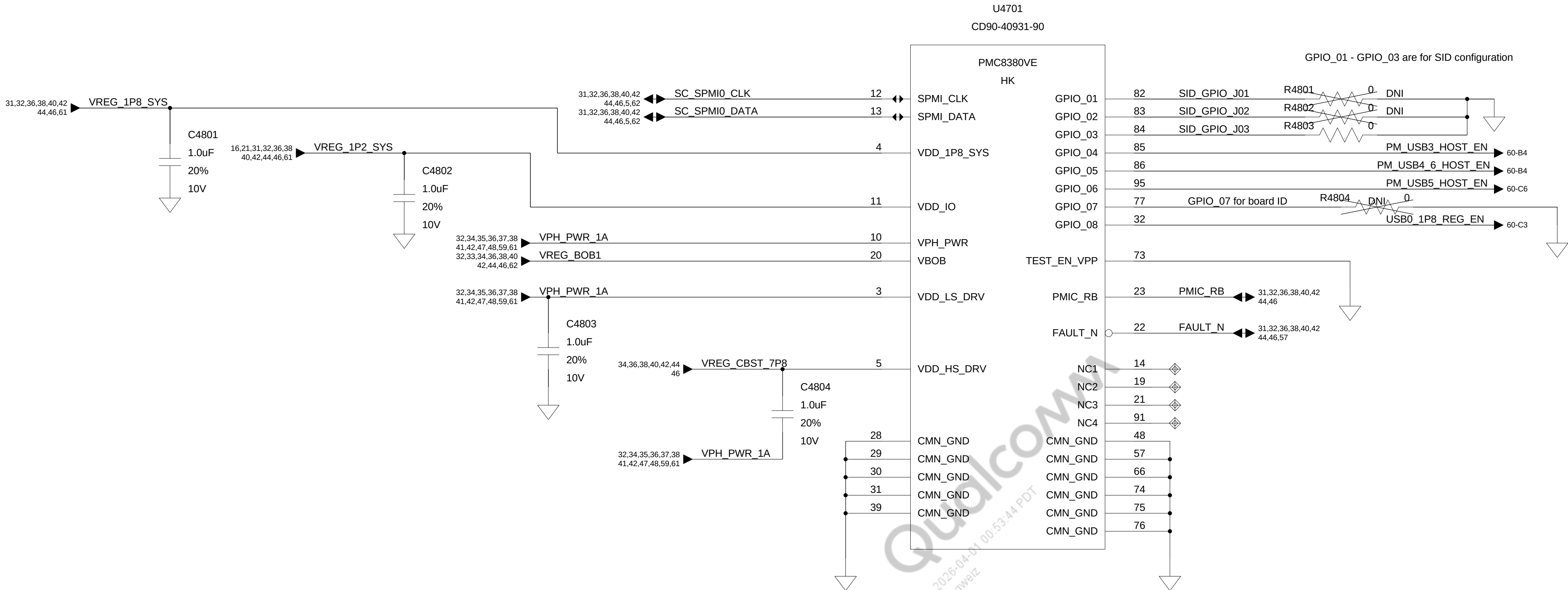


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PMC8380VE_J

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

SID	PMIC	GPIO3	GPIO2	GPIO1
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3	C8D	GND	GND	open
4	C8E	GND	GND	GND
5	C8F	open	GND	GND
6	C8G	GND	open	open
7	C8H	GND	open	GND
8	V8I	GND	open	GND
9	V8J	GND	open	open
A	C8K	open	open	open
B	C8L	open	open	GND
D	C8N	open	GND	open



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SCALE	NONE	SHEET 49	OF 62

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RESERVED_06

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SCALE	NONE	SHEET 51 OF 62	

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RESERVED_08

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SCALE	NONE	SHEET 52	OF 62

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SCALE	NONE	SHEET 53	OF 62

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RESERVED_10

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SCALE	NONE	SHEET 54	OF 62

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REVISIONS			
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Qualcomm
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RESERVED_11

SIZE	DRAWING NO		REV
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SCALE	NONE	SHEET 55	OF 62

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REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		



RESERVED_12

SIZE D	DRAWING NO LD20-90411-764		REV A
SCALE NONE		SHEET 56	OF 62

BGA PAD

D

D

C

C

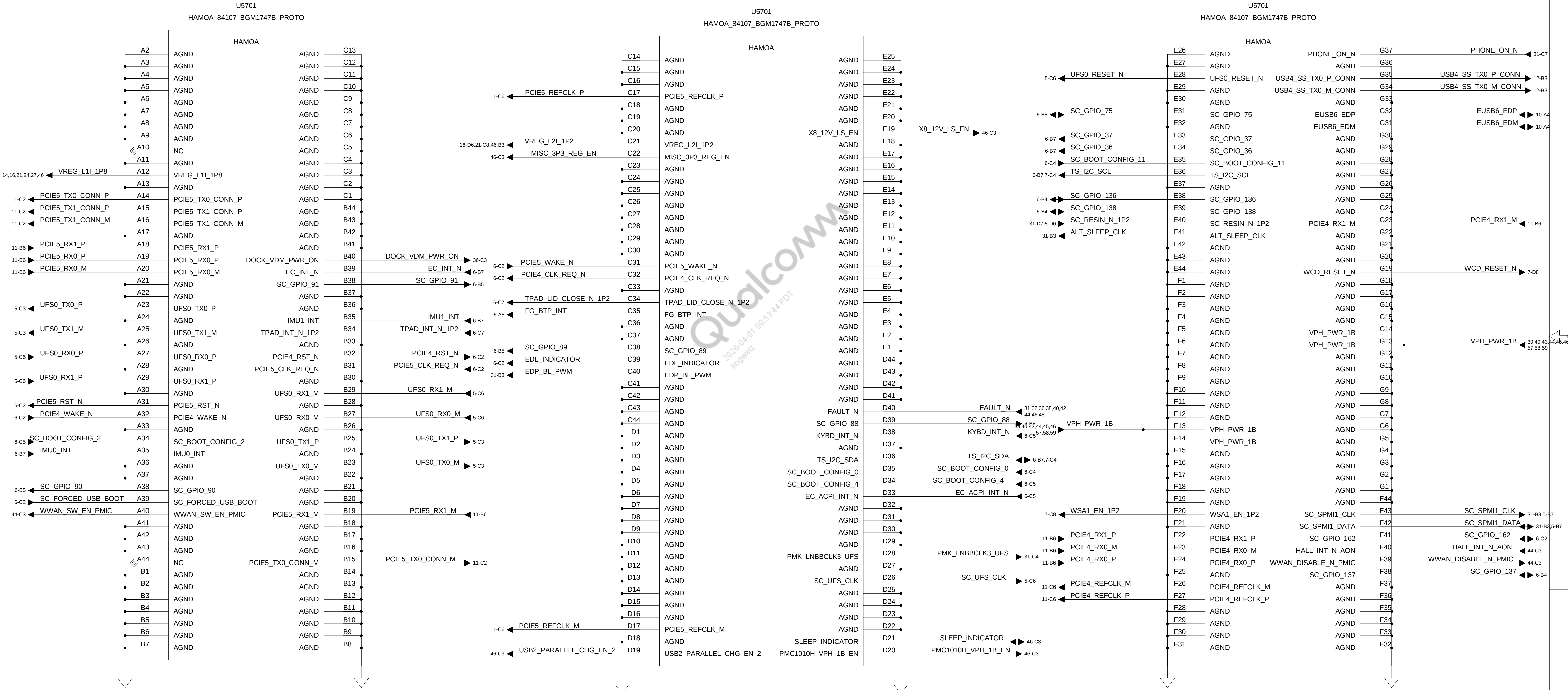
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B

A

A

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		



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SCALE NONE	SHEET 57 OF 62	

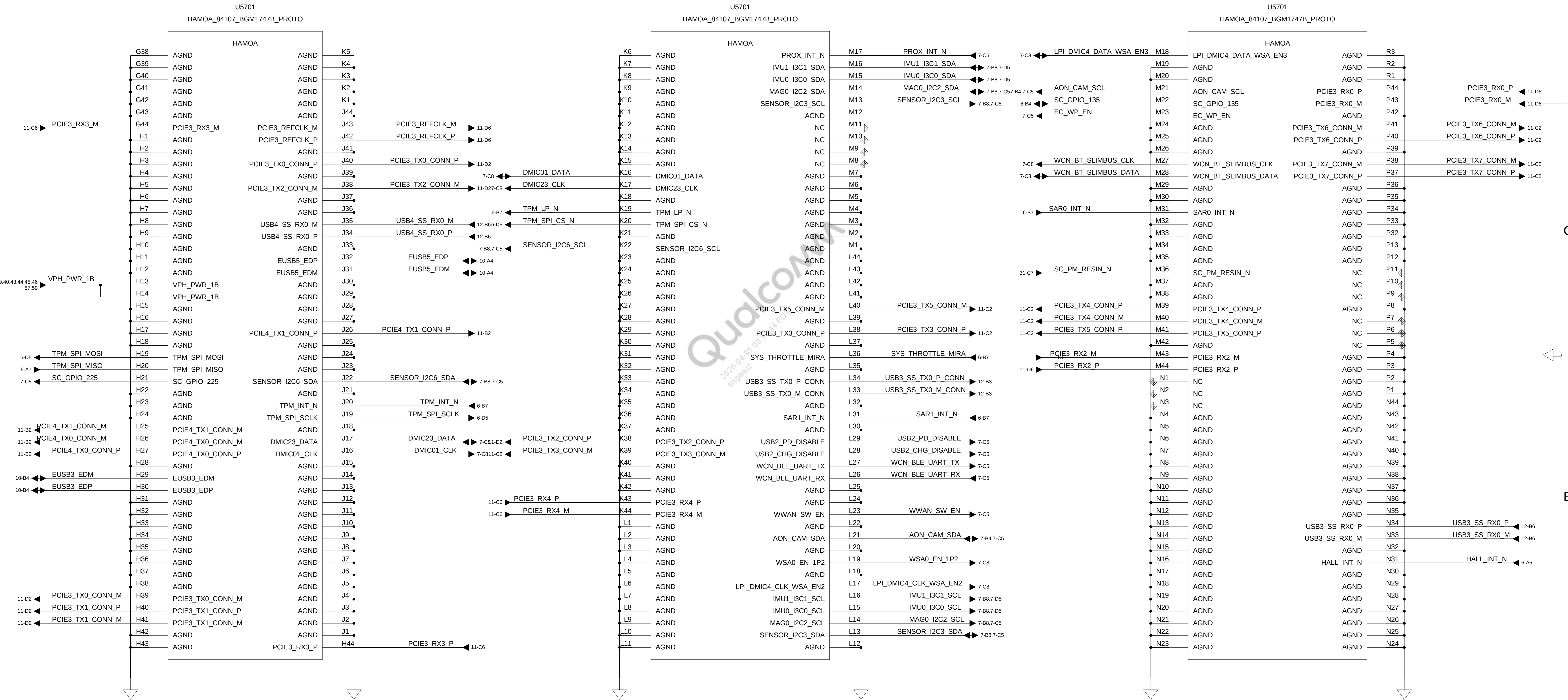
BGA PAD

D

C

B

A



MODULE_IO_2

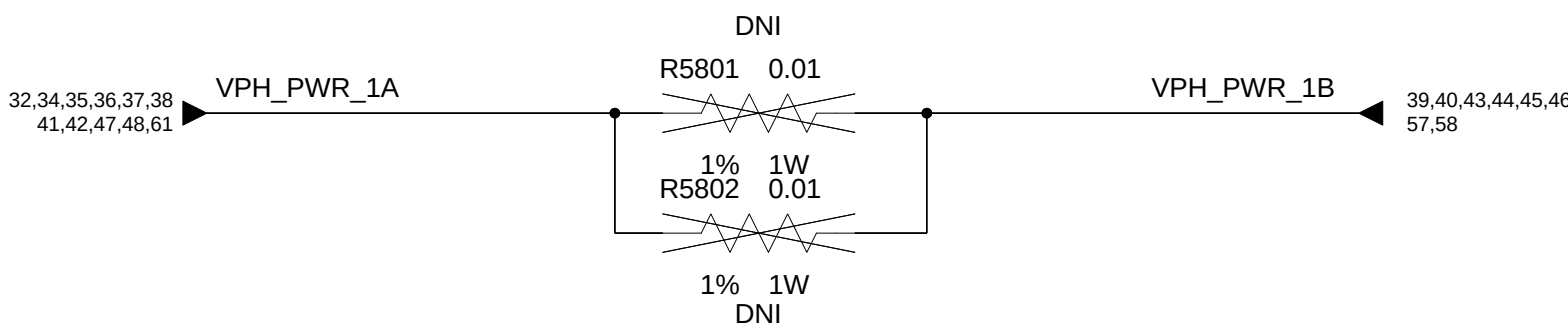
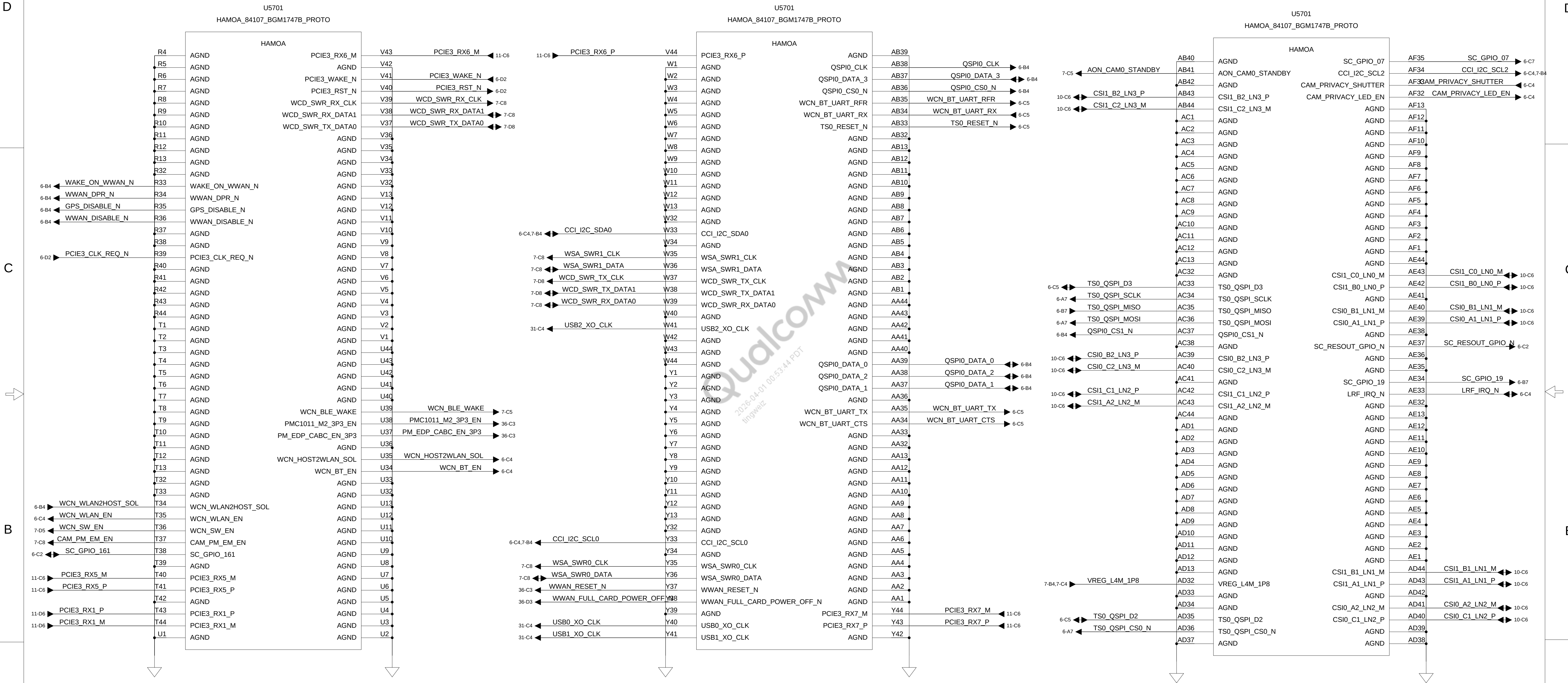
REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	INITIAL RELEASE		

SIZE	DRAWING NO	REV
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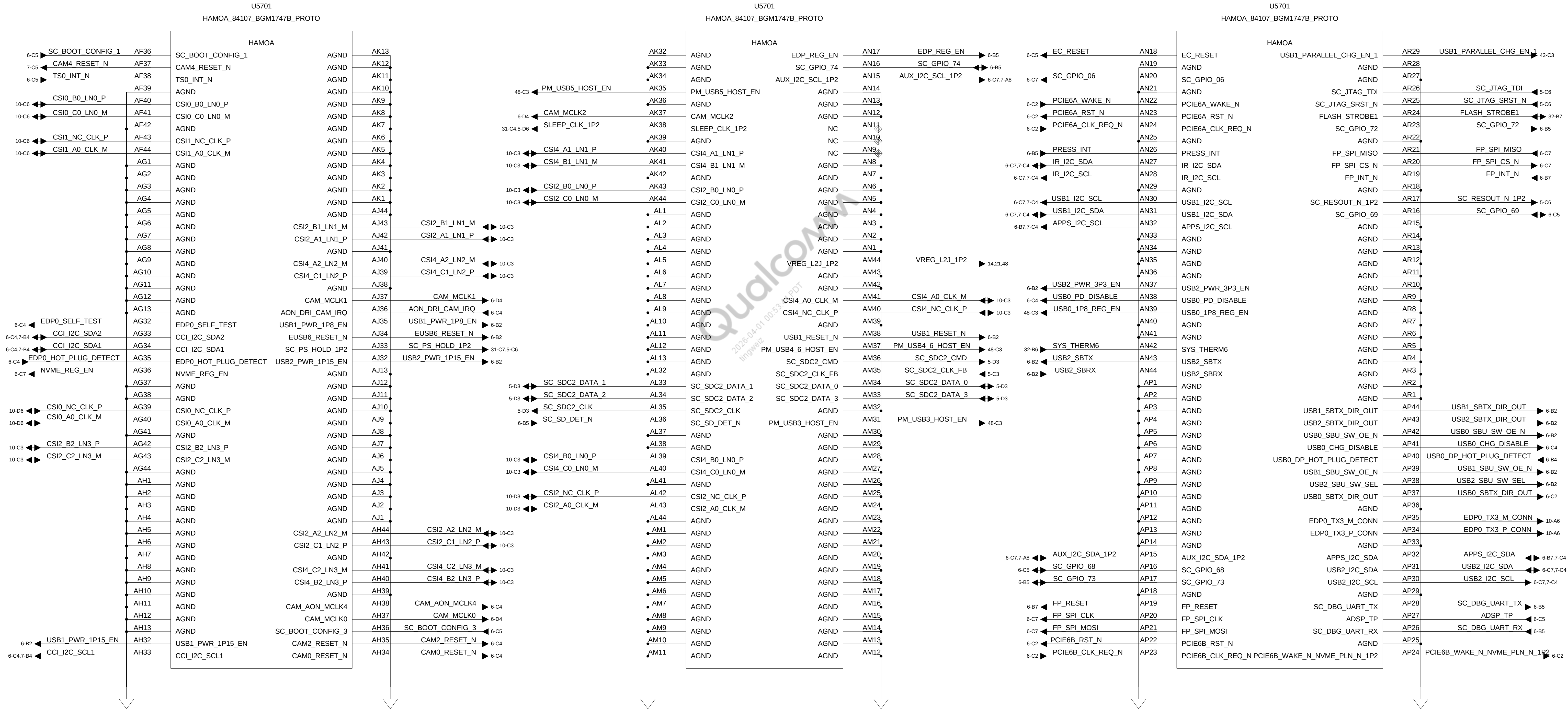
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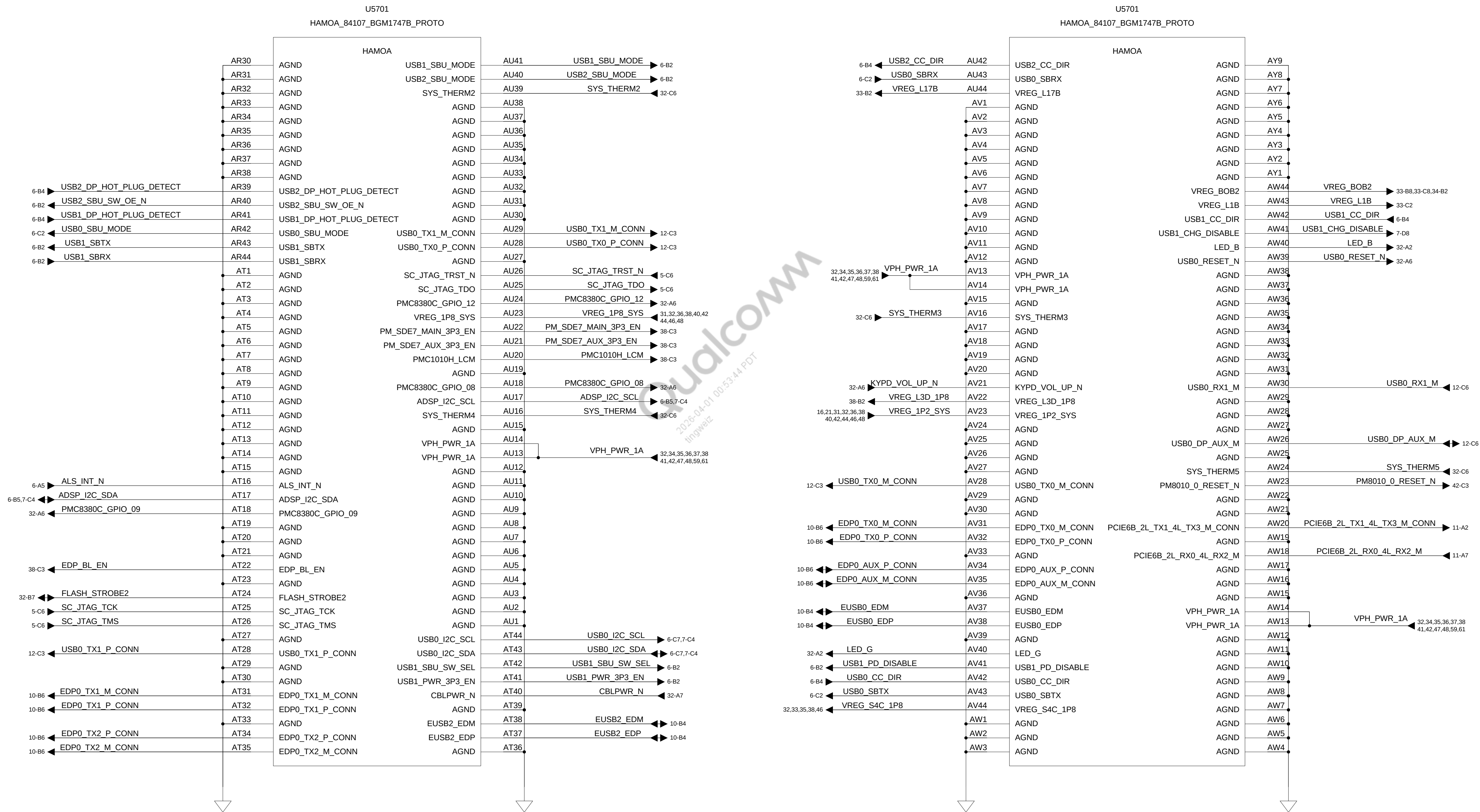
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